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Virtuoso® Layout Pro: T6 Constraint-Driven Flow and Power Routing

Course Version IC 25.1

Lab Manual **Revision 1.0**

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Virtuoso Layout Pro: T6 Constraint-Driven Flow and Power Routing

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Module 1: About This Course

There are no labs in the module

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Module 2: Constraint-Driven Flow

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Lab 2-1 Analyzing VSE XL and VLS XL Features: Usability Improvements in the Constraint-Driven Flow

Objective: To explore the usability improvements in the Constraint-Driven Flow.

Many new features and enhancements have been introduced in the Constraint-Driven Flow to improve the usability and make the flow more seamless.

In this lab, you explore the following features:

- Rail Constraint supported by Analog Automatic Placement (AAP) and Constraint-Aware Editing (CAE).
- Cell Boundary Constraint that includes Area Utilization.
- Usability improvements in the Constraint Manager (CM) (Show Selected and Constraint Name Display).

Opening the Software

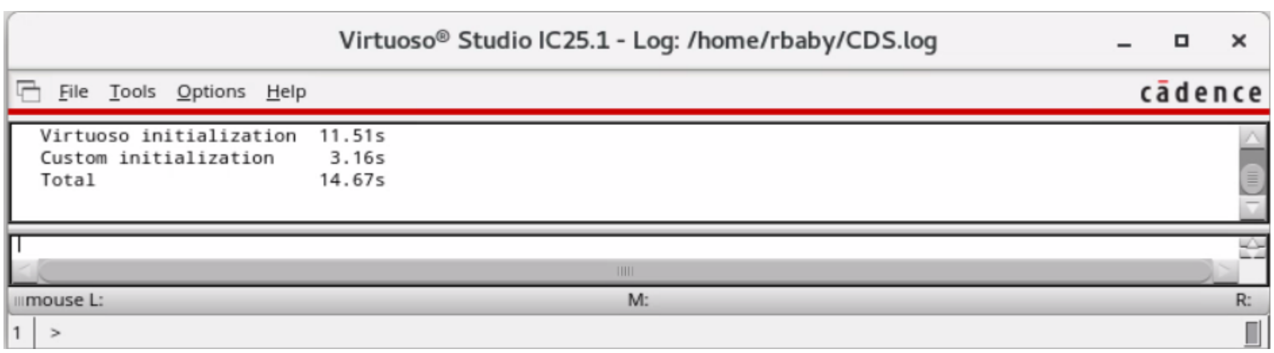
1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

```
cd ./VLPT6_IC_25_1/mod02_Constraint-DrivenFlow
```

2. To start the Cadence® Virtuoso® software, enter this command in the same terminal window:

```
virtuoso &
```

The Command Interpreter Window (CIW) appears.



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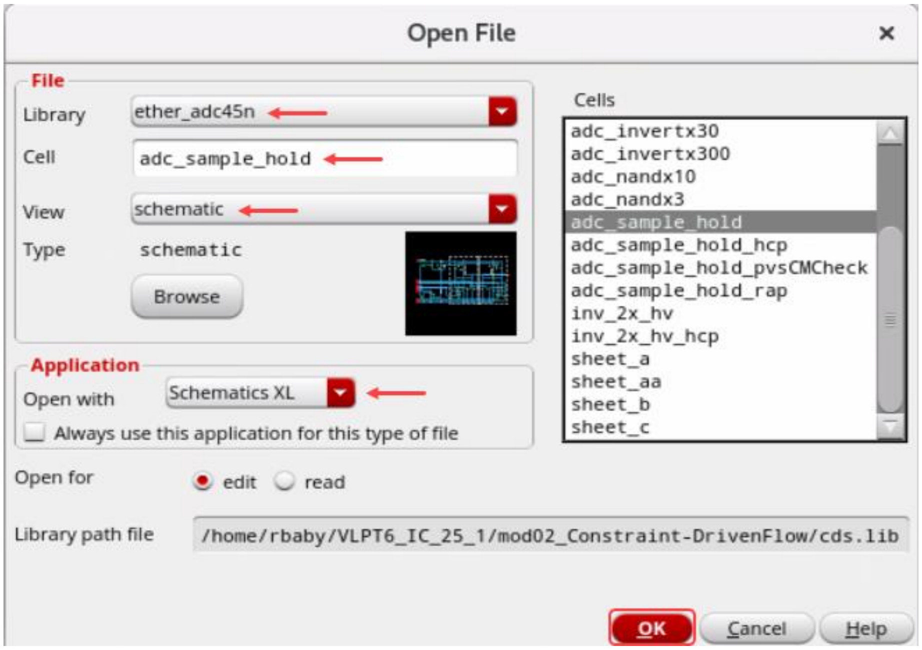
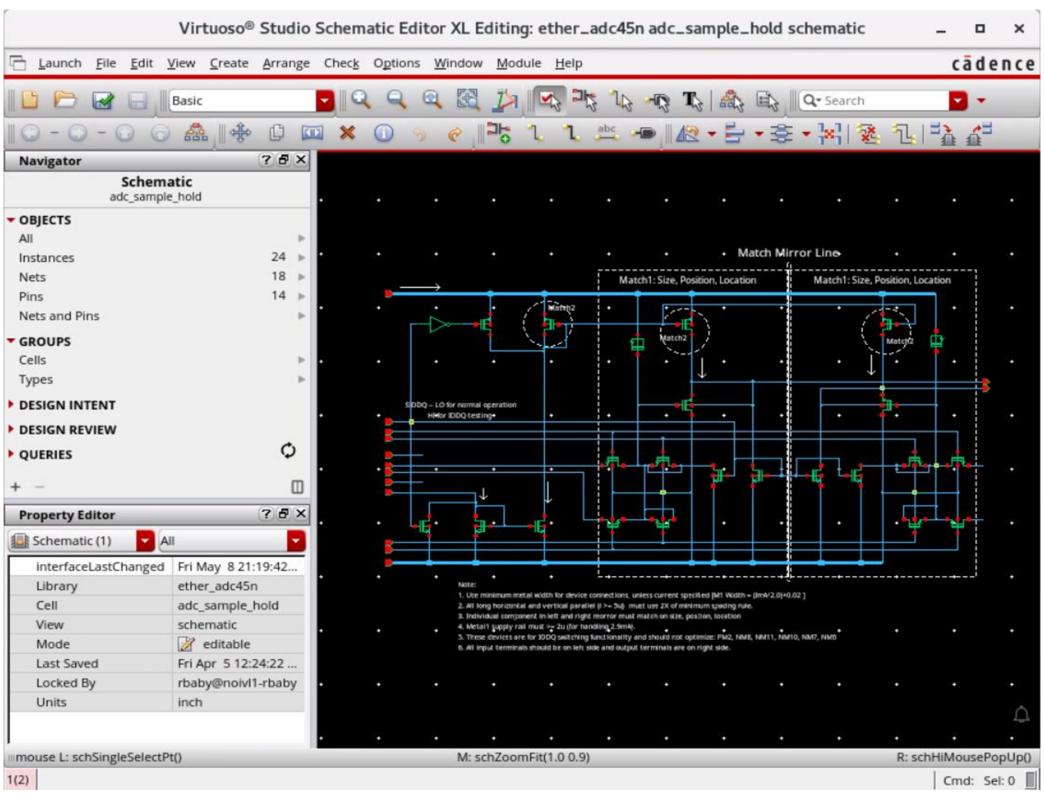
3. In the CIW, choose File – Open menu command and enter these values in the form.

Library	ether_adc45n
Cell	adc_sample_hold
View	schematic



The schematic cellview opens in **XL mode**.



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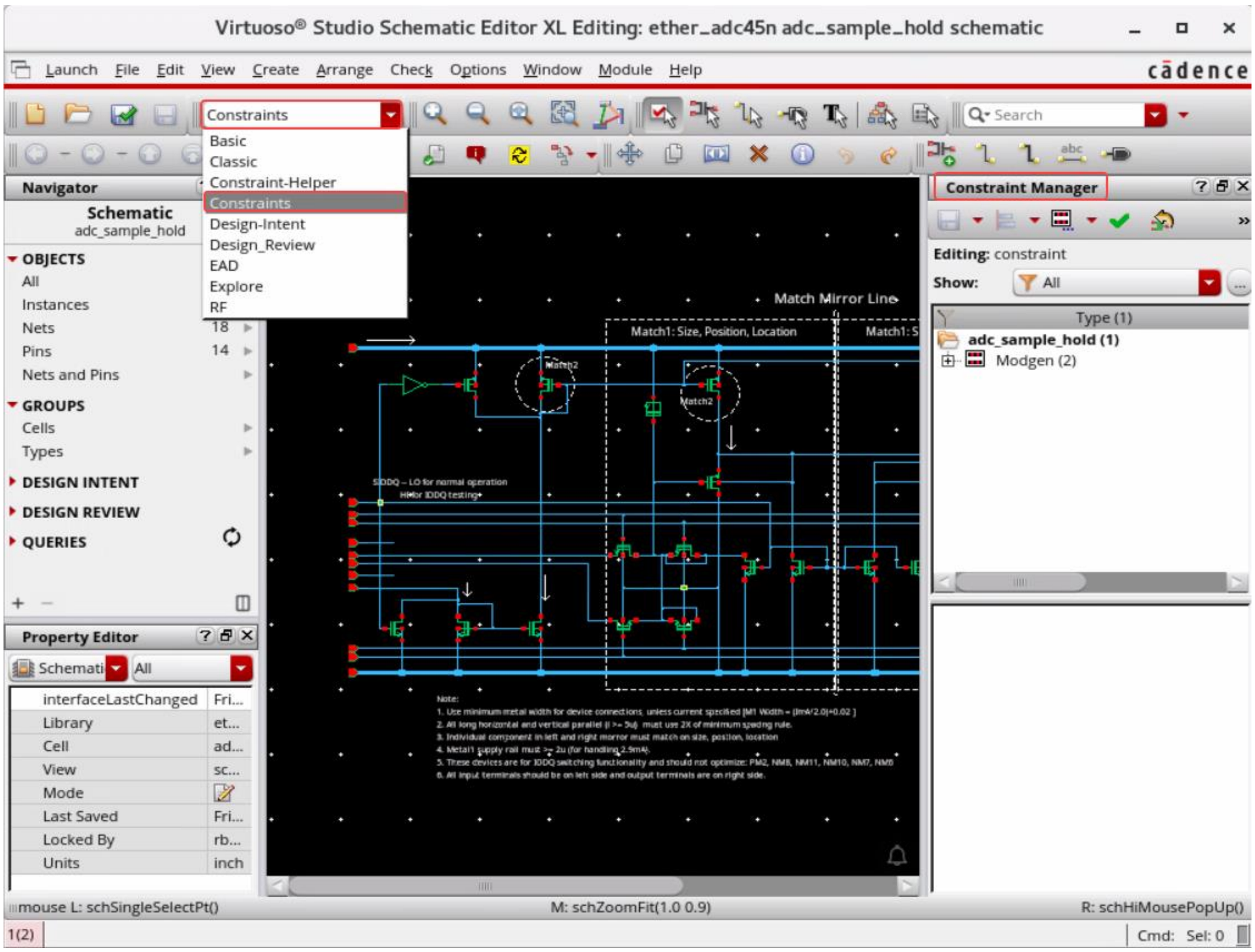
A. Rail Constraint Supported by AAP and CAE

The **Rail** constraint allows users to specify rails in the layout for nets such as *power* and *ground*. A **Rail** constraint can be specified in the Constraint Manager either in the schematic or in the layout view.

In the schematic window, do the following:

1. In the **Workspace Configuration**, change the workspace to *Constraints*.



Navigator, Property Editor, and Constraint Manager (CM) assistants are displayed in the schematic window.

2. Choose the **Load From** pull-down cyclic field from the Constraint Manager.

   The Load Constraints form opens.

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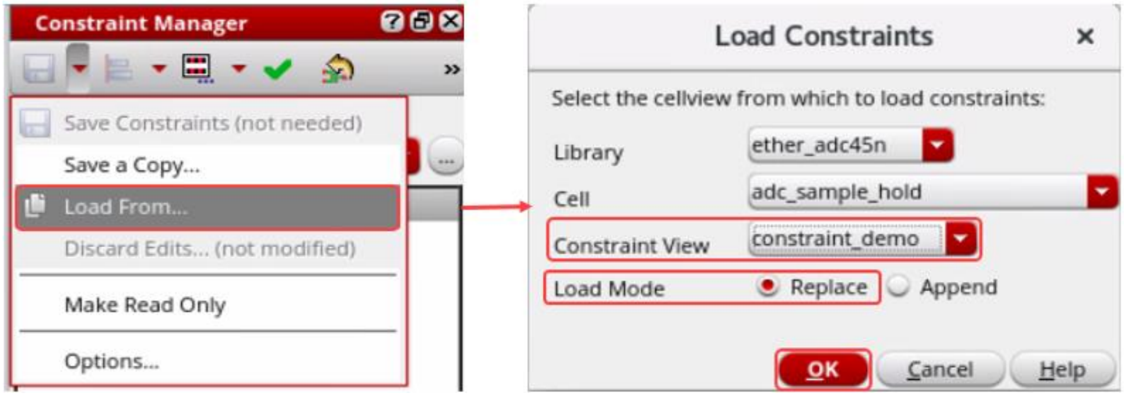
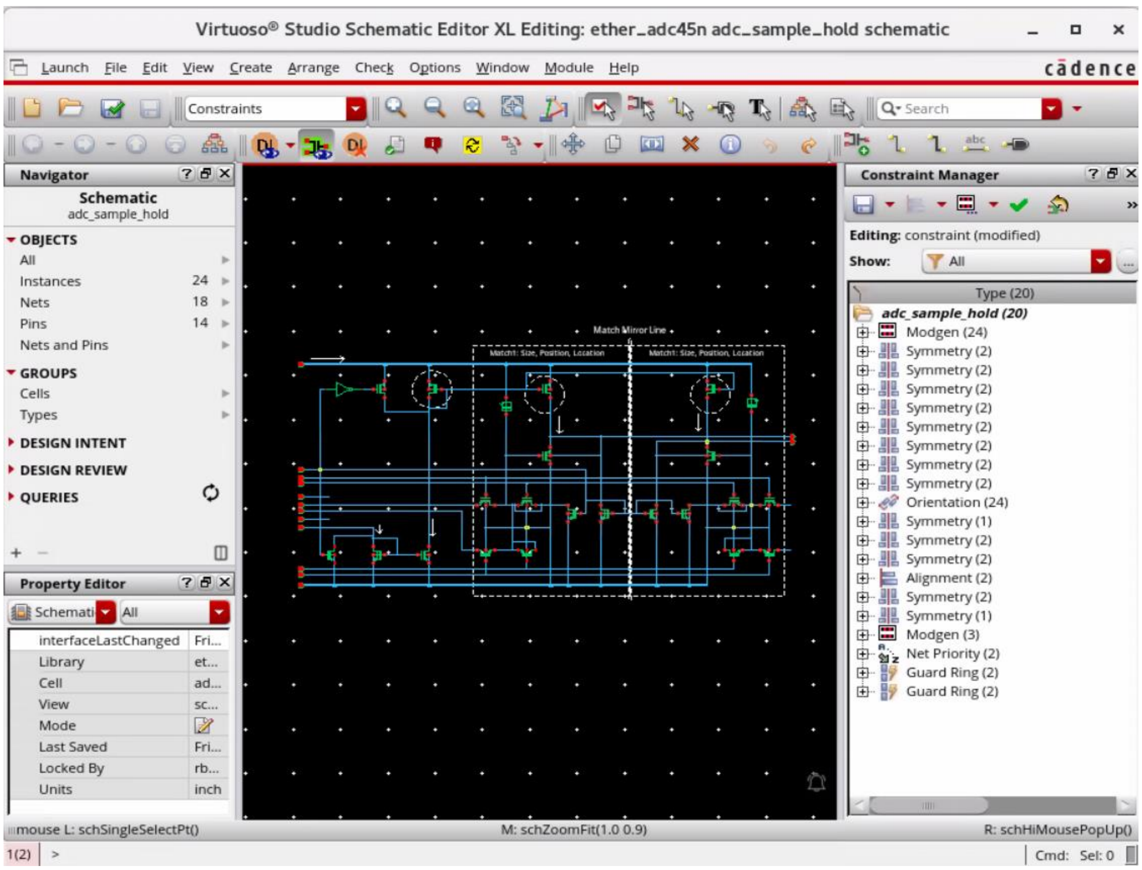
3. Load the pre-saved constraints from the **Constraint View:** *constraint_demo* in the Replace mode in the **Load Constraints** form, as shown here.



4. Click **OK**.



The Constraint Manager assistant gets populated with the constraints loaded from the **Constraint View**: *constraint_demo*.

5. In the schematic window, set the **OBJECTS** category in the Navigator assistant to **Nets**.

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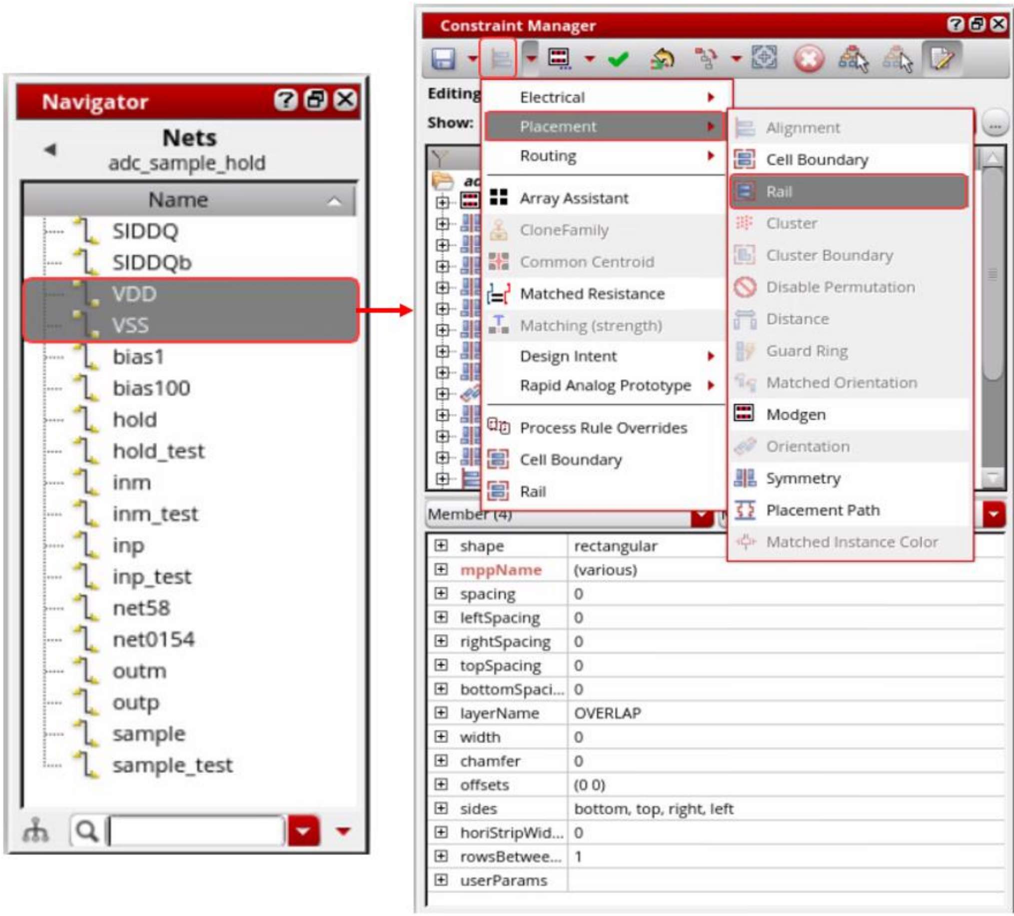
6. In the schematic window, select the nets **VDD** and **VSS** in the Navigator assistant and create the **Rail** constraint on both the nets from the Constraint Manager pull-down, as shown here.



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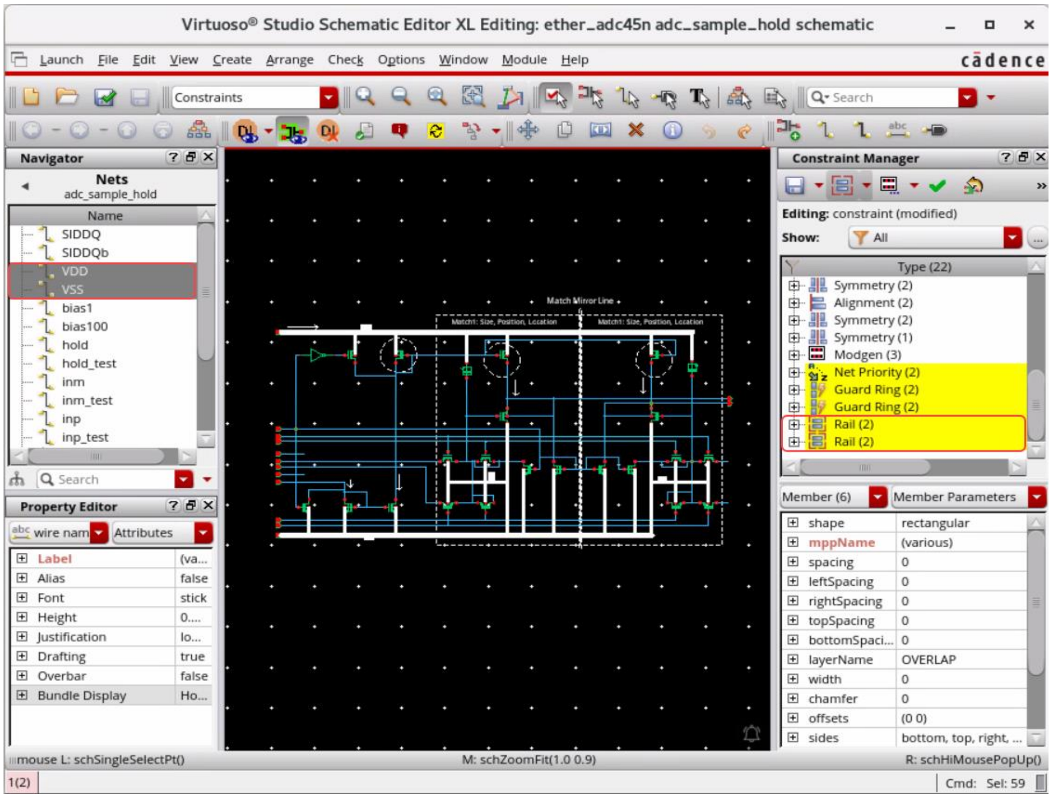
The **Rail** constraint is created on both the nets **VDD** and **VSS,** as shown here.



7. Modify the Rail constraint parameters as follows:

For **VDD** net:

layerName: Metall

OffsetType: top

For **VSS** net:

layerName: Metall

OffsetType: bottom

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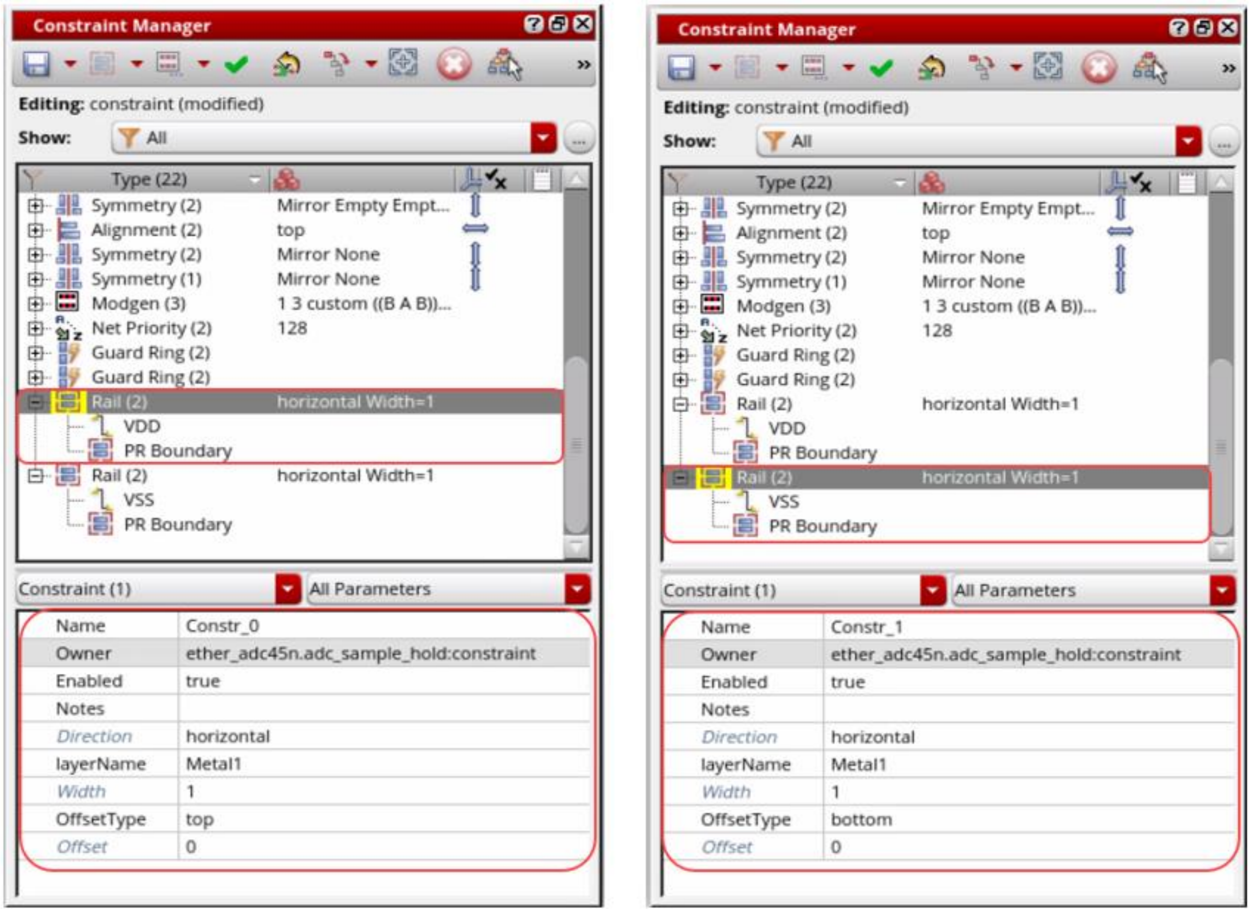

In the following section, you will review the results once a layout is generated and *constraints* are transferred to the layout side.

B. Cell Boundary Constraint That Includes Area Utilization

The **Boundary Area** constraint has been revamped and consolidated with the **Area Utilization** constraint to the new **Cell Boundary** constraint.

The *mode* parameter allows users to specify feasible combinations of the following as range (min / max): Width, Height, Area, Aspect Ratio, and Area Utilization.

The constraint is followed by the Analog Placer (and the Block Placer) and the *CAE* flags any violation. However, the *CAE* does not enforce it.

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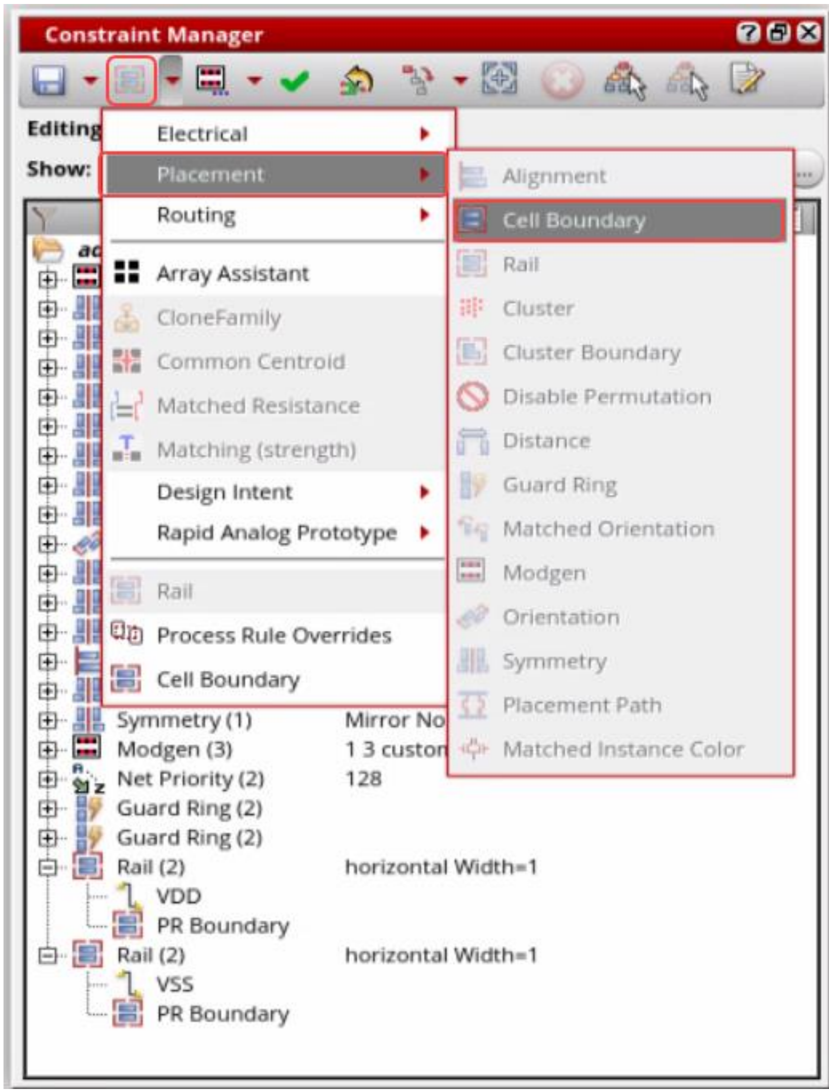
1. From the Constraint Manager pull-down, create the **Cell Boundary** constraint as shown here. No explicit selection is required except for the L1 softblock boundaries. (*prBoundary* is the default member.)



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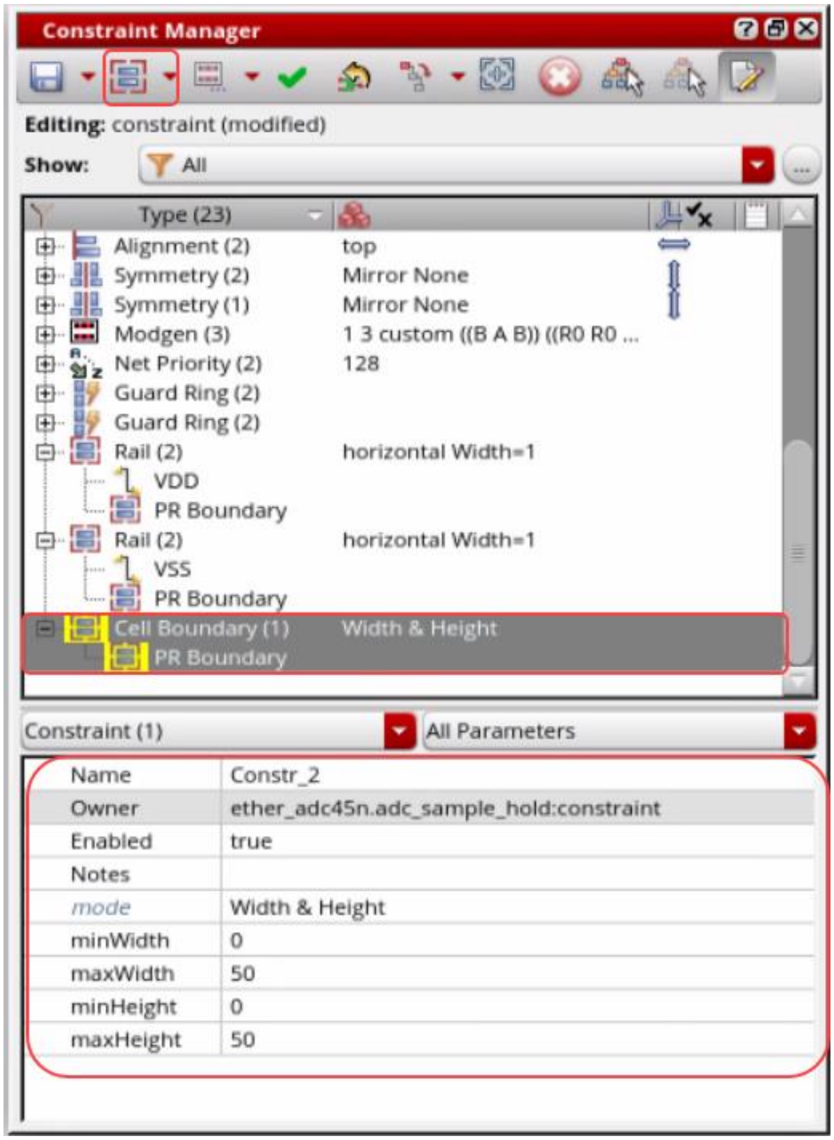
The **Cell Boundary** constraint is created, as shown here.



2. Specify the **Cell Boundary** constraint parameters:
   - Width & Height
   - Width & AreaUtilization
   - Width & AspectRatio
   - Width & Area
   - Height & AreaUtilization
   - Height & AspectRatio
   - Height & Area
   - AreaUtilization & AspectRatio
   - AspectRatio & Area

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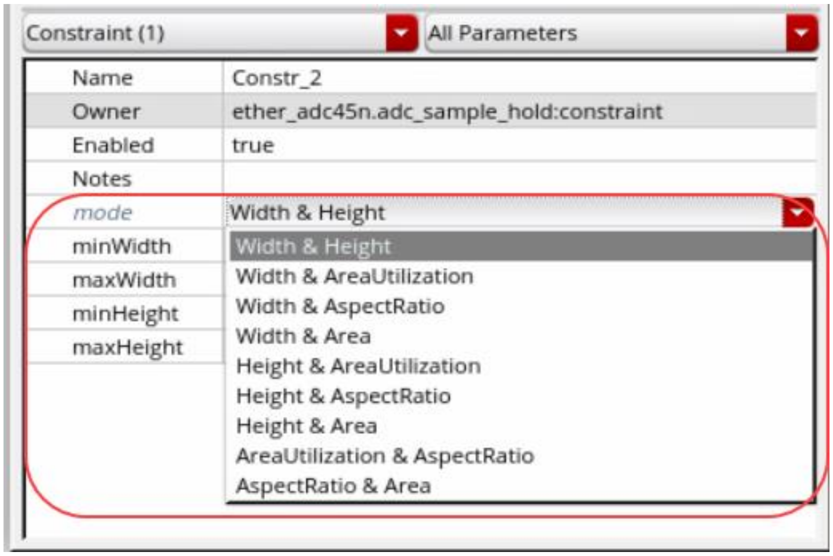
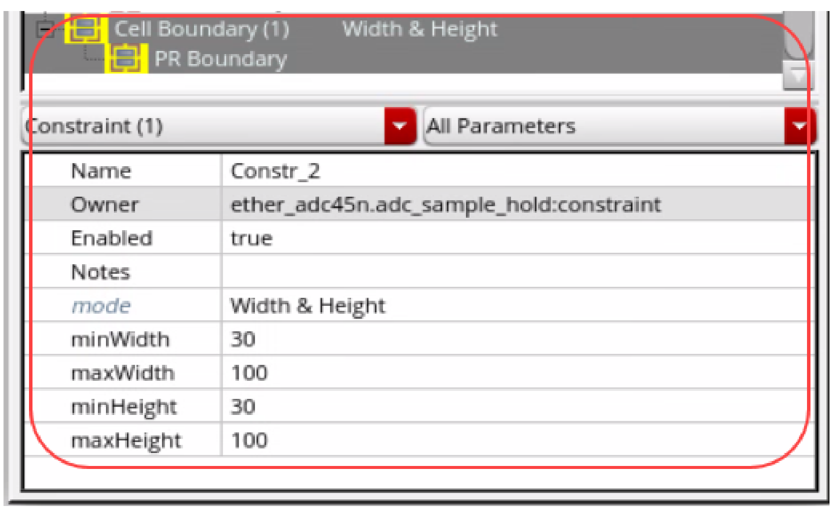
Allowable parameter combinations for *mode* are as shown here.



3. Set the **Cell Boundary** constraint as the following:
   - *mode* – Width & Height
   - minWidth – 30
   - maxWidth – 100
   - minHeight – 30
   - maxHeight – 100



Creating New Layout Cellview

1. In the schematic, choose **Launch – Layout EXL** to bring up the *Startup Option* form.

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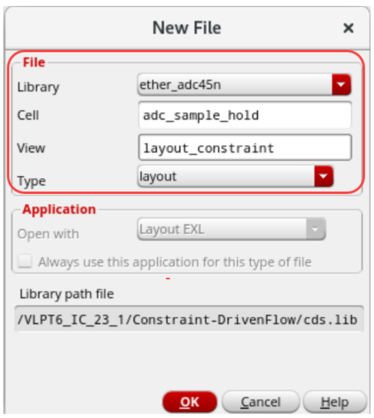
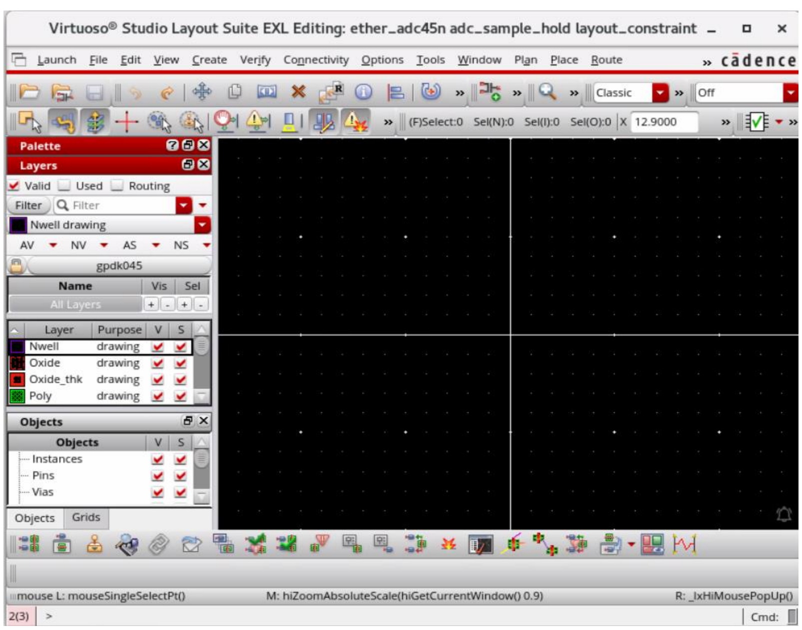
2. Select **Create New** in Layout and **Automatic** in Configuration to create a new layout view and click **OK** to open the **New File** form.

3. Fill out the **New File** form as shown by clicking in the **View** field and changing the view name to *layout_constraint*.



4. Click **OK**.

   The Virtuoso Layout Suite EXL Palette opens.



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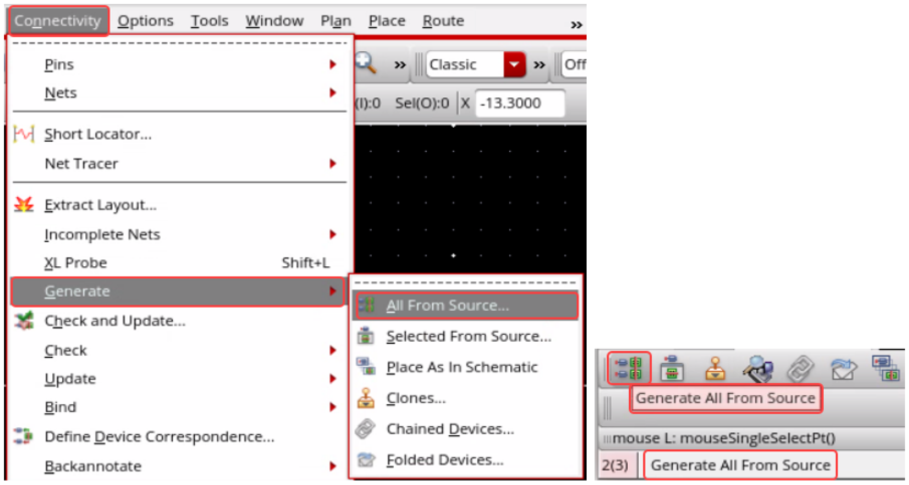
5. Generate all from the source by choosing either the **Connectivity – Generate – All From Source** menu entry or by clicking the **Generate All From Source** (**GFS**) icon.



The Generate Layout form appears, as shown here.

6. Click **OK** in the Generate Layout form with the default settings.

   The placement looks as shown in the next step.

7. Switch the **Workspace** to **Constraints** to turn on the Constraint Manager assistant.

   All the constraints including the **Rail** and **Cell Boundary** constraints get transferred to the Layout (*Status – not checked*).

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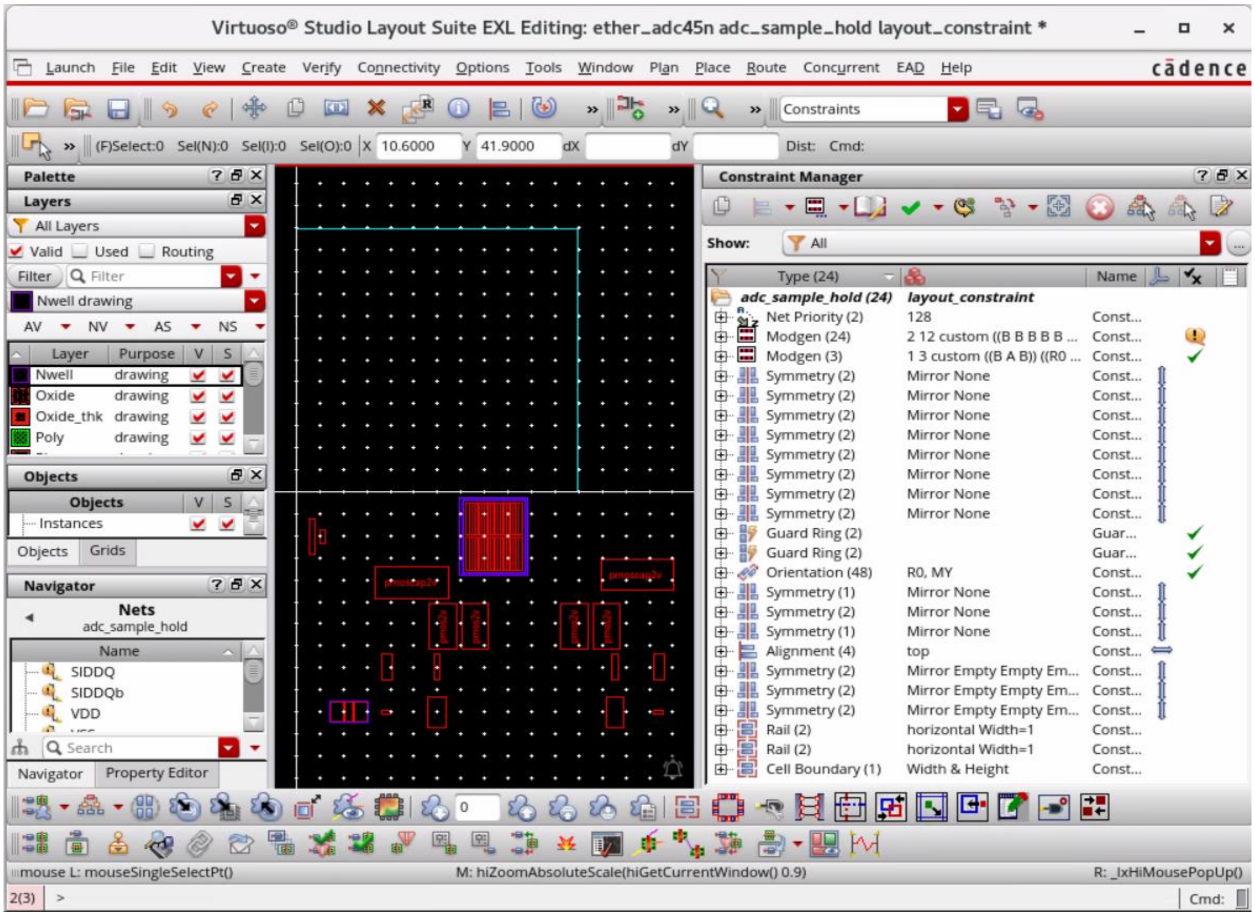
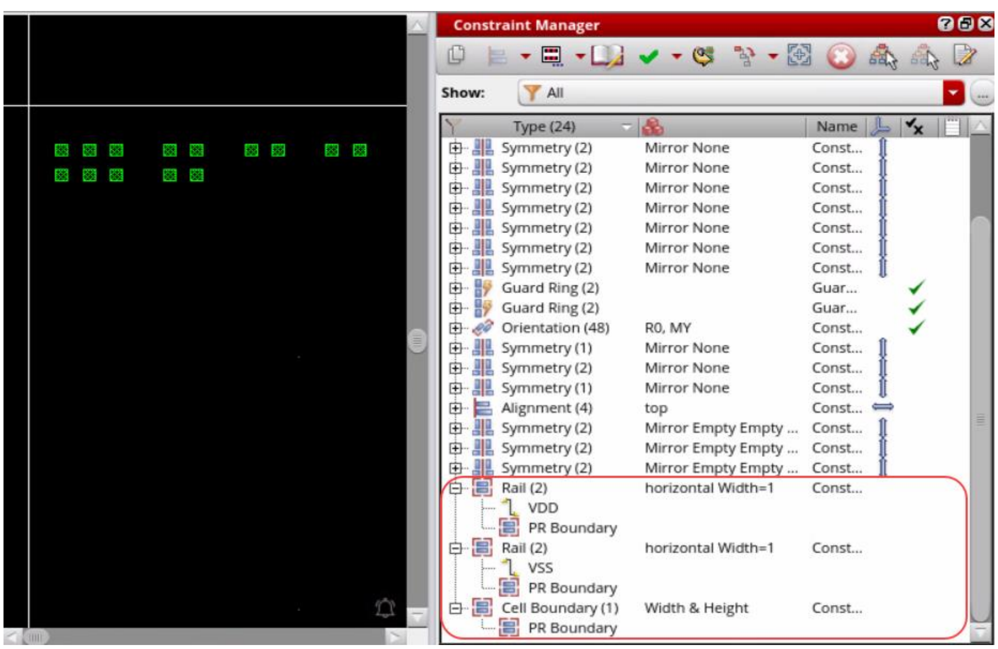
Note: Both constraints can also be added directly on the layout side.



8. Expand the **Rail** and **Cell Boundary** constraints to review the constraint members.



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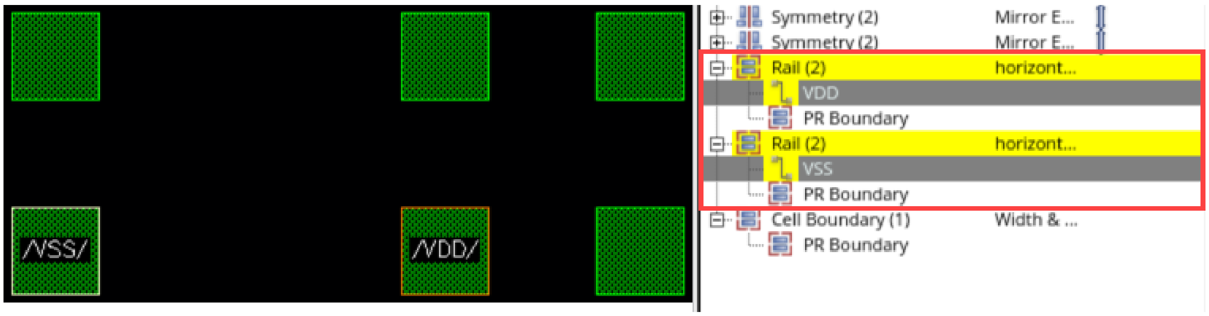
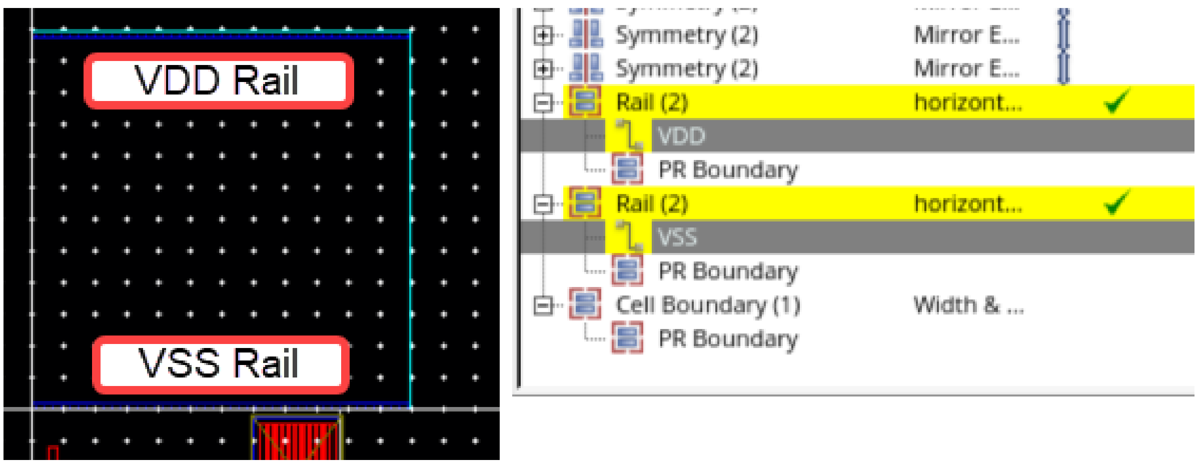
9. Select the **VDD** and **VSS** pins using the Navigator assistant or directly from the layout canvas and move the pins.

   Rails are created when the pins are moved, and the constraints show up as passing, which can be verified in the Constraint Manager (CM).

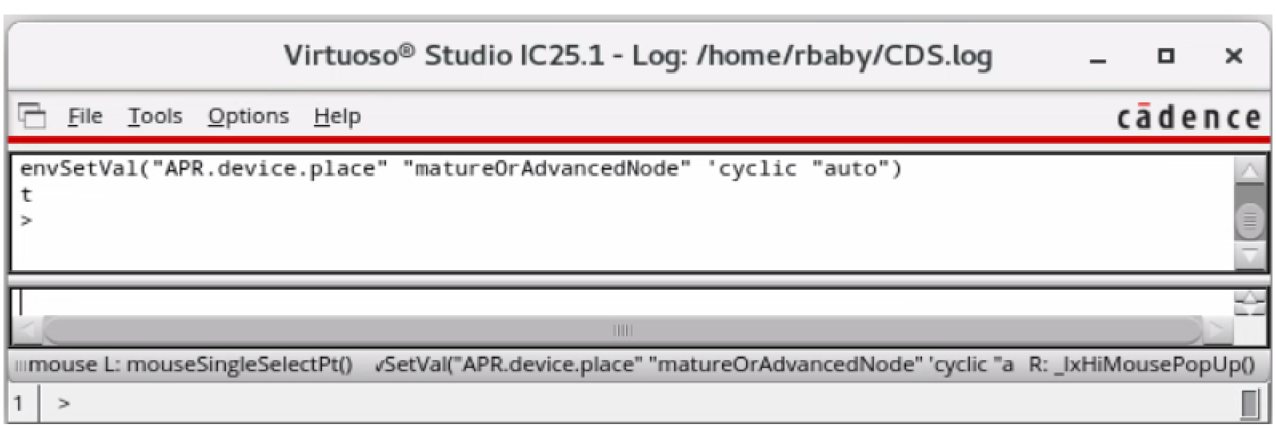




10. Discard all the edits done in the layout window by using the **File – Discard Edits** menu command and clicking **Yes** in the Discard Edits pop-up form.

11. To run the Analog Automatic Placement, execute the following command in the CIW window:

```
envSetVal("APR.device.place" "matureOrAdvancedNode" 'cyclic "auto")
```



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12. In the layout window, choose **Window – Assistants – Auto P&R** or select the workspace **Auto_Place_Route** from the Workspace Configuration.

    The Auto P&R assistant is turned on.

13. You will run the Automatic Analog Placement, as shown here.

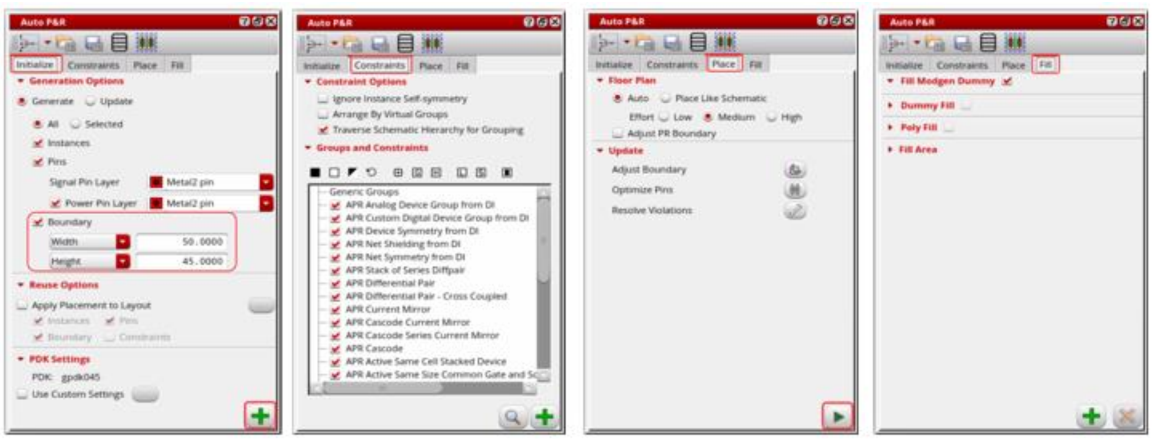
    a. In the **Initialize** tab, keep the following:

    - In the **Boundary** field, keep *Width* as **50** and *Height* as **45**.

    Note: Try different **Boundary** settings to achieve optimum constraints-based placement.

    - Keep the other values as default.

    b. Click the **Apply**  icon to initialize the design.



    c. In the **Constraints** and the **Fill** tabs, keep the default settings.

    d. In the **Place** tab, with the default settings, click the **Run Analog Placer**  icon.

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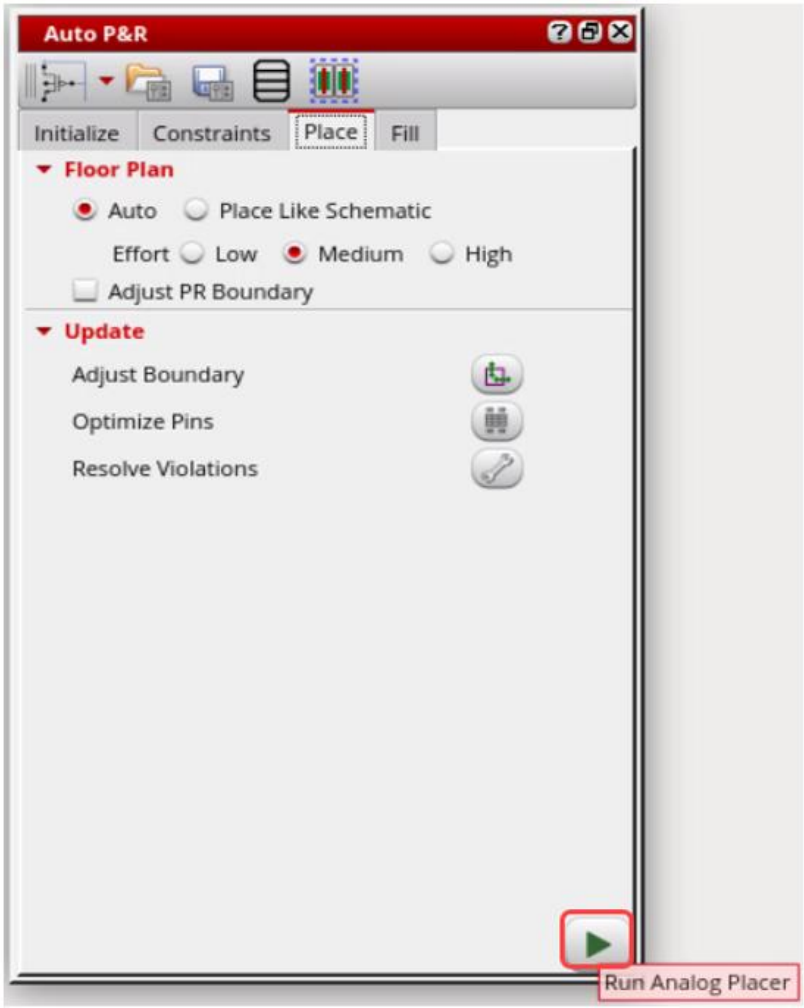
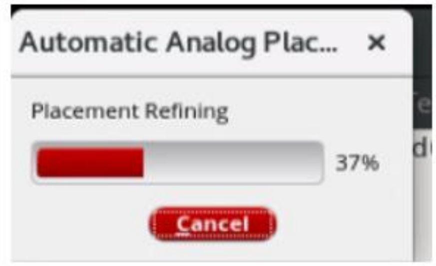
Automatic Analog Placer runs, as shown here.





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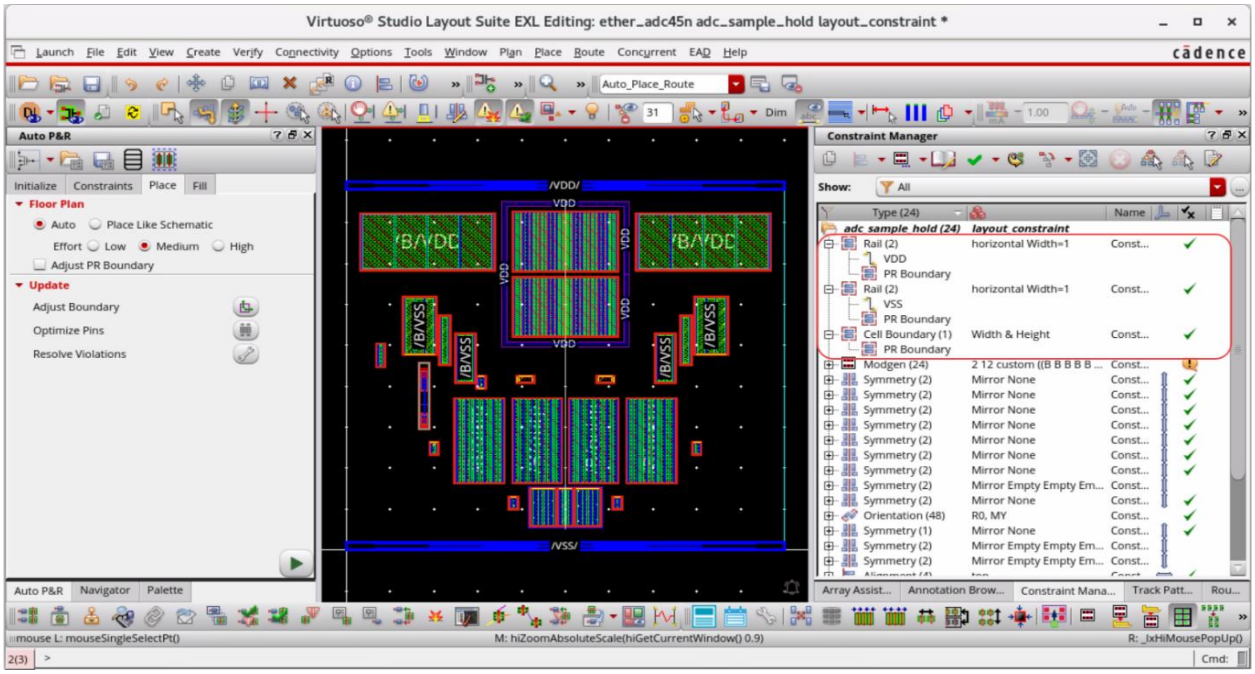
14. An example placement is shown here. All the constraints including the **Rail** and **Cell Boundary** constraints are transferred to the layout and the constraints show up as passing which can be verified in the Constraint Manager (CM).



15. Modifying the constraint in the CM dynamically updates its status. Try changing the **VDD Rail** constraint as follows:

    a. Change the **Rail** layer in the Constraint Manager on the schematic side, say for example from **Metal1** to **Metal2**.

    b. Followed by **Update layout constraints from schematic**  on the layout side.

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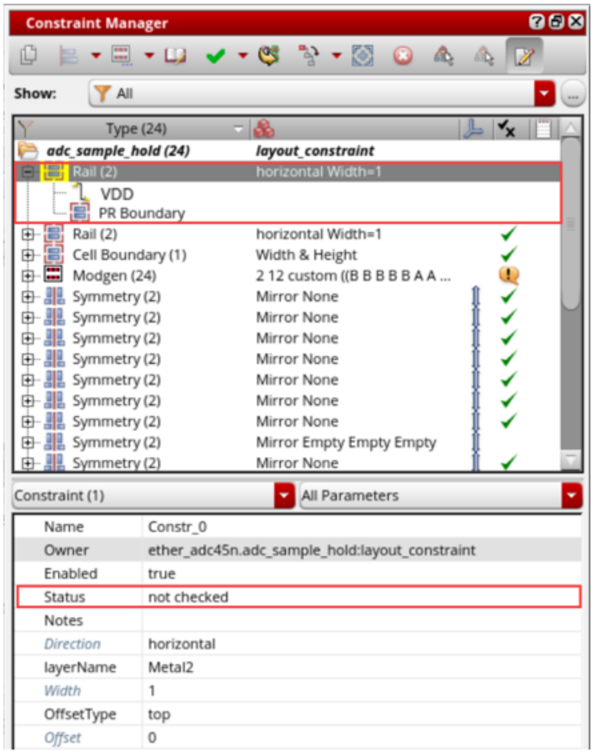
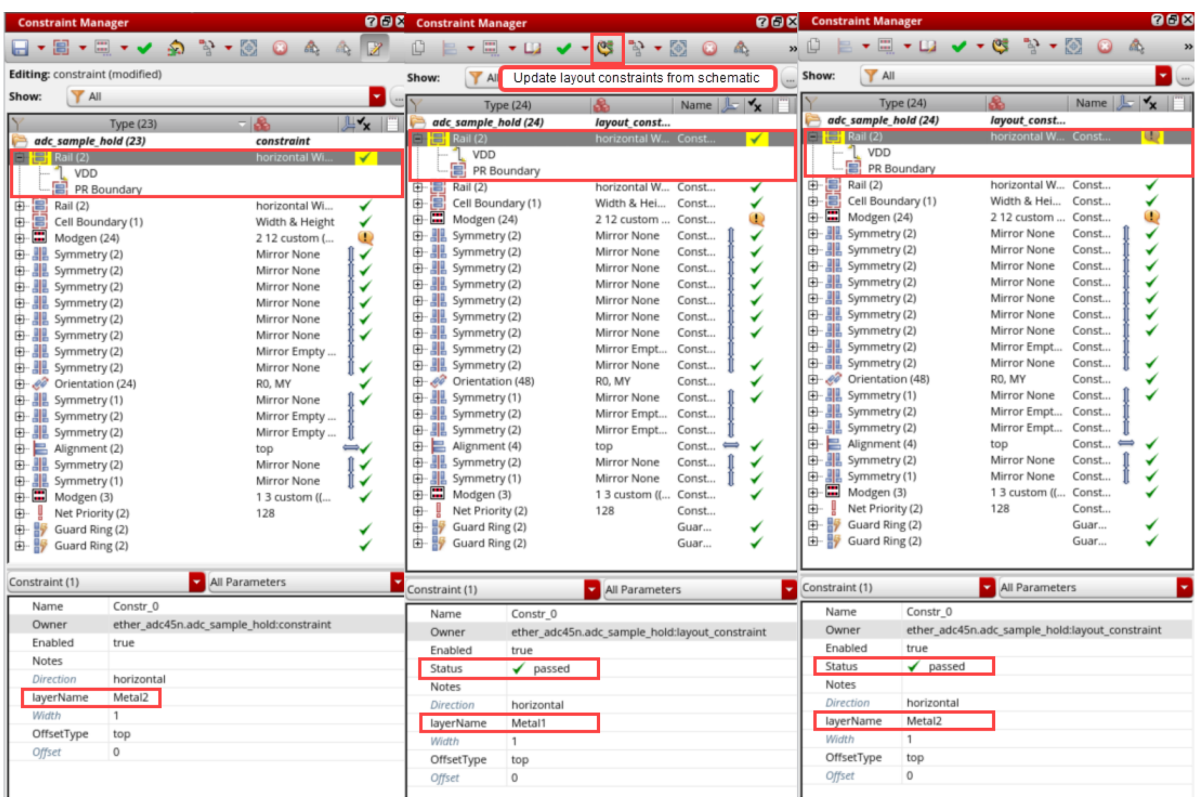
Constraint *Status* now shows up as **not checked** in the layout side, as shown here.



Note: The modified **Rail** constraint can be implemented and fixed by rerunning the Analog AutoPlacer.

Constraint *Status* now shows up as **passed** in the layout side, as shown here.



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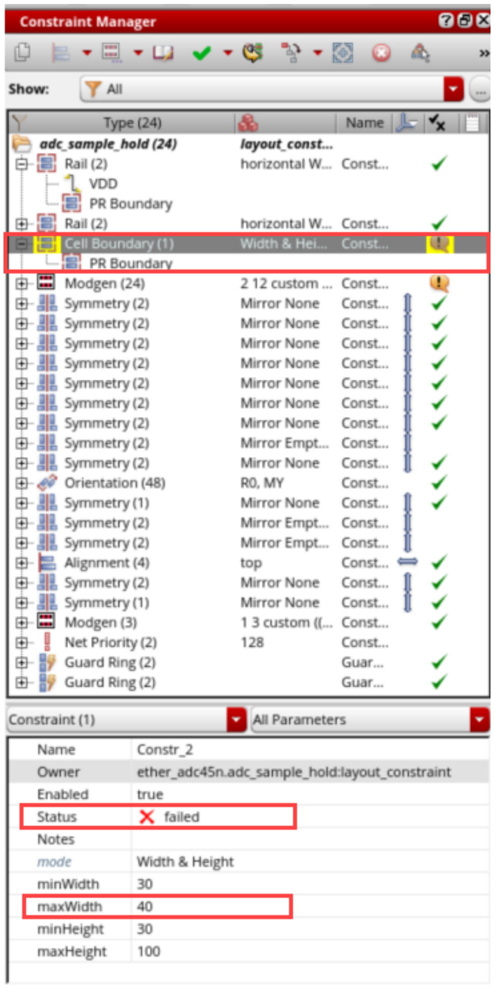
16. Now, try changing the *maxWidth* of the **Cell Boundary** constraint on the layout side from **100** to **40** as shown here and run the placer again.



17. Constraint *Status* is dynamically updated as **failed** in the Constraint Manager assistant and reported by the Annotation Browser assistant.

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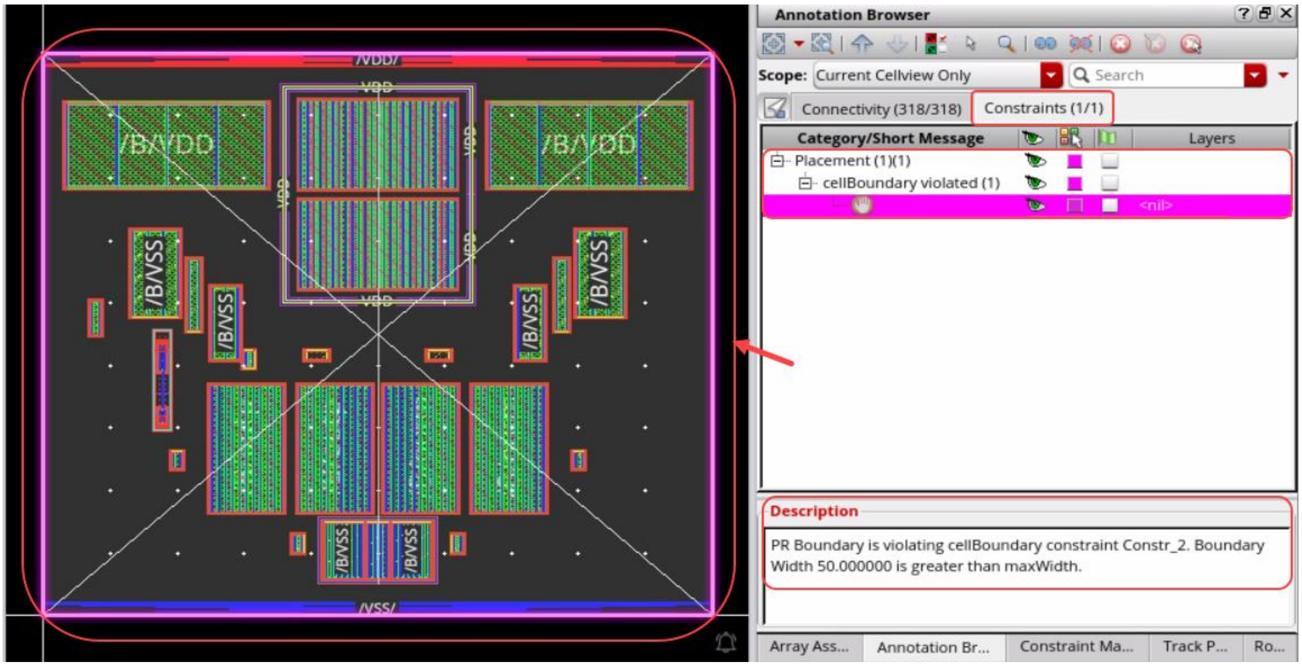
18. In the layout canvas, choose the **Window – Assistants – Annotation Browser** menu command to turn on the Annotation Browser assistant (if it is not turned on) to view the violations.



19. Try changing the *maxWidth* of the **Cell Boundary** constraint and run the Analog Auto Placer again to reimplement and fix the constraint violation.

20. Discard all the edits done in the layout window by using the **File – Discard Edits** menu command and clicking **Yes** in the *Discard Edits* pop-up form.

21. In the Layout EXL Editing window, again generate all from the source by choosing either the **Connectivity – Generate – All From Source** menu entry or by clicking the **Generate All From Source** (**GFS**) icon.

22. Click **OK** in the Generate Layout form with the default settings.

    All the constraints get transferred to the Layout (*Status – not checked*).

C. Usability Improvements in the CM: *Show Selected* and *Constraint Name Display*

1. In the design canvas, set the **OBJECTS** category in the Navigator assistant to **Instances**.

2. In the layout canvas, select the instances **|NM7** and **|NM10** in the Navigator assistant.

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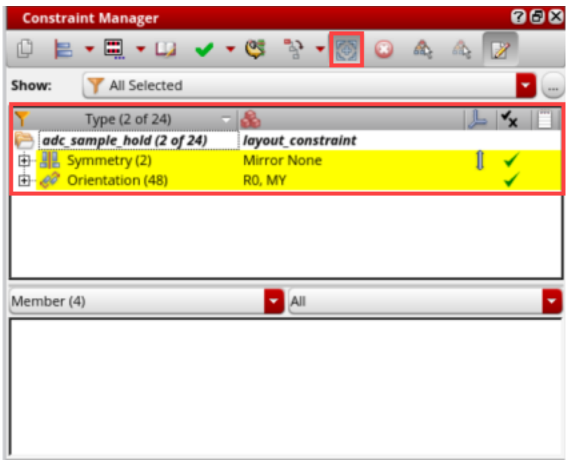
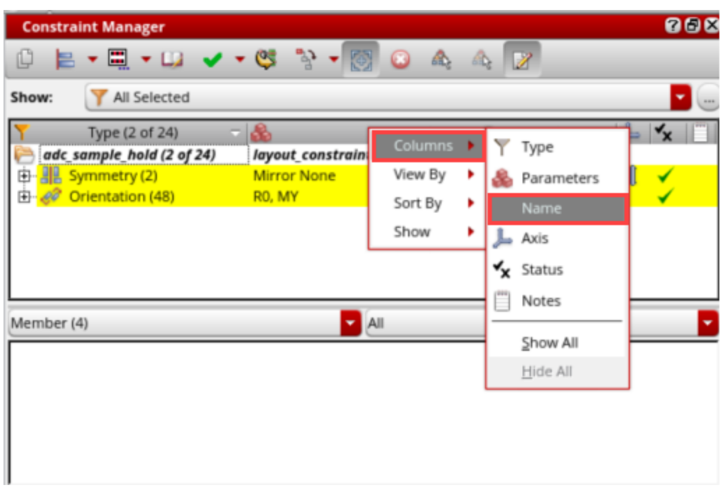
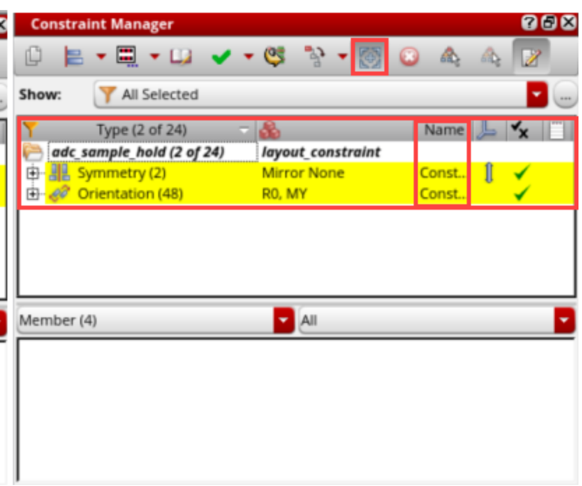
3. You can quickly check whether there is already a constraint set on this or not. Constraints on selected objects can be viewed by clicking the **Show Selected** toggle button in the Constraint Manager toolbar which in turn controls the corresponding filter.



4. Constraint names can be displayed as a column in the Constraint Manager through the **Columns** customization menu if it is not visible already.

From the Constraint Manager assistant, turn on **Columns – Name** as shown here.





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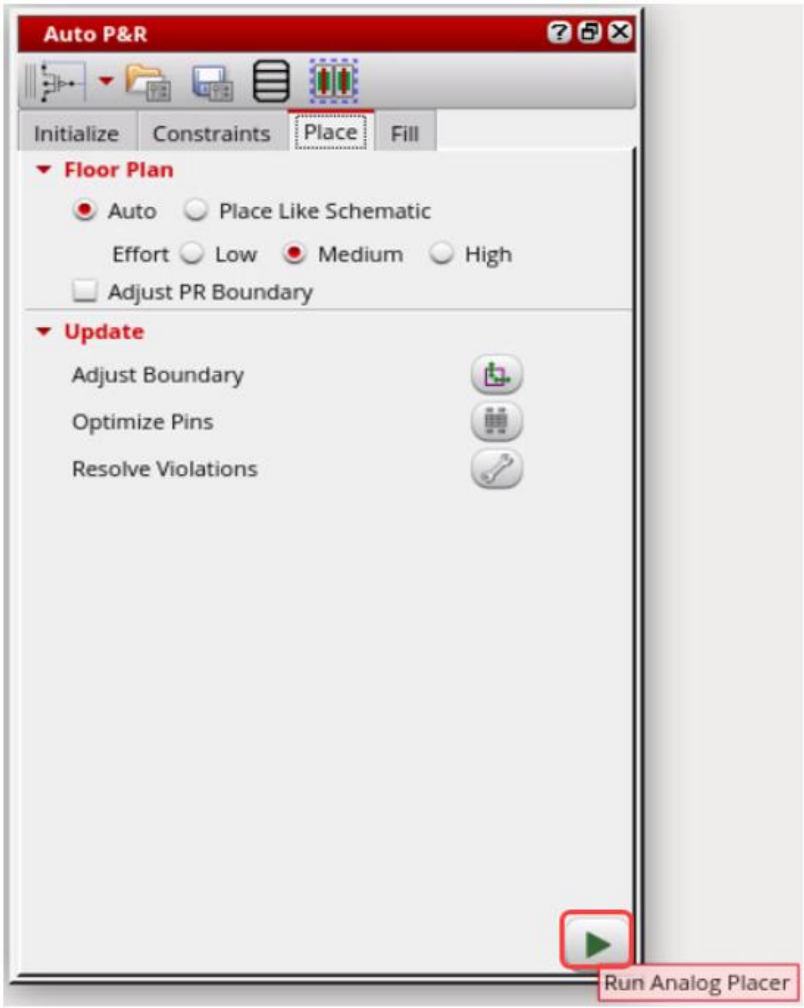
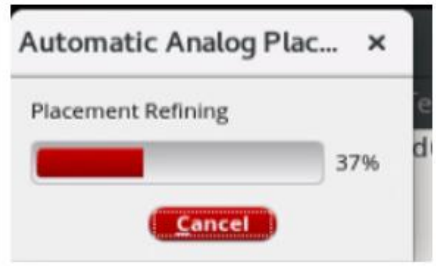
5. In the layout canvas, from the Auto P&R assistant, click the **Run Analog Placer** icon to run the Automatic Analog Placer.





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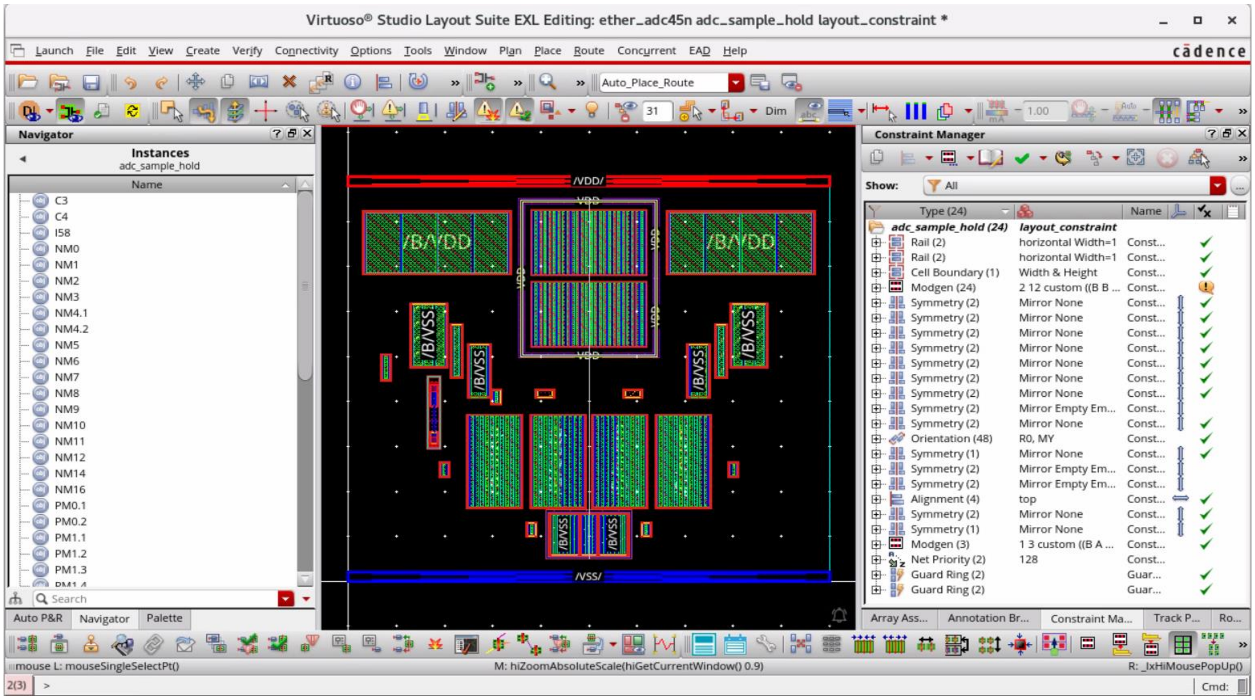
The Analog Placement is done, as shown here.

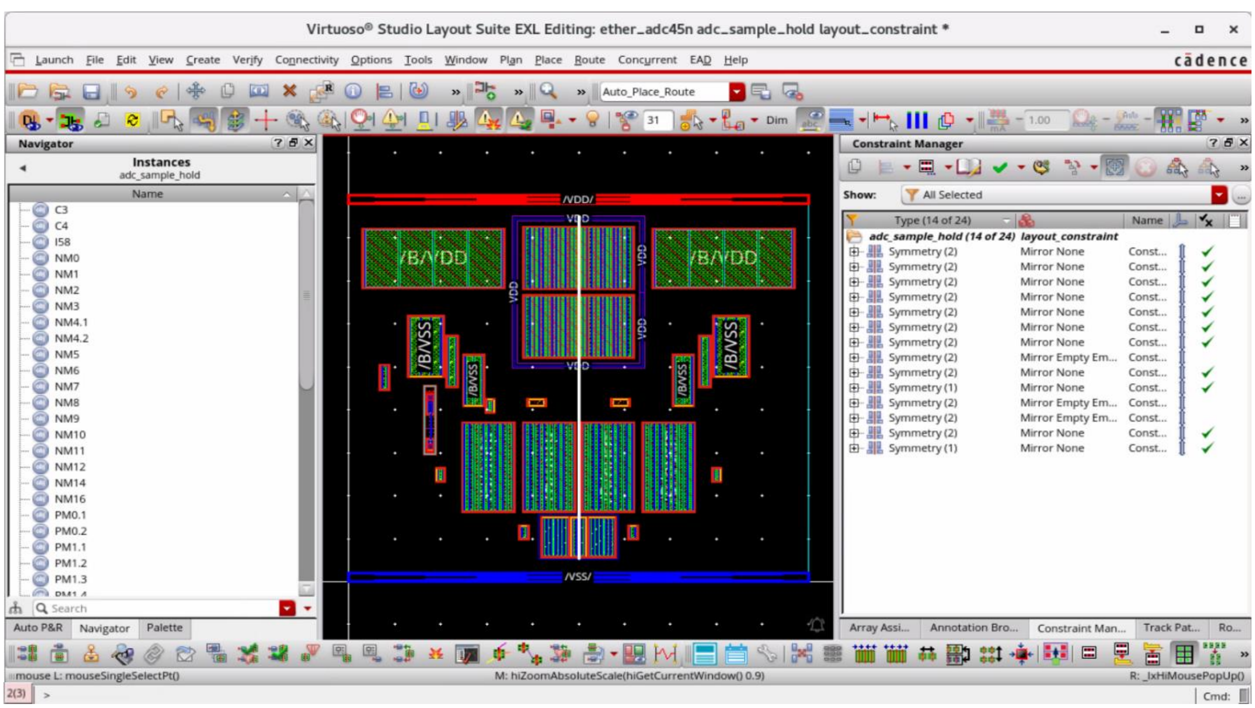


6. Selecting **Axis** in the layout canvas displays the associated constraints with the **Show Selected** filter.

   a. Click to select the **Axis**.

      The Constraint Manager gets populated with the related **Symmetry** constraints.

      Axis is also highlighted in the layout when a constraint is selected.



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7. Close all the views. **DO NOT** save. Keep the Virtuoso session running.

Summary

In this lab, you learned about the following:

- Rail Constraint support by Analog Automatic Placement (AAP) and Constraint-Aware Editing (CAE).
- Cell Boundary constraint that includes Area Utilization.
- Usability improvements in the Constraint Manager (CM) (*Show Selected* and *Constraint Name Display*).

End of Lab

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Lab 2-2 Performing Hierarchical Constraint Propagation

Objective: To explore the hierarchical constraint propagation.

The Hierarchical Constraint Propagation feature allows propagating net-based constraints up and down the hierarchy (to the top level or to the lower-level blocks) as it is often required in a top-down or concurrent design flow for floorplanning and block integration.

Propagation of net-based constraints across the hierarchy now also works in schematic. In IC 6.1.5, this feature was available only in the layout. An identical pull-down menu for constraint propagation is available in the Schematic Constraint Manager, and the use model is the same as that in the layout.

Constraints can now be propagated recursively (all the way to the bottom or the top of the hierarchy) from the Constraint Manager. In IC 6.1.5, this was possible only through SKILL®.

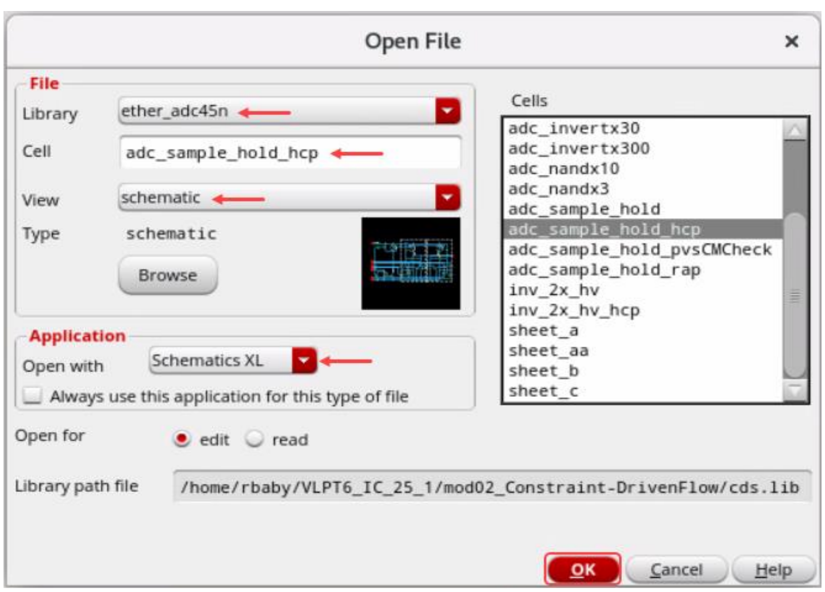
Transfer of propagated constraints in the lower levels of the hierarchy from the schematic to layout follows the same use model (transfer occurs during editing of the lower-level cell in VLS-XL).

Propagation of Net-Based Constraints Across the Schematic Hierarchy

1. Ensure that your working directory is **./VLPT6_IC_23_1/Constraint-DrivenFlow** and Virtuoso is running.

2. In the CIW, choose **File – Open** and enter the following values in the form.

Library	ether_adc45n
Cell	adc_sample_hold_hcp
View	schematic



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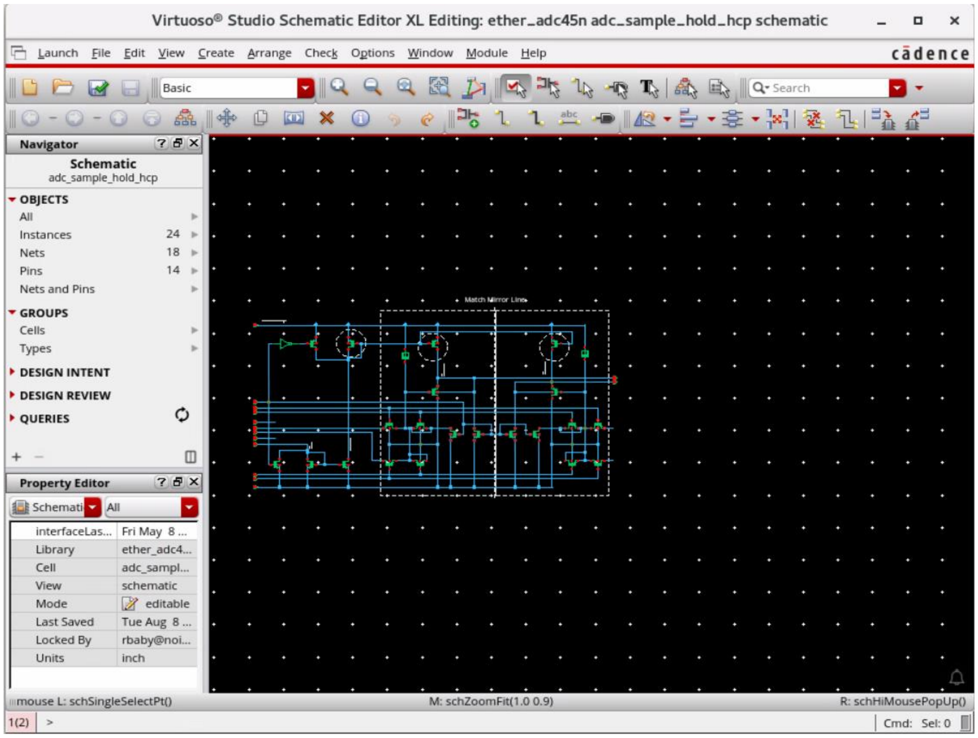
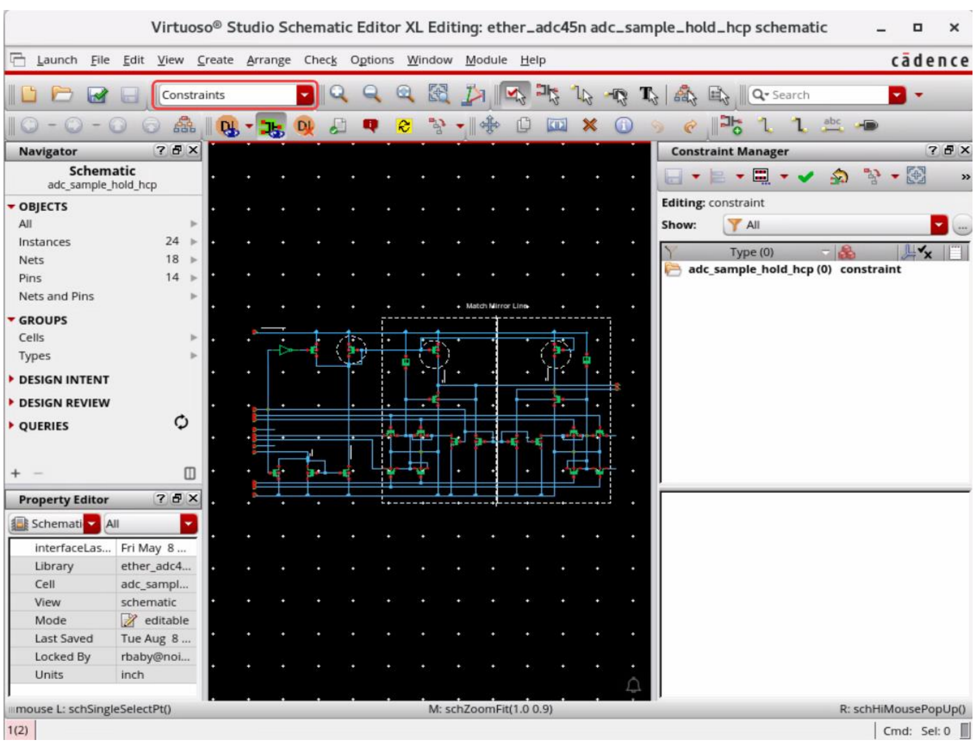
The schematic cellview opens in XL mode.



3. In the **Workspace Configuration**, change the workspace to *Constraints*.



Navigator, Property Editor, and Constraint Manager (CM) assistants are displayed in the schematic window.

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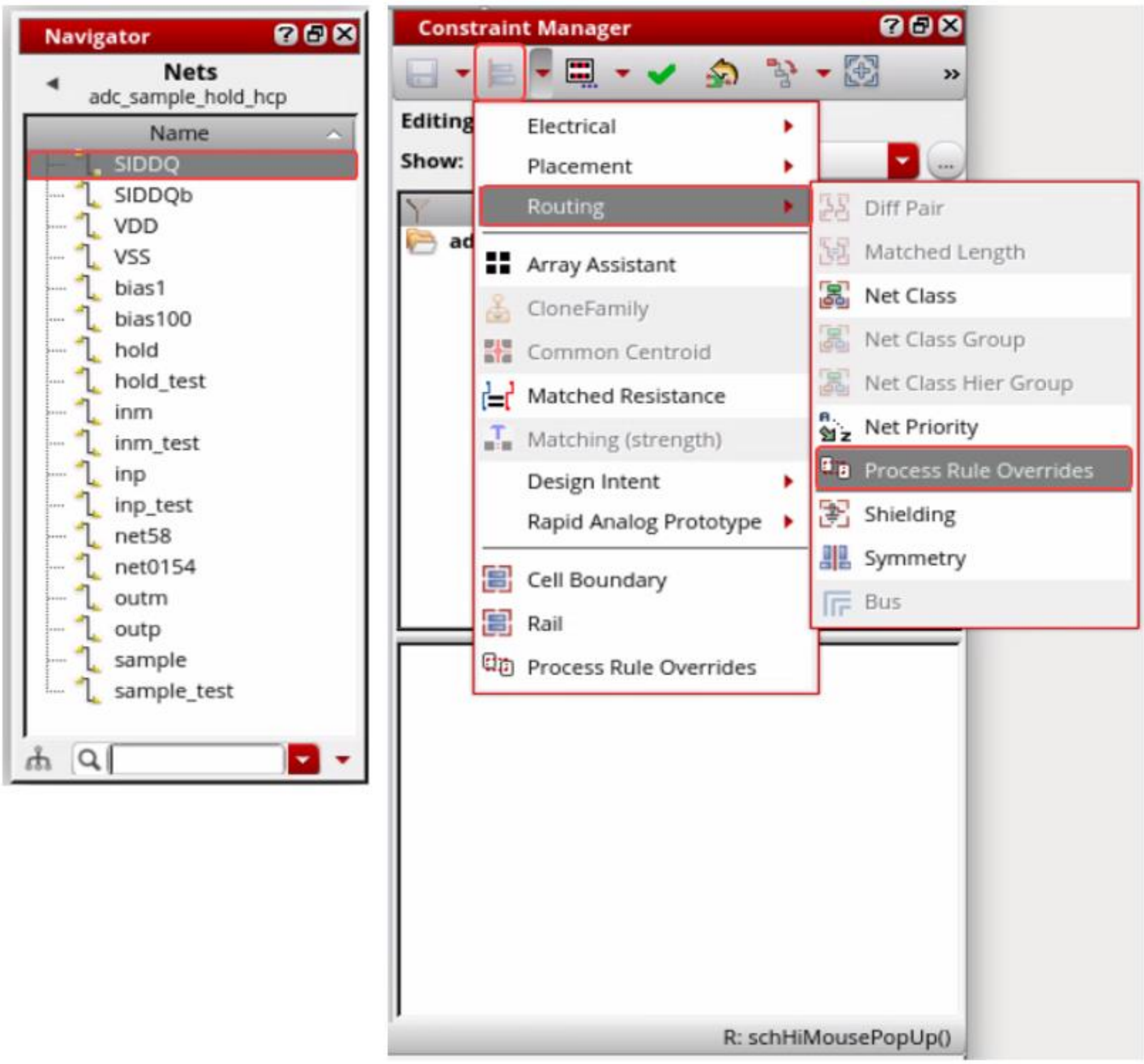
Creating a Process Rule Overrides (PRO) Constraint

1. In the schematic window, set the **OBJECTS** category in the Navigator assistant to **Nets**.

2. In the schematic window, select the net **SIDDQ** in the Navigator assistant and create a **PRO** (**Process Rule Overrides**) constraint on it from the Constraint Manager pull-down, as shown here.



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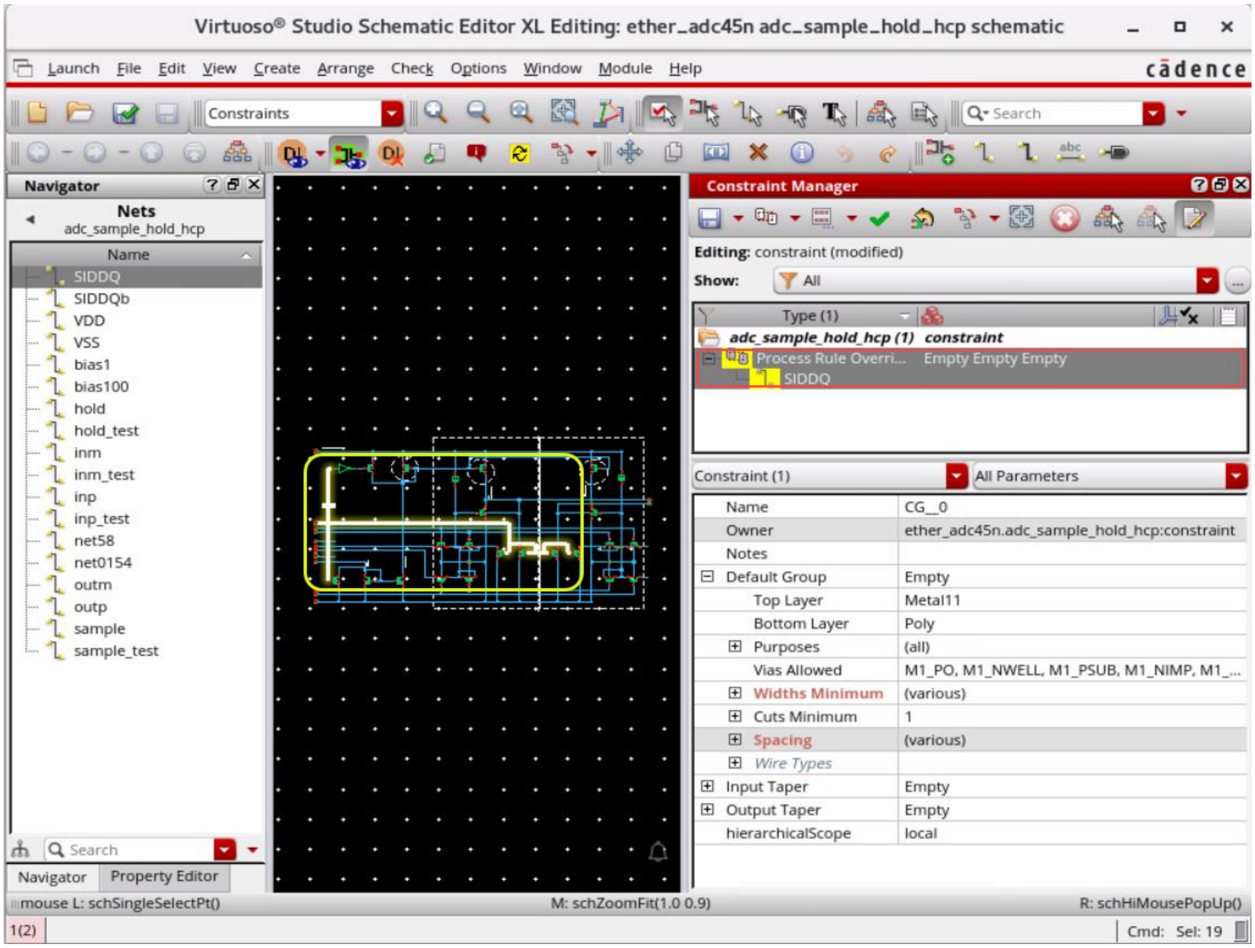
The **PRO** constraint is created on the net **SIDDQ,** as shown here.



Editing a Process Rule Overrides (PRO) Constraint

1. Select the group **2xWidth (Generate)** in the cyclic field under the *Default Group*. Refer to the graphic shown here.

2. Select the *Top Layer* to be **Metal3** and the *Bottom Layer* to be **Poly**.

3. In the Parameters section, click on the field for **hierarchical scope** (**hierarchicalScope**) which brings up the *Edit hierarchicalScope* form.

   Here, you can specify the scope as desired. Currently, it shows up as **local** (applies locally).

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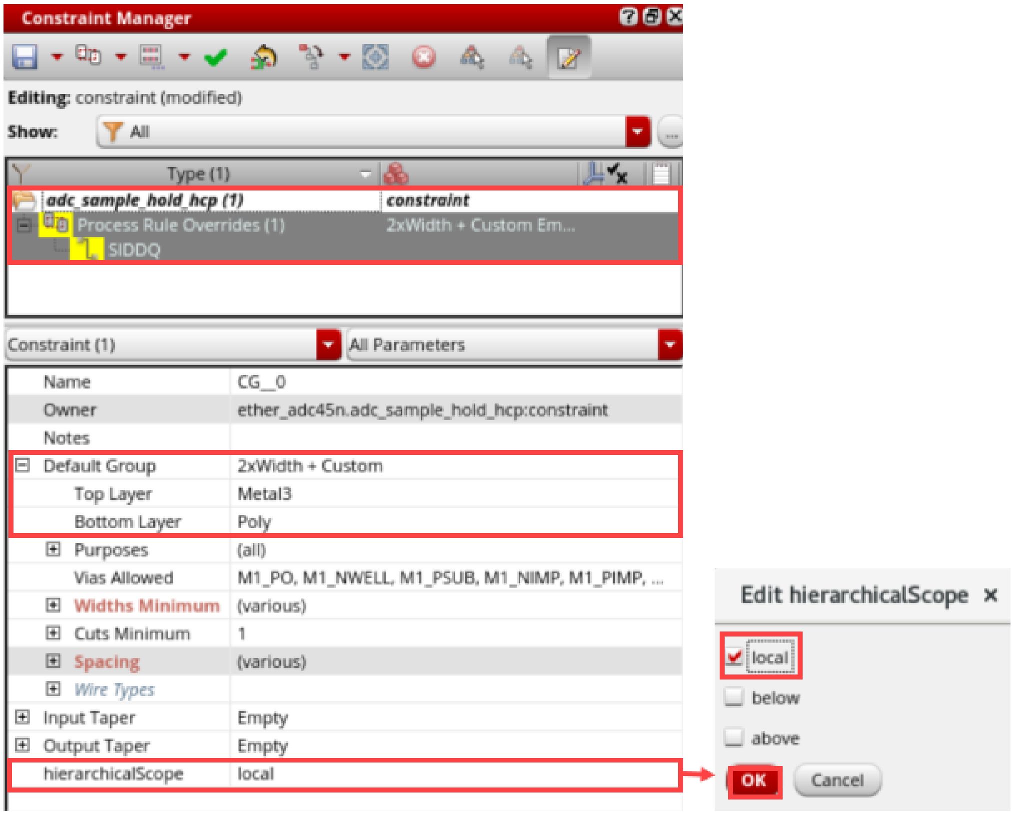
Constraints can be tagged automatically for propagation, without having to set explicitly the scope parameter. So, leave it unchanged. Cancel the *Edit hierarchicalScope* form.

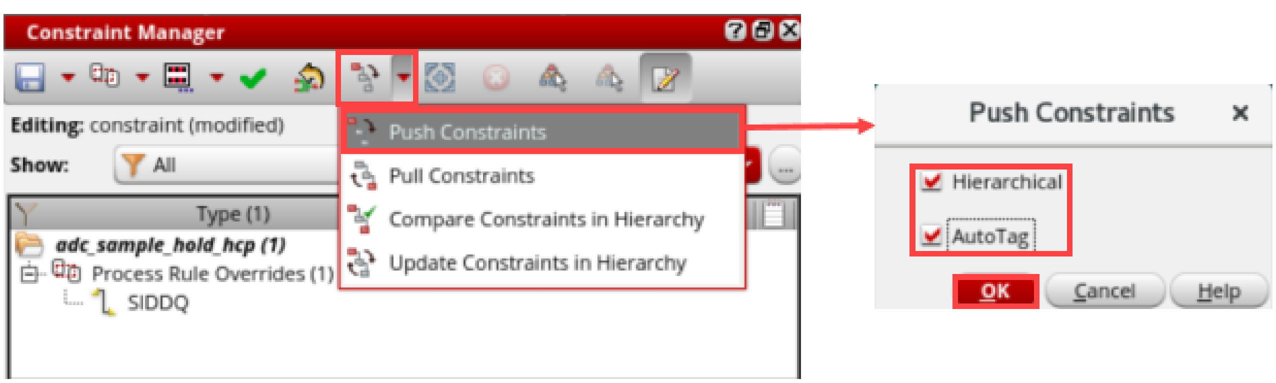


4. Select the instance **I58** (an instance of an inverter) in the schematic, and from the Constraint Manager toolbar, click on the *Constraint Propagation* pull-down and select the **Push Constraints** option to display the *Push Constraints* form.

   Hierarchical: Constraints can now be propagated recursively (all the way to the bottom or to the top of the hierarchy) from the Constraint Manager.

   AutoTag: Constraints can be tagged automatically for propagation, without having to set explicitly the scope parameter.

5. Turn on both **Hierarchical** and **AutoTag** options in the *Push Constraints* form and click **OK**.



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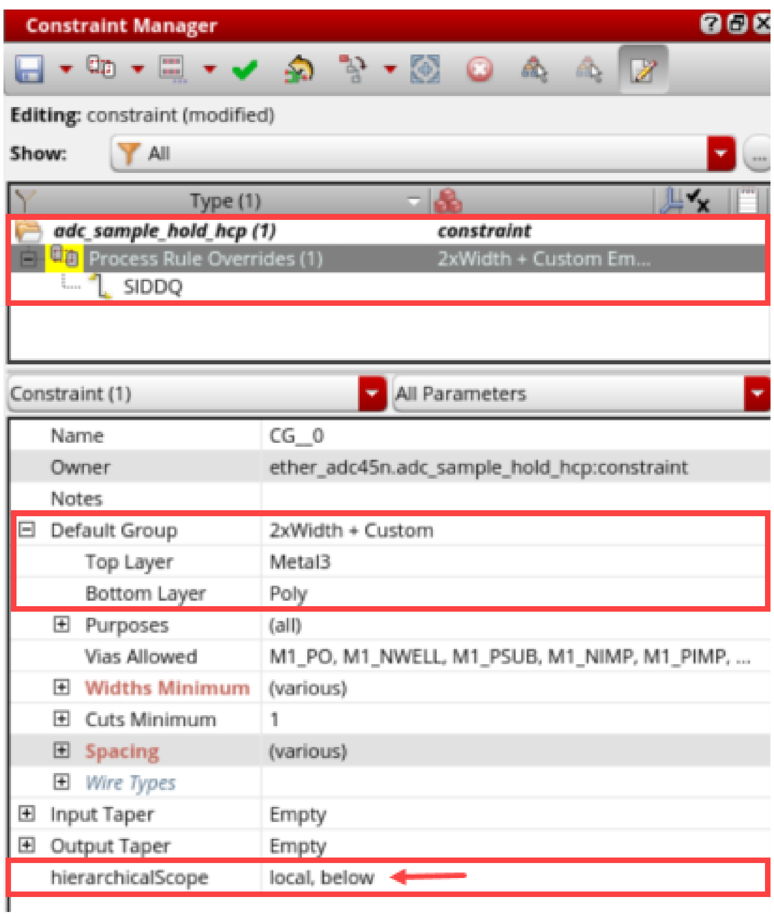
6. Messages in CIW indicate that constraints are pushed into the selected lower-level block.

INFO (CMGR-5020): Created constraint of type 'processOverride' in cache 'ether_adc45n inv_2x_hv_hcp schematic' with name 'CG__0'.

7. Observe that *hierarchicalScope* gets updated to **local**, **below** as shown here.



8. From the Constraint Manager assistant, click the **Options** button to invoke the *Constraint Manager Options* form.

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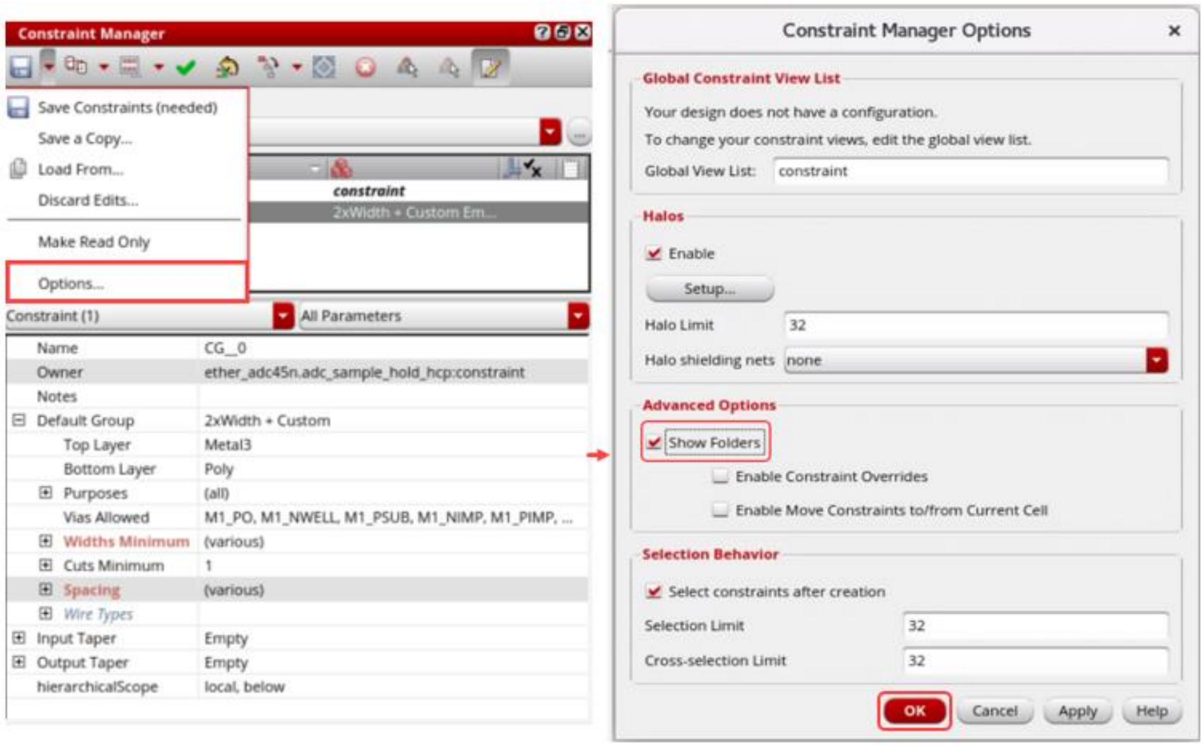
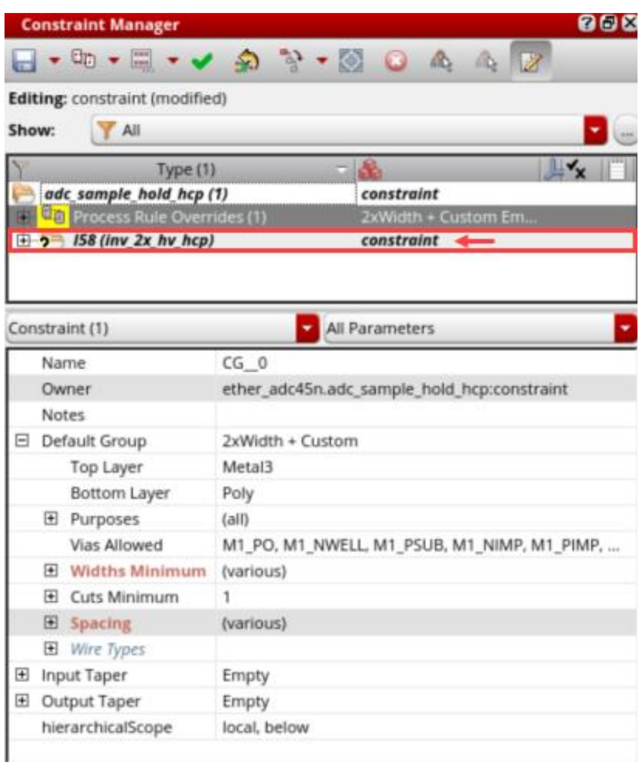
9. In the *Constraint Manager Options* form, enable the **Show Folders** button and click **OK** as shown here.



10. The lower-level constraints are not displayed as shown here, until you expand the folder to avoid any performance impact.



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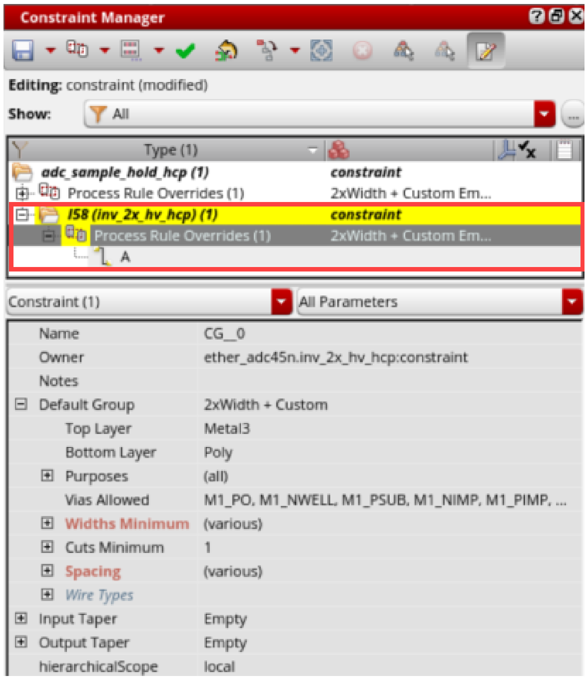
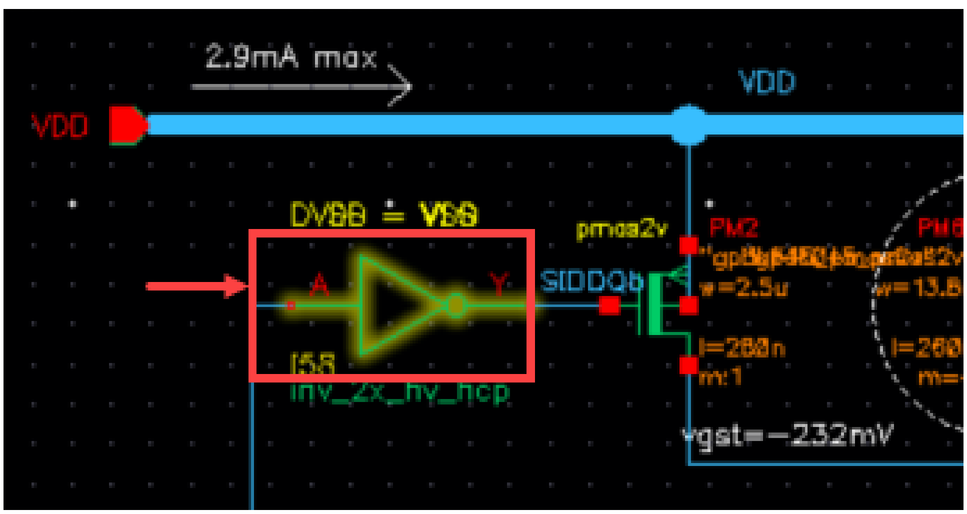
11. Expand the folder to see the pushed constraint.



The selected schematic instance **I58** is shown here in the graphic.



12. Close all the views. Do **NOT** save. Keep the Virtuoso session running.

Summary

In this lab, you learned about the following:

- The propagation of net-based constraints across the schematic hierarchy.

End of Lab

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Lab 2-3 Analyzing the Rapid Analog Prototype (RAP) Category in the Circuit Prospector and Modgen

Objective: To explore the Rapid Analog Prototype (RAP) category in the Circuit Prospector and Modgen.

The Rapid Analog Prototyping solution in Virtuoso involves generation of a placed and routed layout using automatic tools, following schematic entry, test setup and pre-layout simulation.

A dedicated new category has been introduced in the *Circuit Prospector* for rapid prototyping. It comprises finders including those for identifying and creating optimized Modgens. Associated with each finder is a generator to create relevant constraints for the found structures, including Modgens. The idea is to get the user started with a basic set of constraints to drive placement and routing. Modgen constraints are created automatically for structures such as Differential Pairs, Current Mirrors, Arrays of Resistors/Capacitors, Large M-factored Devices, etc.

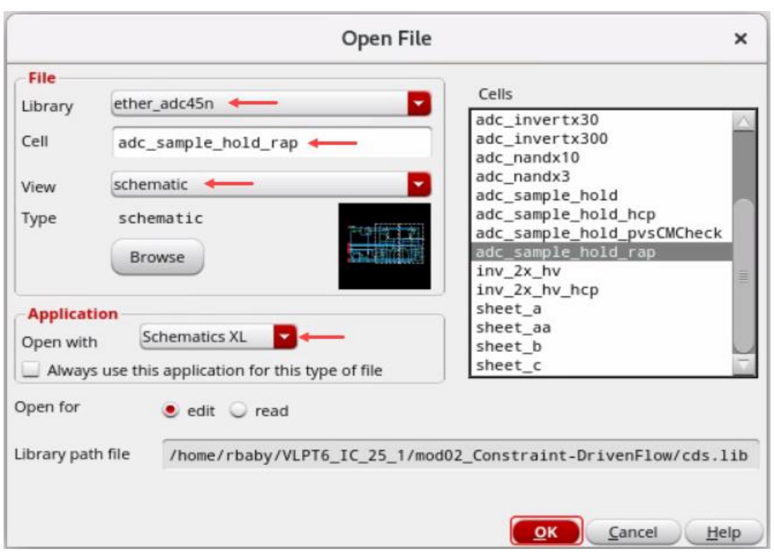
The finders/generators in the *Rapid Analog Prototype* (*RAP*) category will generate all the constraints required to achieve a good initial (prototype) layout.

Rapid Analog Prototype (RAP) Flow and Modgen

1. Ensure that your working directory is **./VLPT6_IC_23_1/Constraint-DrivenFlow** and Virtuoso is running.

2. In the CIW, choose **File – Open** and enter the following values in the form.

Library	ether_adc45n
Cell	adc_sample_hold_rap
View	schematic



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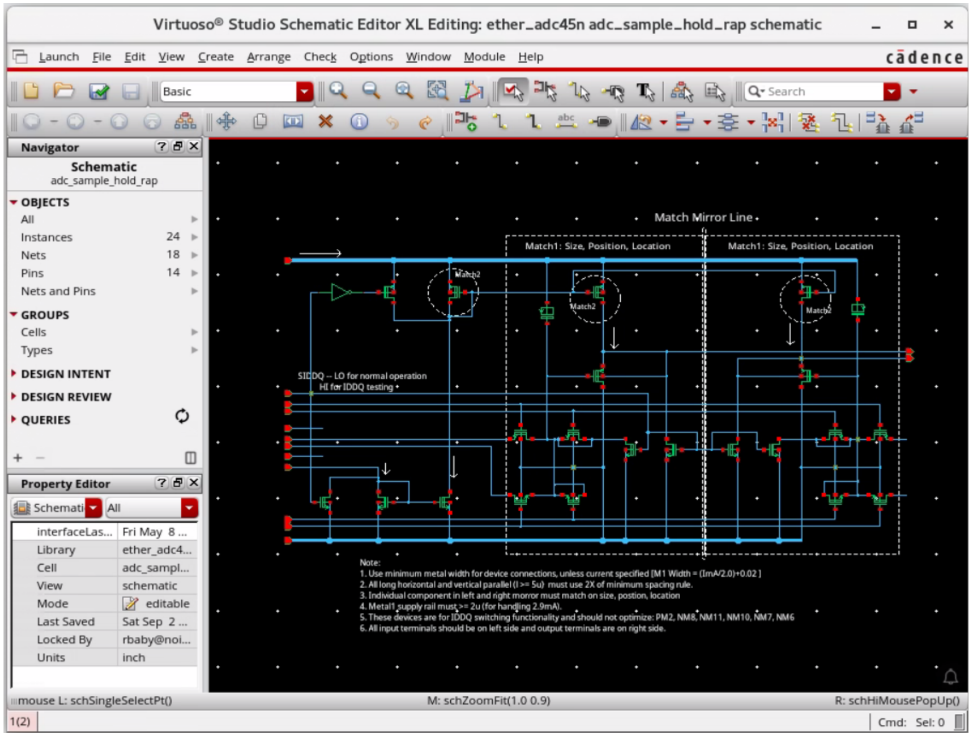
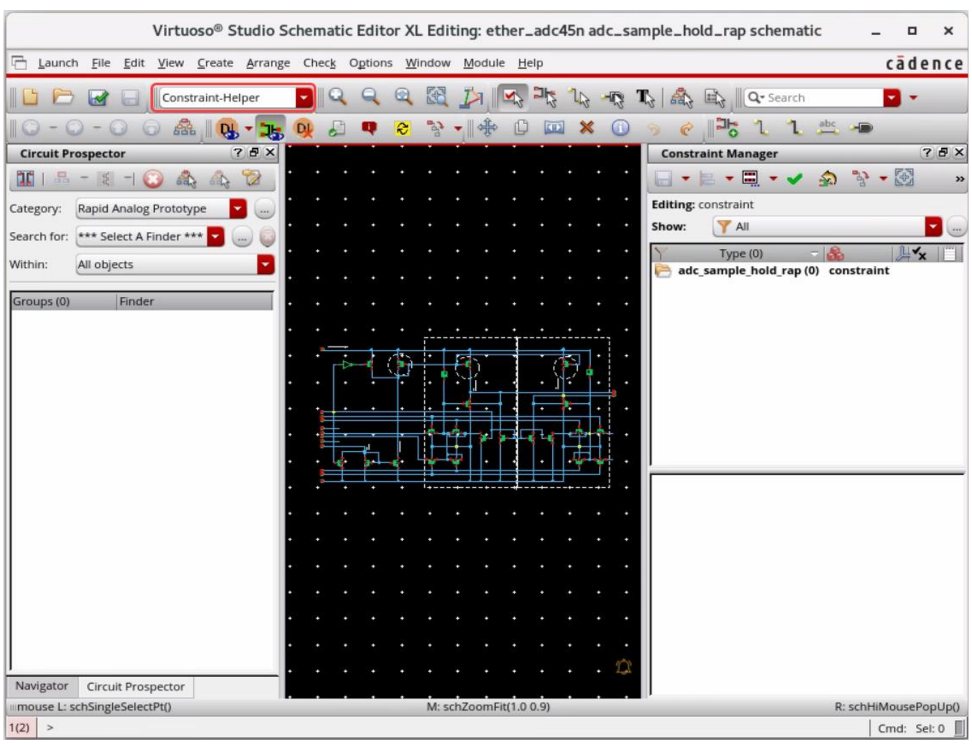
The **Schematic** cellview opens in XL mode.



3. In the **Workspace Configuration**, change the workspace to *Constraint-Helper*.



Navigator, Circuit Prospector, and Constraint Manager (CM) assistants are displayed in the schematic window.

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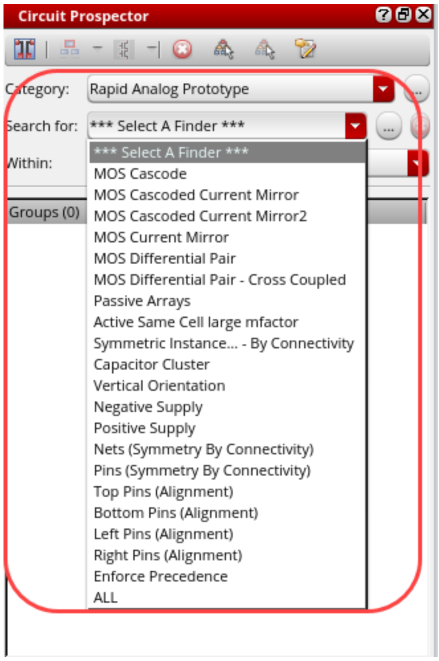
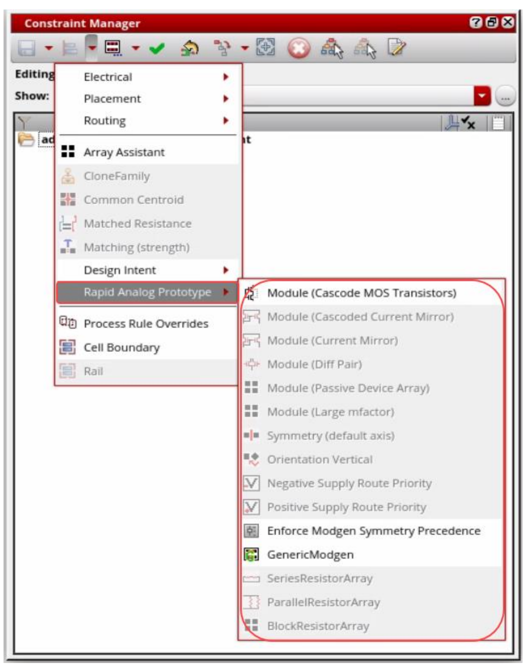
4. The *Rapid Analog Prototype* (*RAP*) category in the Circuit Prospector contains all the finders/generators required to automatically constrain a design. In the Circuit Prospector, choose the **Category** as *Rapid Analog Prototype* and under **Search for**, review all the available finders.



5. The RAP generators are also available from the **Constraints** menu under the *Rapid Analog Prototype* submenu in the Constraint Manager. This allows you to manually select their structures and run the required RAP generator on them.



In this lab, you will use the *Rapid Analog Prototype* category in the Circuit Prospector assistant for rapid prototyping. You will run all the finders in the RAP category.

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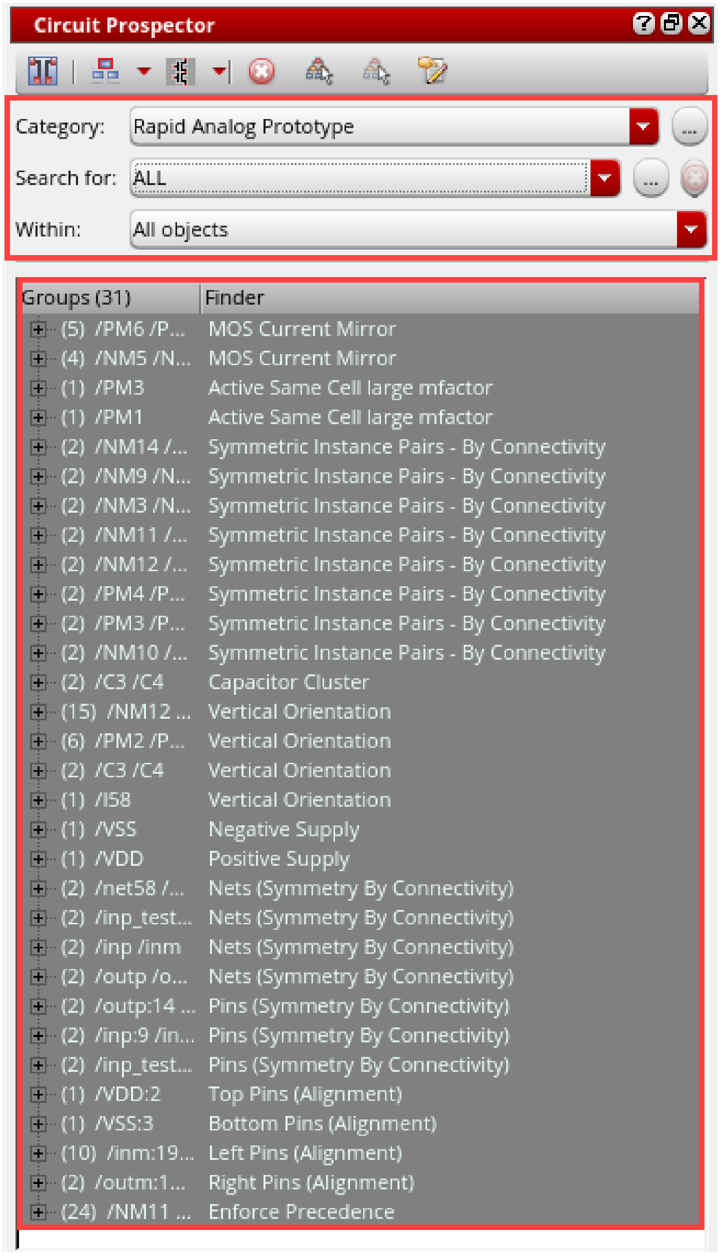
6. In the Circuit Prospector, choose the following options:

   a. Choose the **Category** as *Rapid Analog Prototype*.

   b. Under **Search for**, select *ALL*.

   c. Under **Within**, select *All objects*.

   Ensure that all the groups are selected from the **Finder** results, as shown here.



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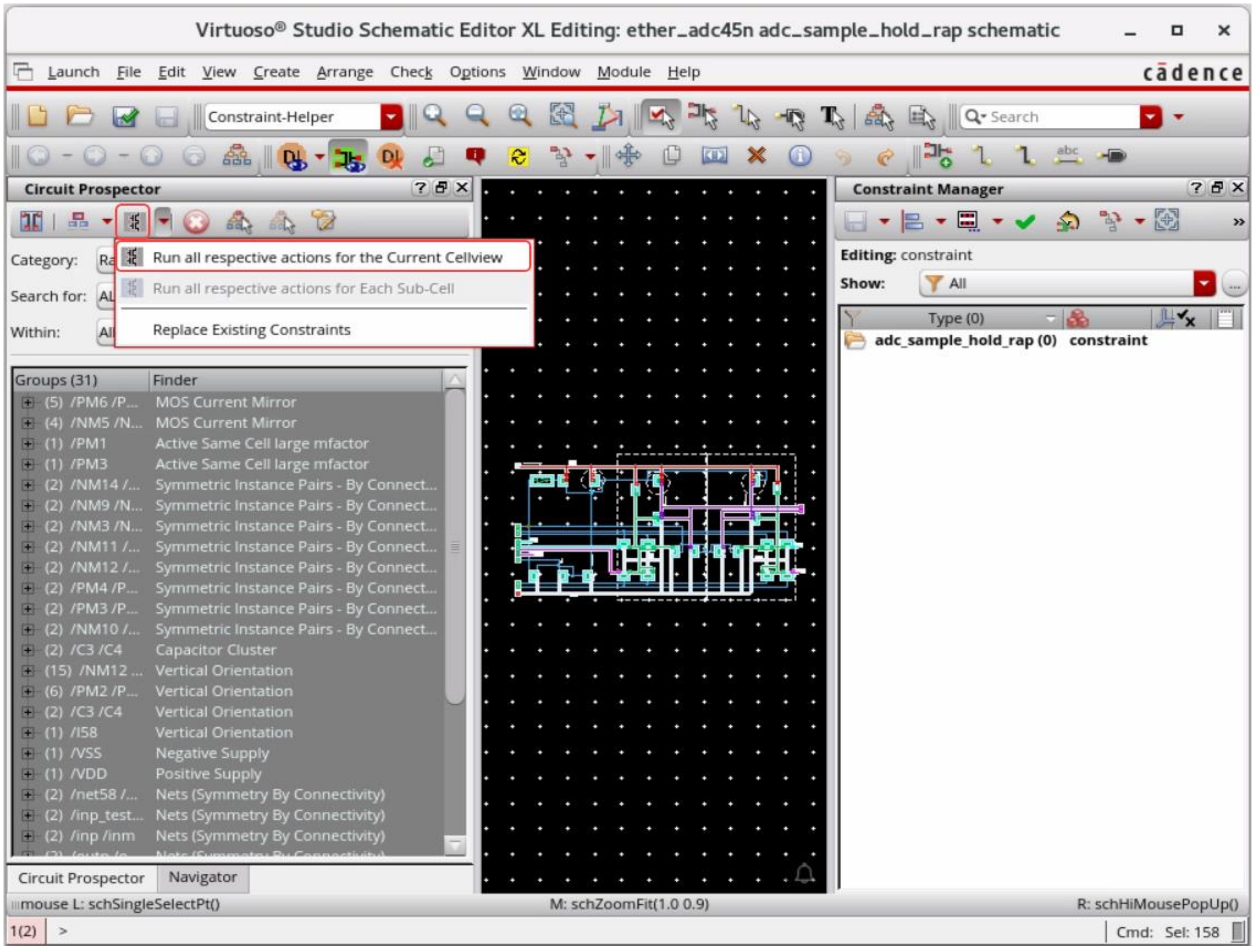
7. In the Circuit Prospector, click on the **Run all respective actions for the Current Cellview** option to add the constraints.



Note: The Constraint Generator dialogs will pop up for each RAP constraint generator allowing you to configure the generator parameters for the associated set of devices.

Note: The device names are displayed in the title of the dialog box. The **OK ALL** button allows you to accept the defaults for all subsequent constraint generators, and the dialogs will not pop up for those. The **Cancel ALL** button cancels the current constraint generator and all the subsequent constraint generators.

The Constraint Generator dialog pops up for *Module (Current Mirror): /PM6 /PM3 /PM1*.

8. Select the options as follows in the Constraint Generator:

   a. **Style**: *SingleRow*

   b. Enable **Add Dummies**

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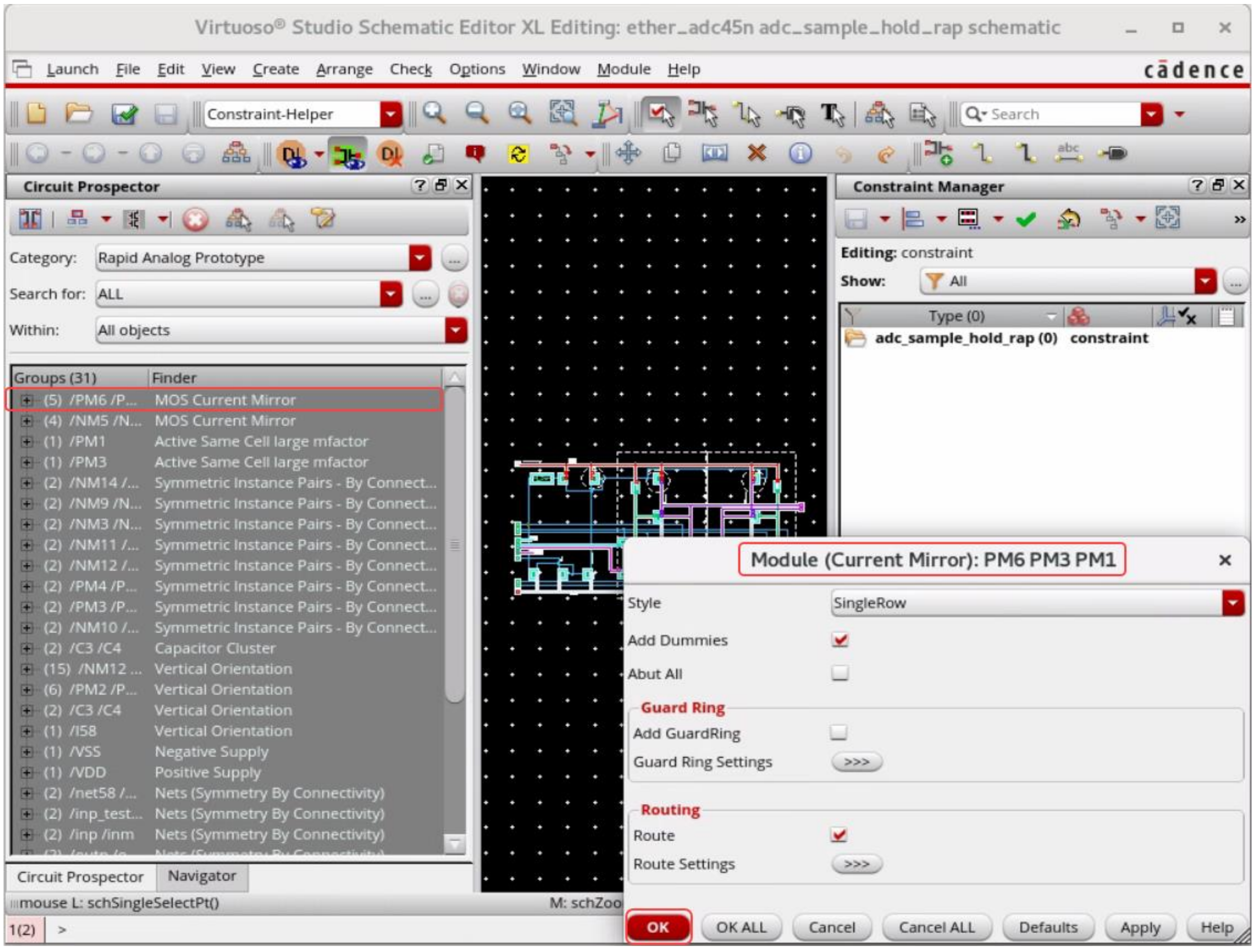
c. Click **OK** in the pop-up for Module (Current Mirror): /PM6 /PM3 /PM1.



9. Next, the Constraint Generator dialog pop ups for *Module (Current Mirror): /NM5 /NM4*.

   In the form, change the options as given in the next step and continue.

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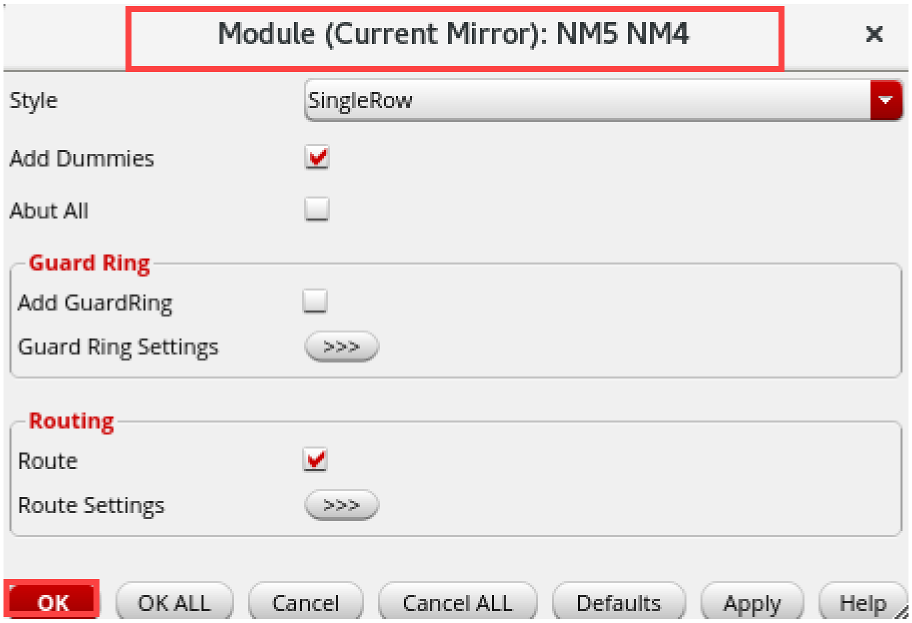
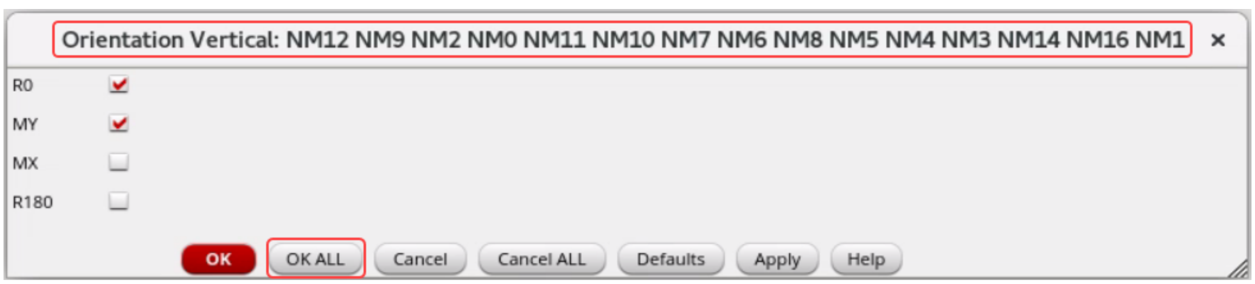
10. In the *Module (Current Mirror): /NM5 /NM4* dialog form, change the **Style** to *SingleRow*, enable **Add Dummies**, uncheck **Abut All**, and click **OK**.



11. Next, the Constraint Generator dialog pop ups as shown here. This time select **OK ALL** (accept the defaults for all subsequent constraint generators).



The constraints show up in the Constraint Manager as shown here.

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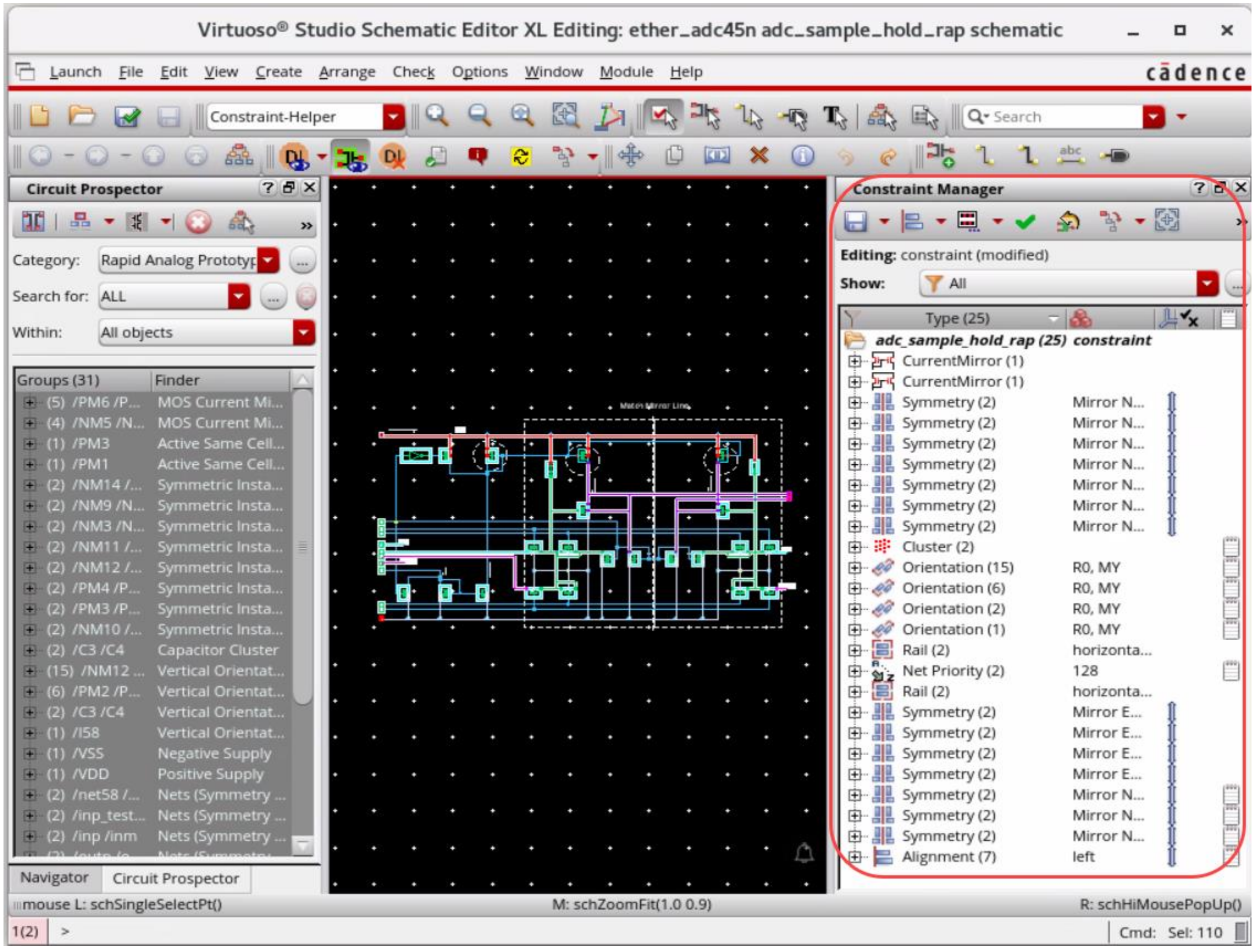
12. Refer to the schematic window with the added constraints in the Constraint Manager, as shown here.



Creating New Layout Cellview

Now, you will create the layout.

1. In the schematic, choose **Launch – Layout EXL** to bring up the *Startup Option* form.

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2. Select **Create New** in Layout and **Automatic** in Configuration to create a new layout view and click **OK** to open the **New File** form.

3. Fill out the **New File** form as *ether_adc45n/adc_sample_hold_rap/layout_constraint*.

4. Click **OK**.

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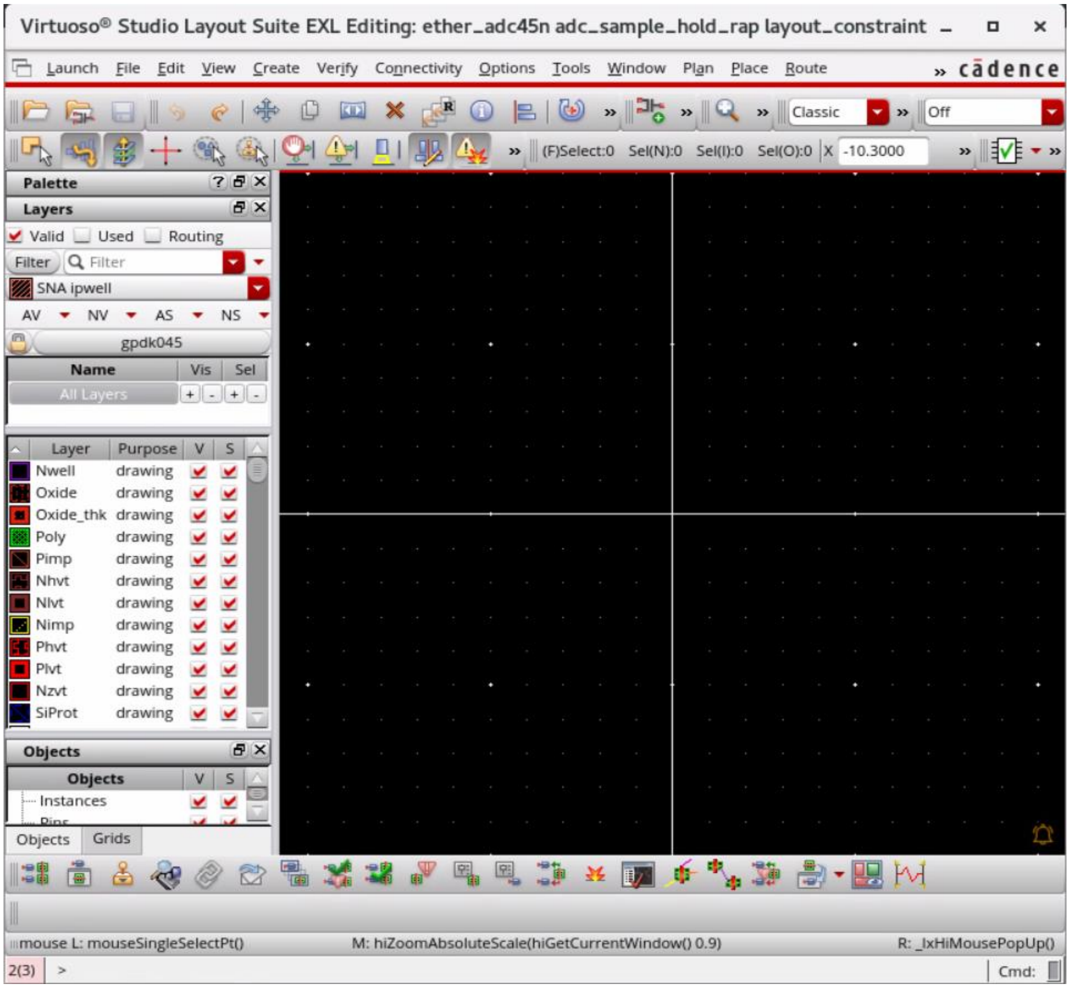
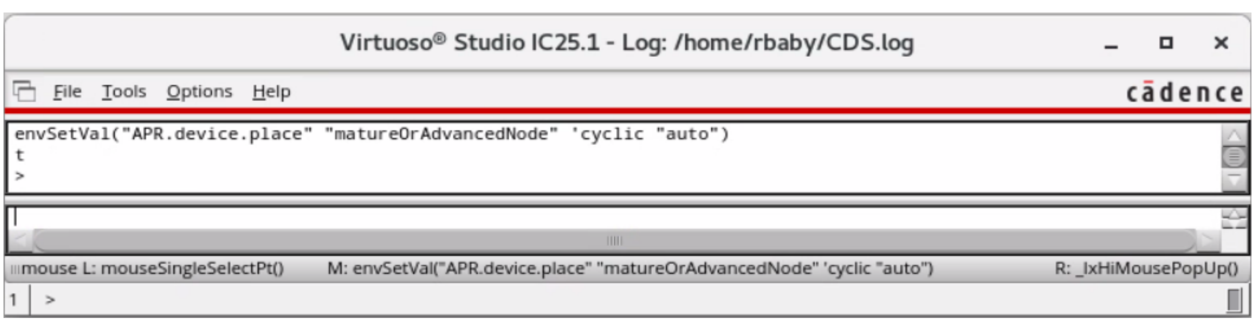
The Virtuoso Layout Suite EXL Palette opens.



5. To run the Analog Automatic Placement, execute the following command in the CIW window.

```
envSetVal("APR.device.place" "matureOrAdvancedNode" 'cyclic "auto")
```



6. In the layout window, choose **Window – Assistants – Auto P&R** or select the workspace **Auto_Place_Route** from the Workspace Configuration.

   The Auto P&R assistant is turned on.

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7. You will run the Automatic Analog Placement, as shown here.

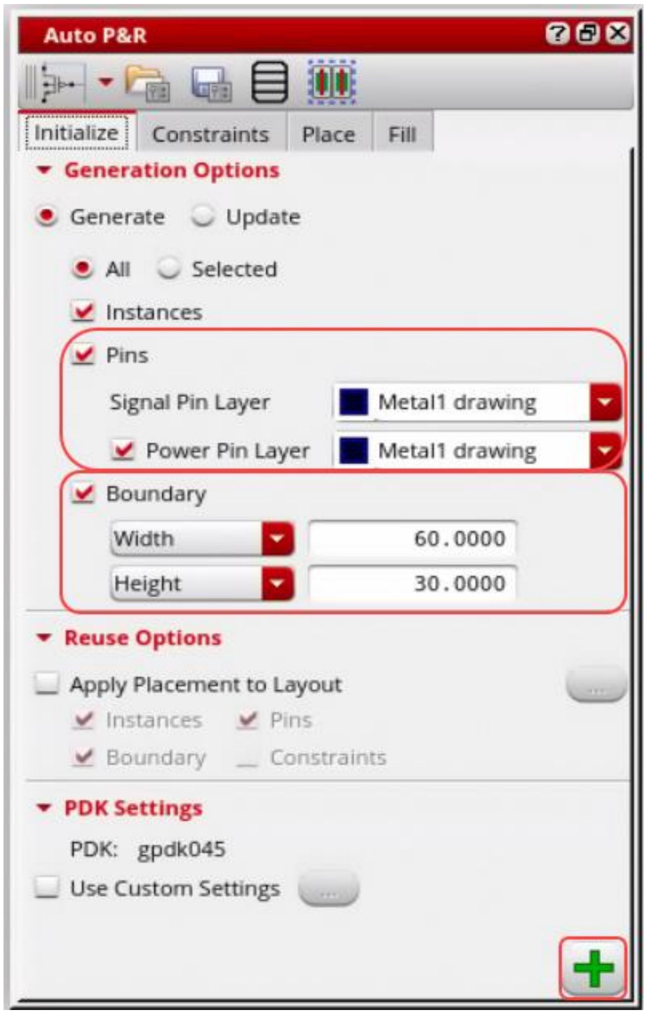
   a. In the **Initialize** tab, keep the following:

      - Change the *Signal Pin Layer* and *Power Pin Layer* to the **Metal1 drawing**.
      - In the **Boundary** field, keep *Width* as **60** and *Height* as **30**.

      Note: Try different **Boundary** settings to achieve optimum constraints-based placement.

      - Keep the other values as default.

   b. Click the **Apply** icon  to initialize the design.



Devices are preplaced as follows, and constraints get transferred from the schematic to the layout view.

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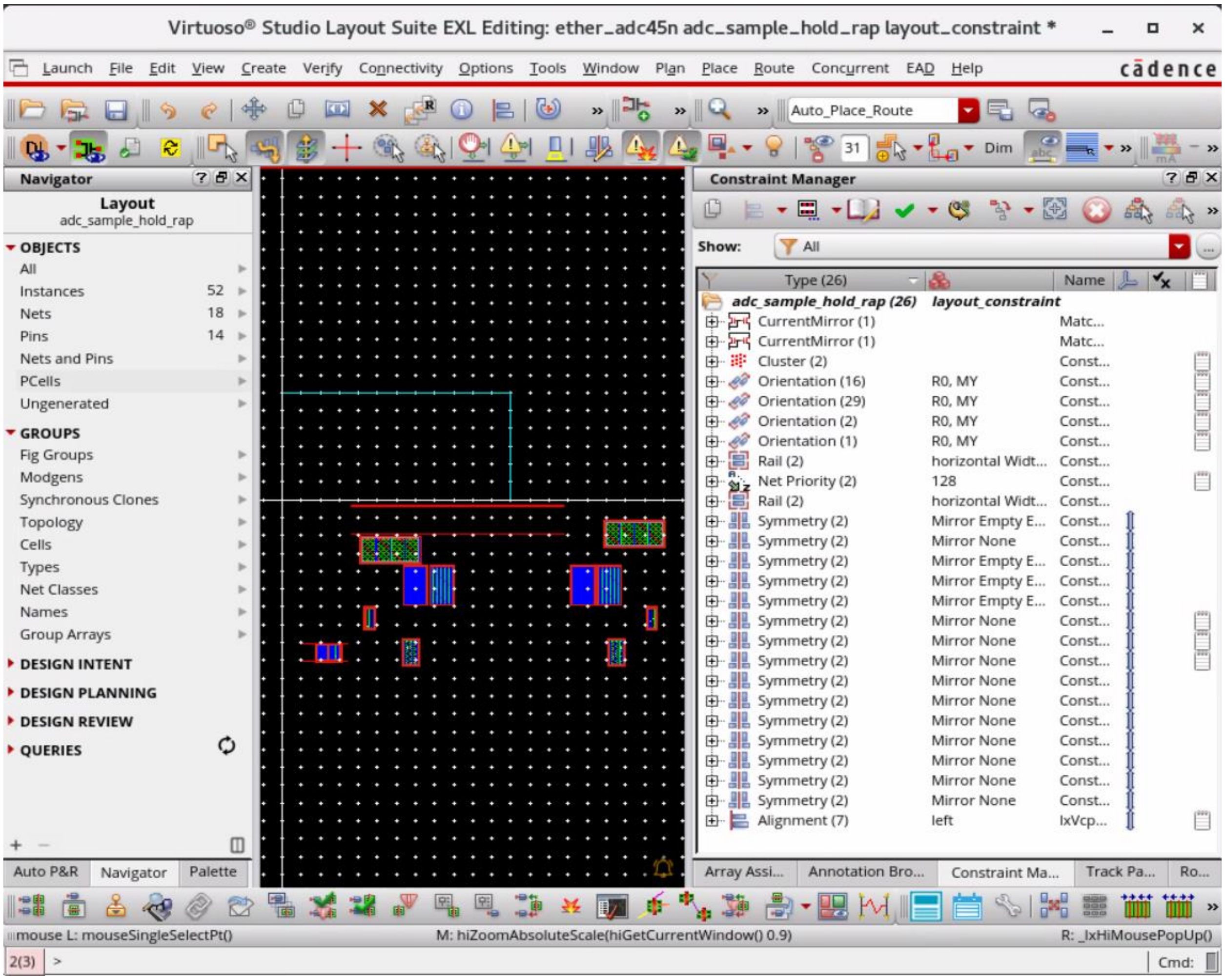
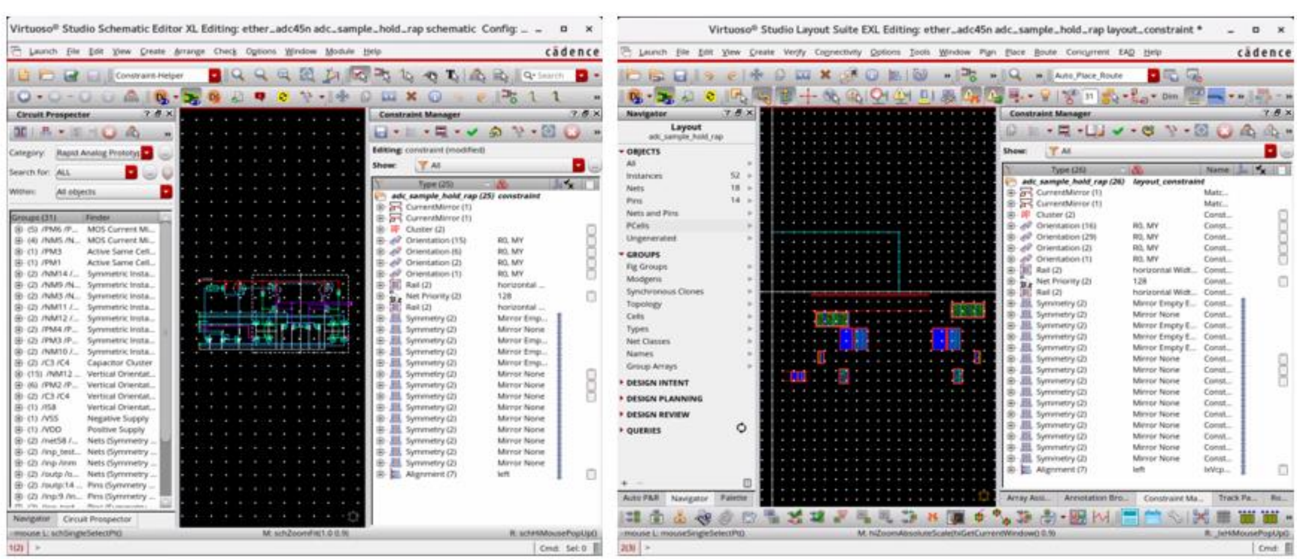
Note: Click on the **Constraint Manager** tab to turn on the Constraint Manager assistant, as shown here.



8. Tile the schematic and the layout windows next to each other, as shown here.



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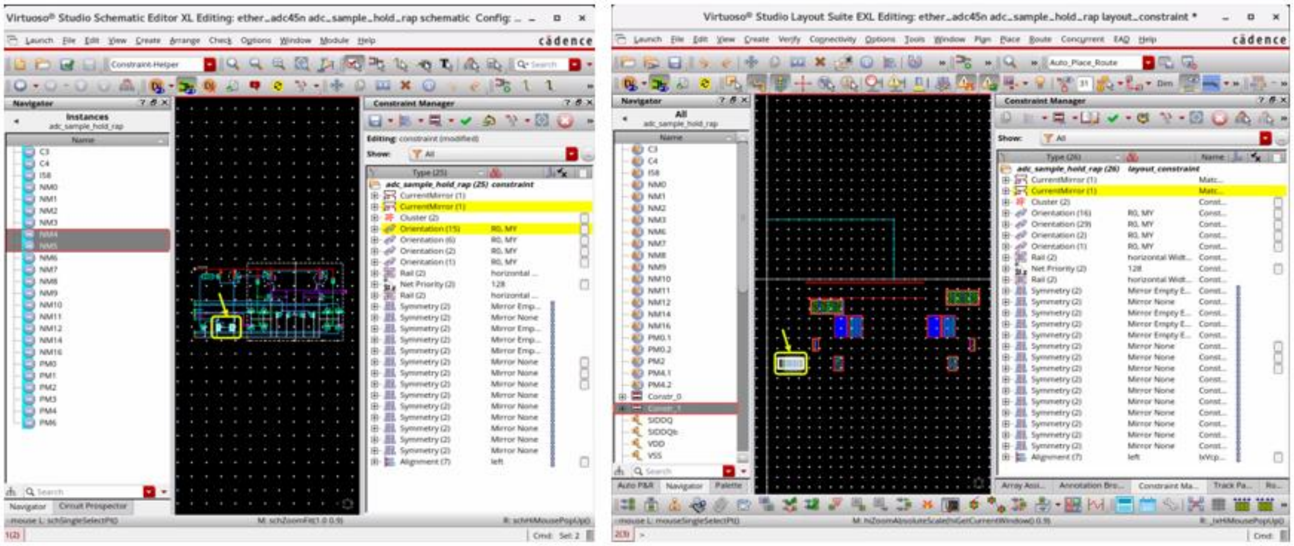
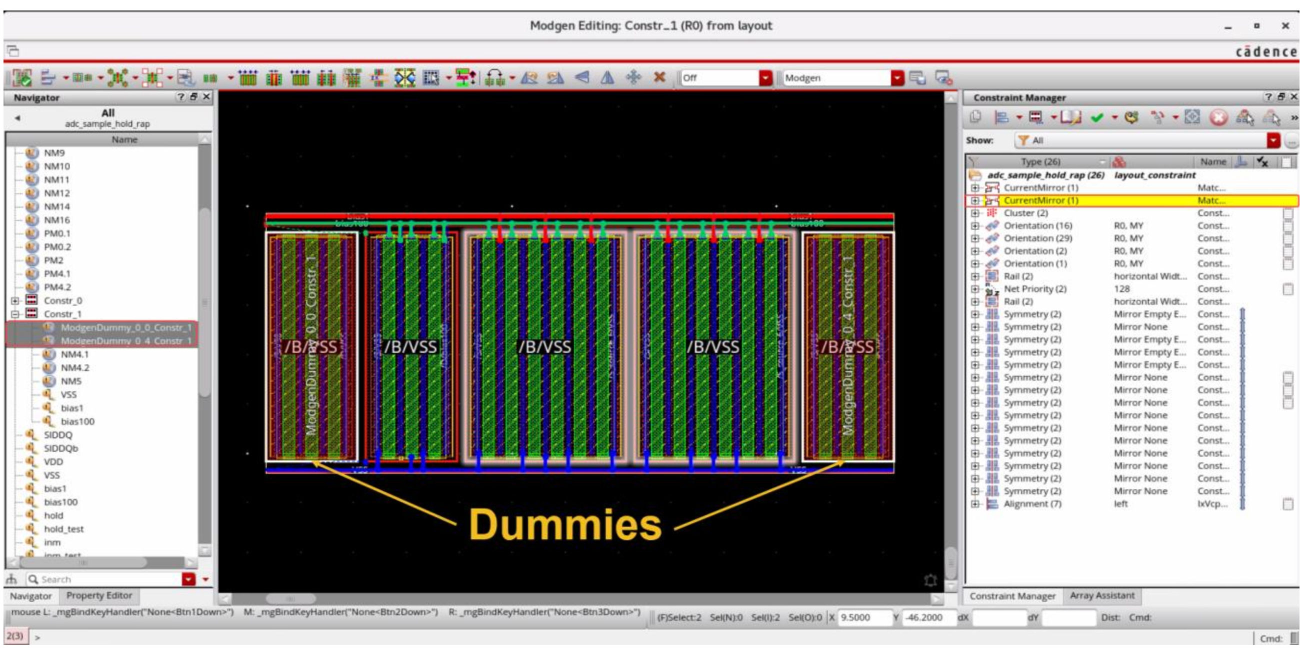
9. Select the instances **/NM5 /NM4** in the Navigator on the schematic side.



Notice that the corresponding **figGroup** gets highlighted in the Navigator assistant and in the canvas on the layout side.

10. In the layout window, **double-**click to descend and open the **figGroup** in the Modgen Editing window (**dummies** are also added – since you enabled the checkbox when running the Constraint Generator).



The Modgen Editing window opens with the Navigator, Property Editor, Constraint Manager, and Array Assistant. To maximize the working area in the canvas, you can close the assistants that you are not using.

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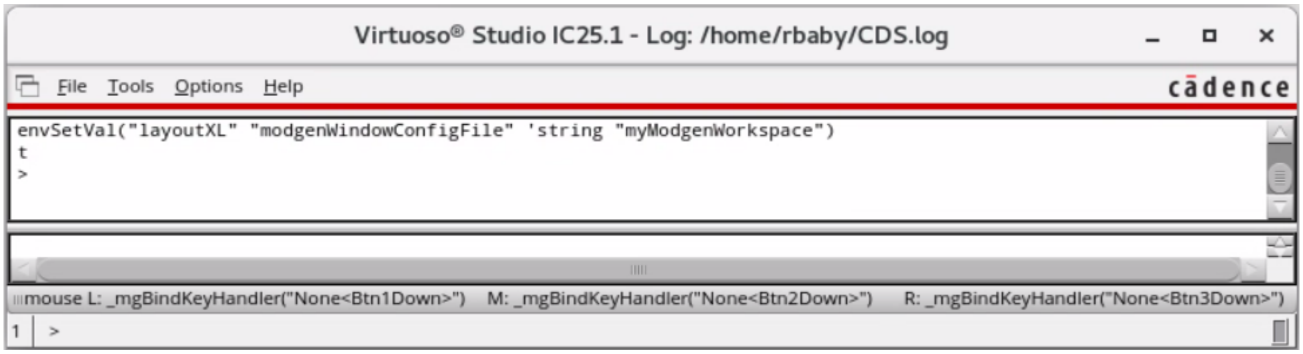
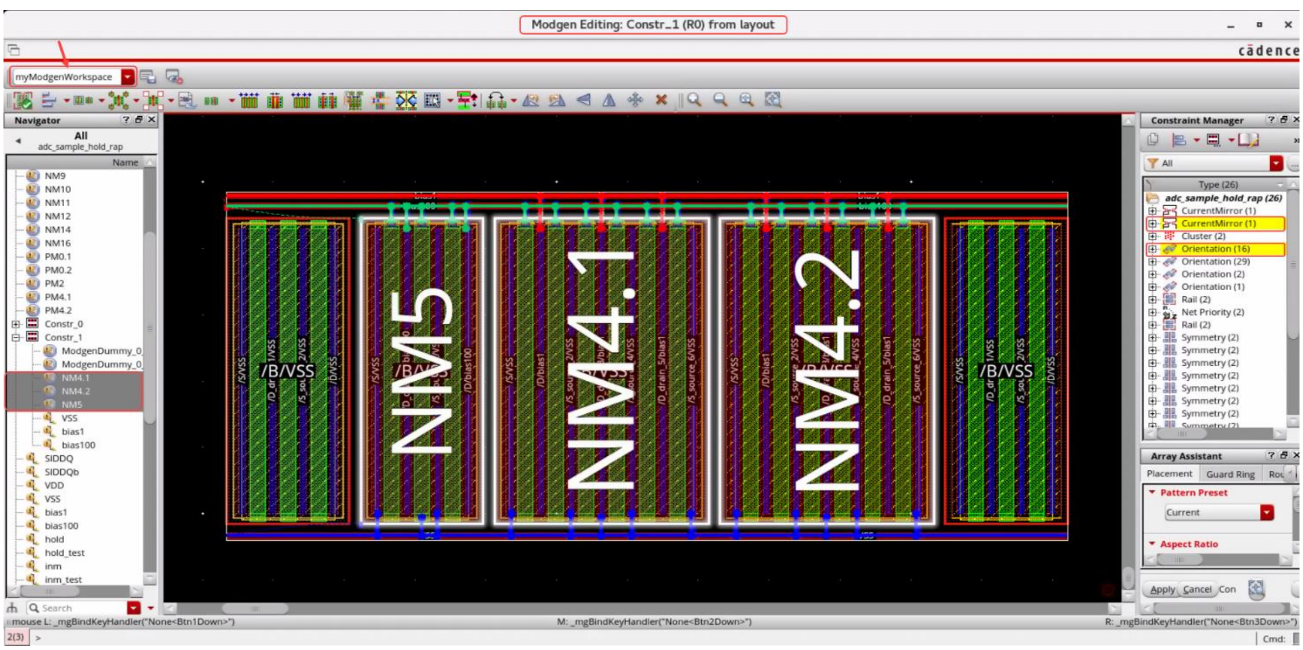
Note: Now, you have the option of specifying the default workspace for the Modgen Editing Session to start with. Set the following environment variable to point to the user workspace with the name “myModgenWorkspace” (Navigator assistant enabled in this workspace).

11. In the CIW, enter the following SKILL, and reinvoke the Modgen Editor, which opens with *myModgenWorkspace* workspace in the **Workspace Configuration** menu.




```
envSetVal(“layoutXL” “modgenWindowConfigFile” ‘string
   “myModgenWorkspace”)
```



Observe the Modgen Editing window banner stating (**R0**) indicates the Modgen orientation.

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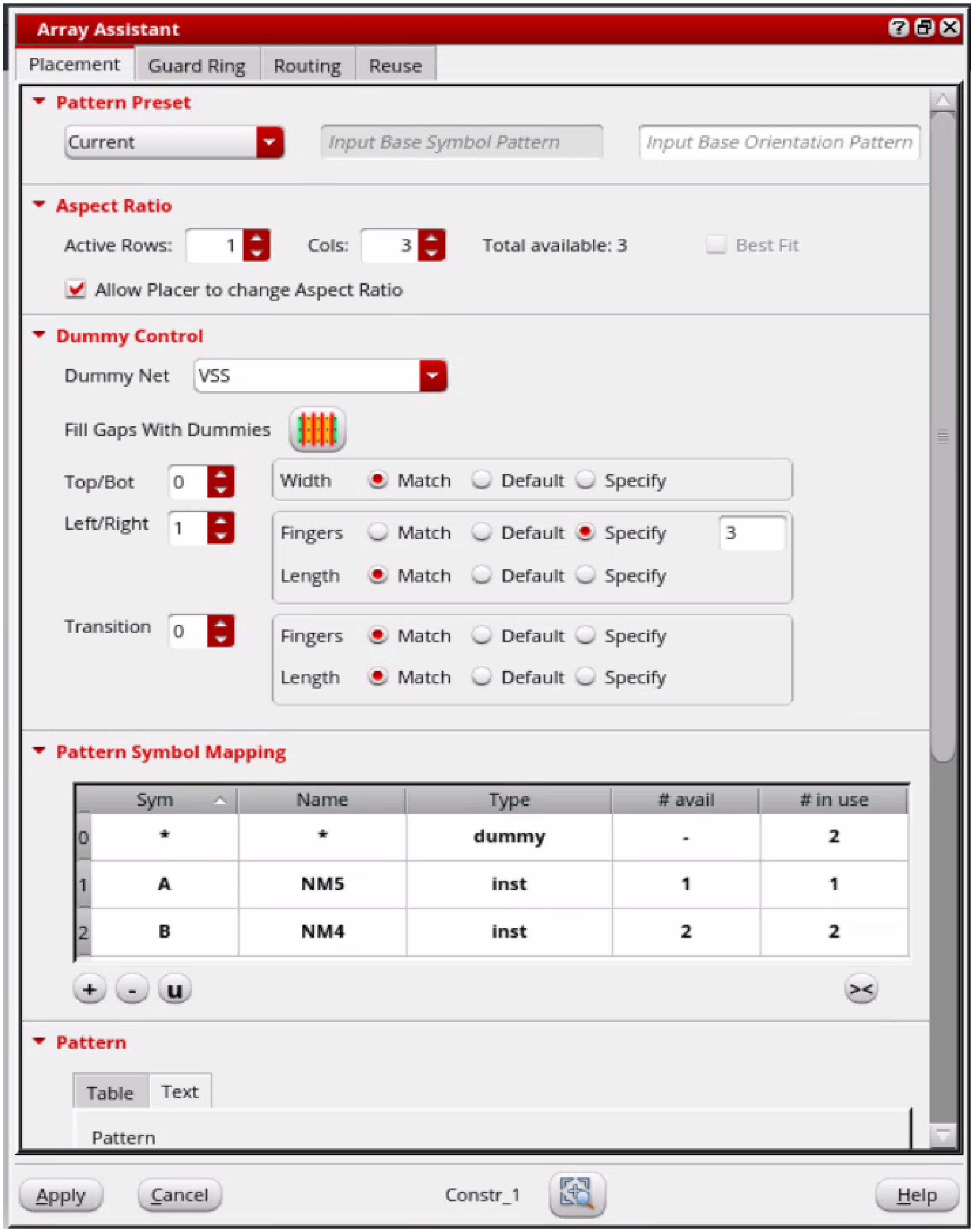
12. The Array Assistant is shown here.



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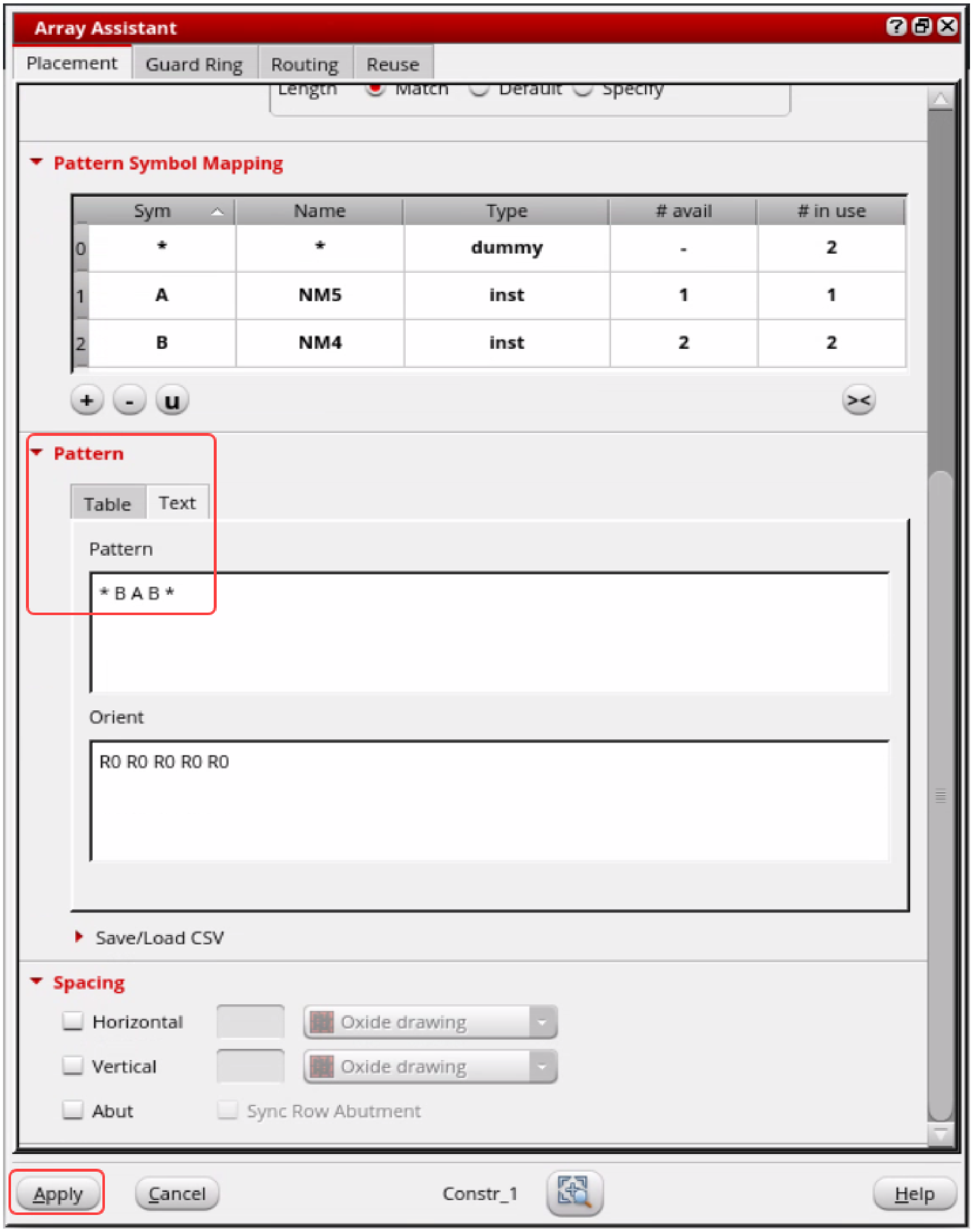
13. In the Array Assistant, in the **Pattern** field-**Text** tab, swap the **Pattern** as follows and click **Apply**.

```
* A B B * into * B A B *
```



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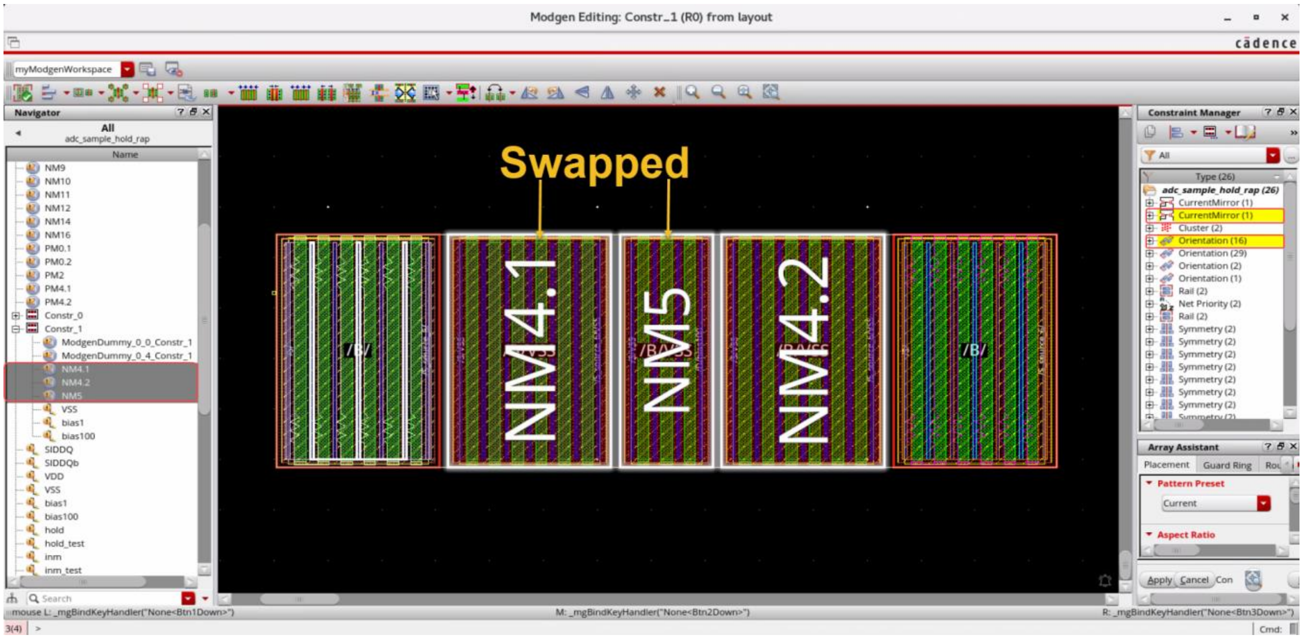
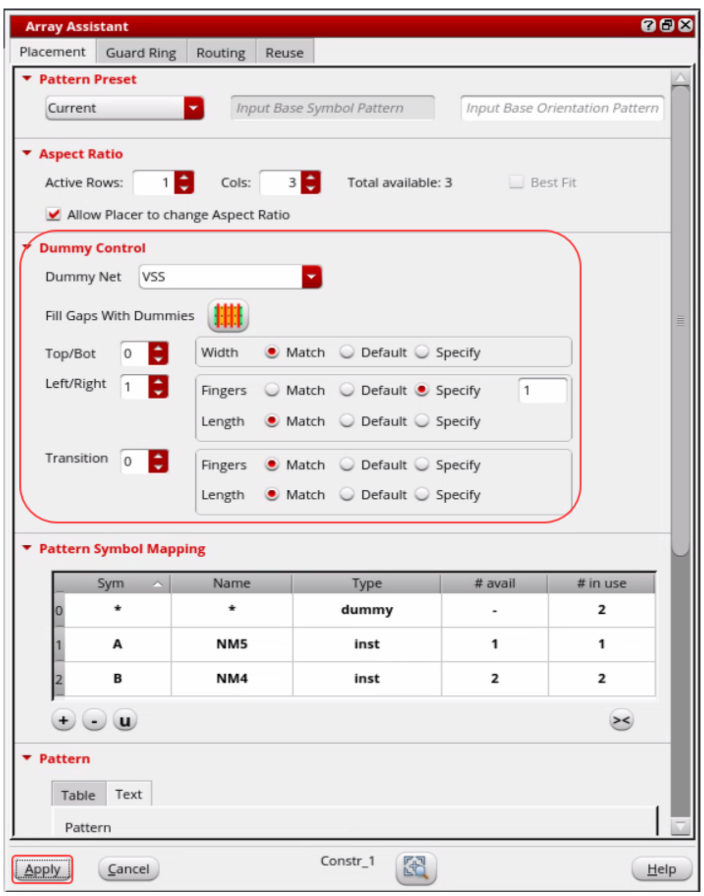
The instances get swapped, as shown here.



14. In the Array Assistant, modify the **Dummy Control** section as follows and click **Apply**.



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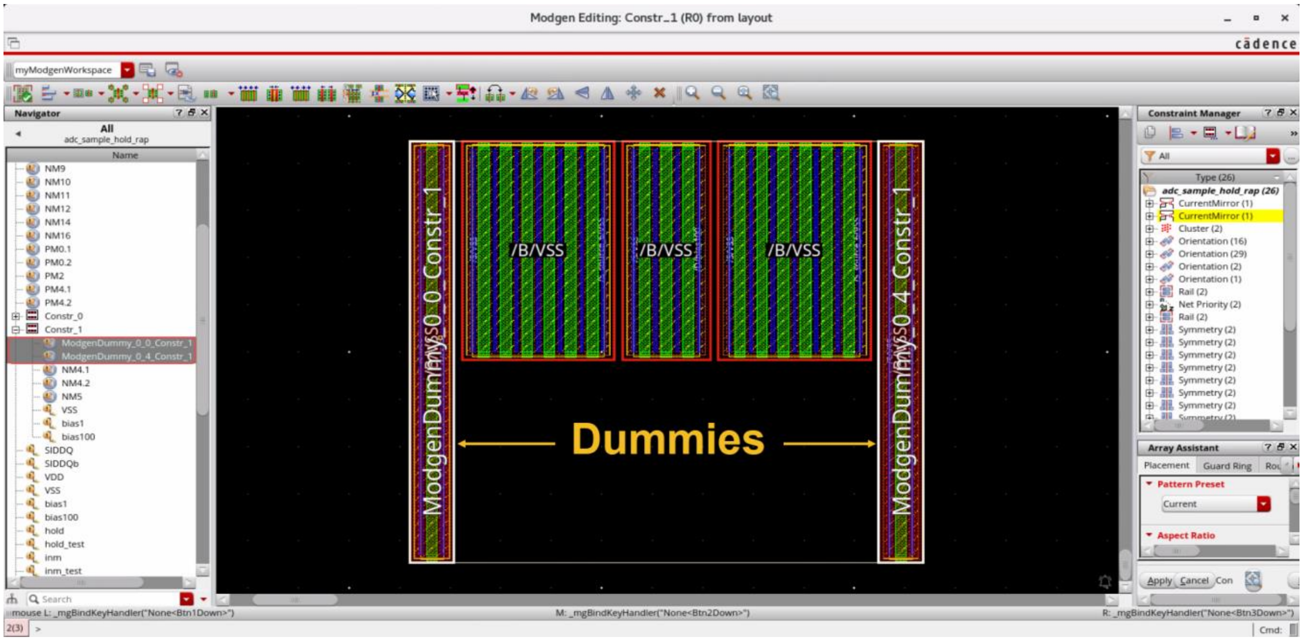
The dummies get updated, as shown here.



15. Exit the Modgen Editor window.

    Next, you will do the constraints-based placement using the Analog Placer.

16. In the layout canvas, turn on the Auto P&R assistant if it is not already turned on.

    a. In the **Constraints** tab, keep the default settings.

    b. In the **Placer** tab, with the default settings, click the **Run Analog Placer** icon to run the Automatic Analog Placer.

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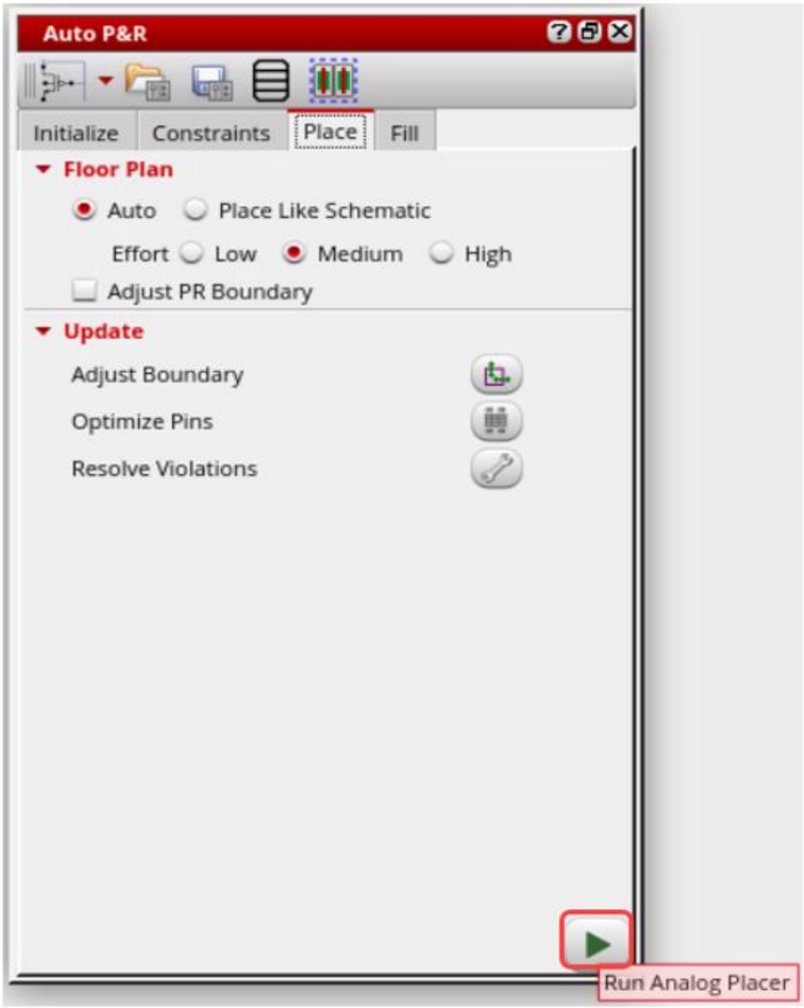
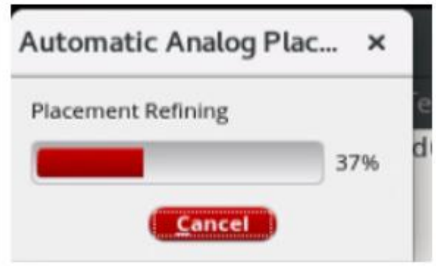
Automatic Analog Placer runs, as shown here.





The devices get placed in the layout, meeting all the constraints, as shown here.

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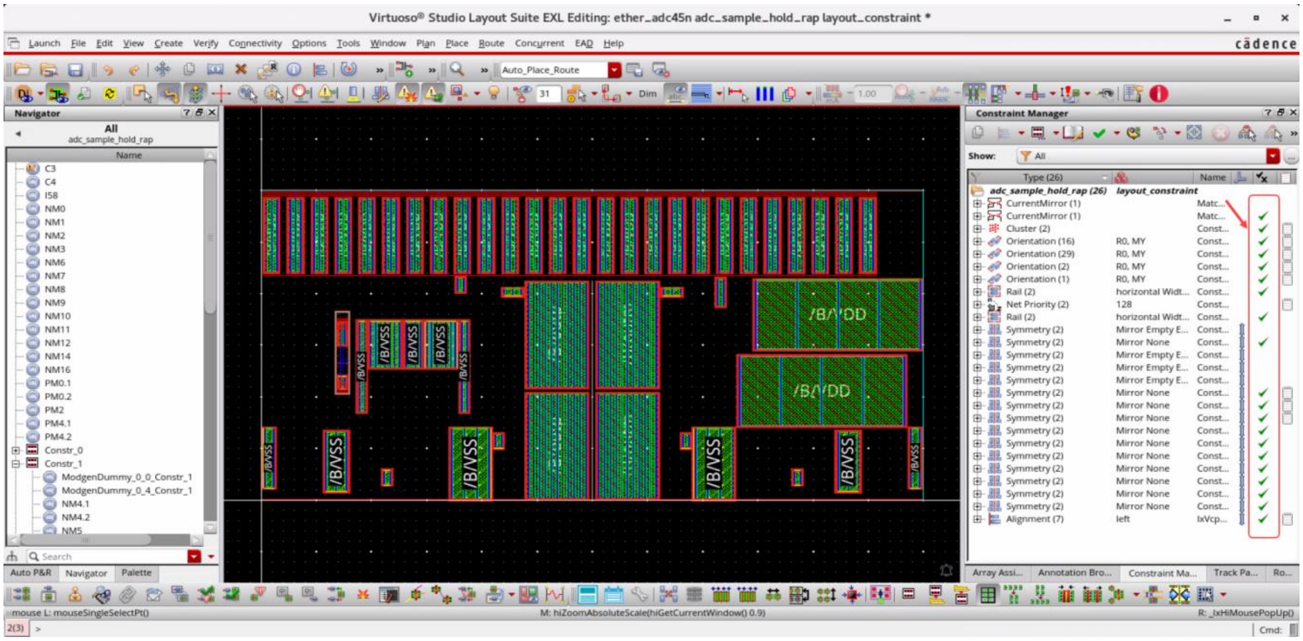
Note: You will get a placement similar to what is shown here.



17. Close all the cellviews without saving and exit the Virtuoso session.

Summary

In this lab, you learned about the following:

- The RAP flow and the Modgen features using the Array Assistant.

End of Lab

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Module 3: Power Routing

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Lab 3-1 Power Routing Using VSR

Objective: To implement a power structure on a given design using VSR.

The automatic power routing will allow the user to implement a power structure on a design. The Power Router has been re-introduced into the Virtuoso® Layout environment in IC 6.1.6. From IC 6.1.3 through IC 6.1.5, the power router was part of the Routing Integrated Development Environment and was not available within the Virtuoso Layout Suite environment.

In this lab, you will:

- Go through all the steps required to create a power structure on a given design.
- Step through the power routing and signal routing of block-level stdcell ASIC flow.

You will start with a placed design with connectivity and placed I/O pins. The power router will work only on power and ground nets. Additionally, if you have critical clock timing, it should be pre-routed after power routing and before signal routing with a digital clock tree tool like in EDI/Innovus™. In this case, you will be routing it within VSR.

You will see how power routing can be done with the Virtuoso Power Routing capability. It is important to note that while this is called a power router, it is really more of a toolbox to allow you to build power structures very quickly. It is not completely automatic.

Opening the Software

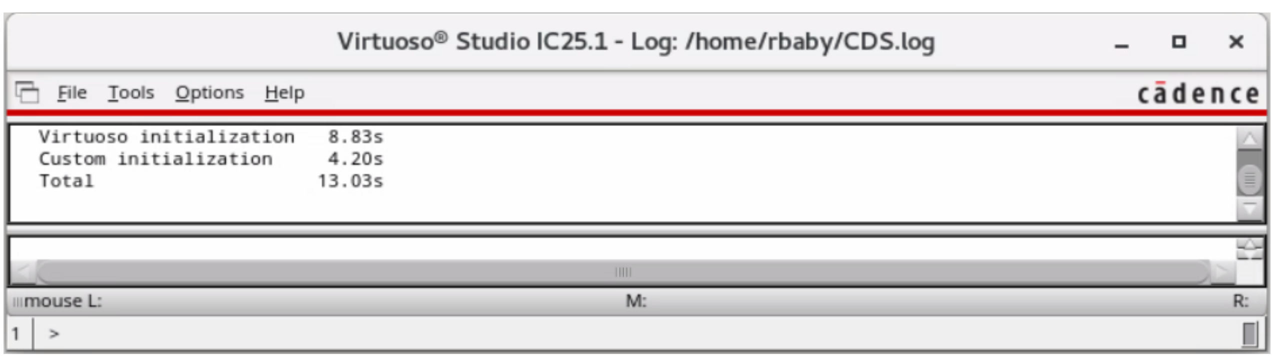
1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

   ```
   cd ./VLPT6_IC_25_1/mod03_PowerRouting
   ```

2. To start the Cadence® software, enter this command in the same terminal window:

   ```
   virtuoso &
   ```

   The Command Interpreter Window (CIW) appears.



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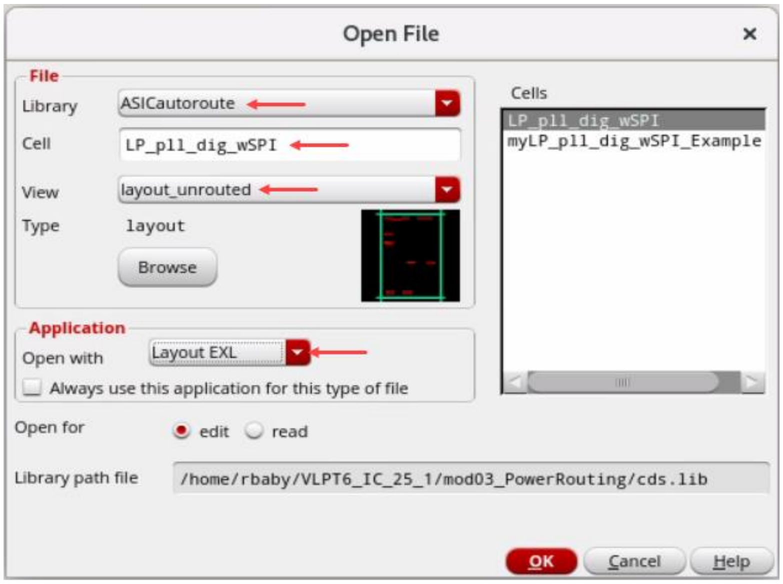
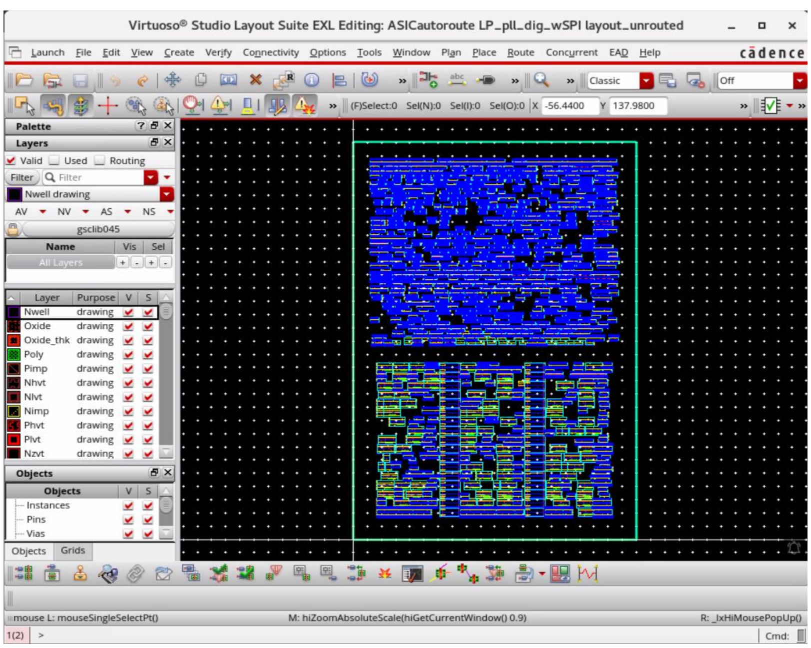
3. In the CIW, choose **File – Open** and enter the following values in the form:

Library	ASICautoroute
Cell	LP_pll_dig_wSPI
View	layout_unrouted



4. Click **OK**.

   The Virtuoso Layout Suite EXL Palette opens.



   You can see that the design is a basic block-level ASIC layout with 2 regions of standard cells and I/O pins. You will see how to use the power routing functions to achieve the power distribution that you want.

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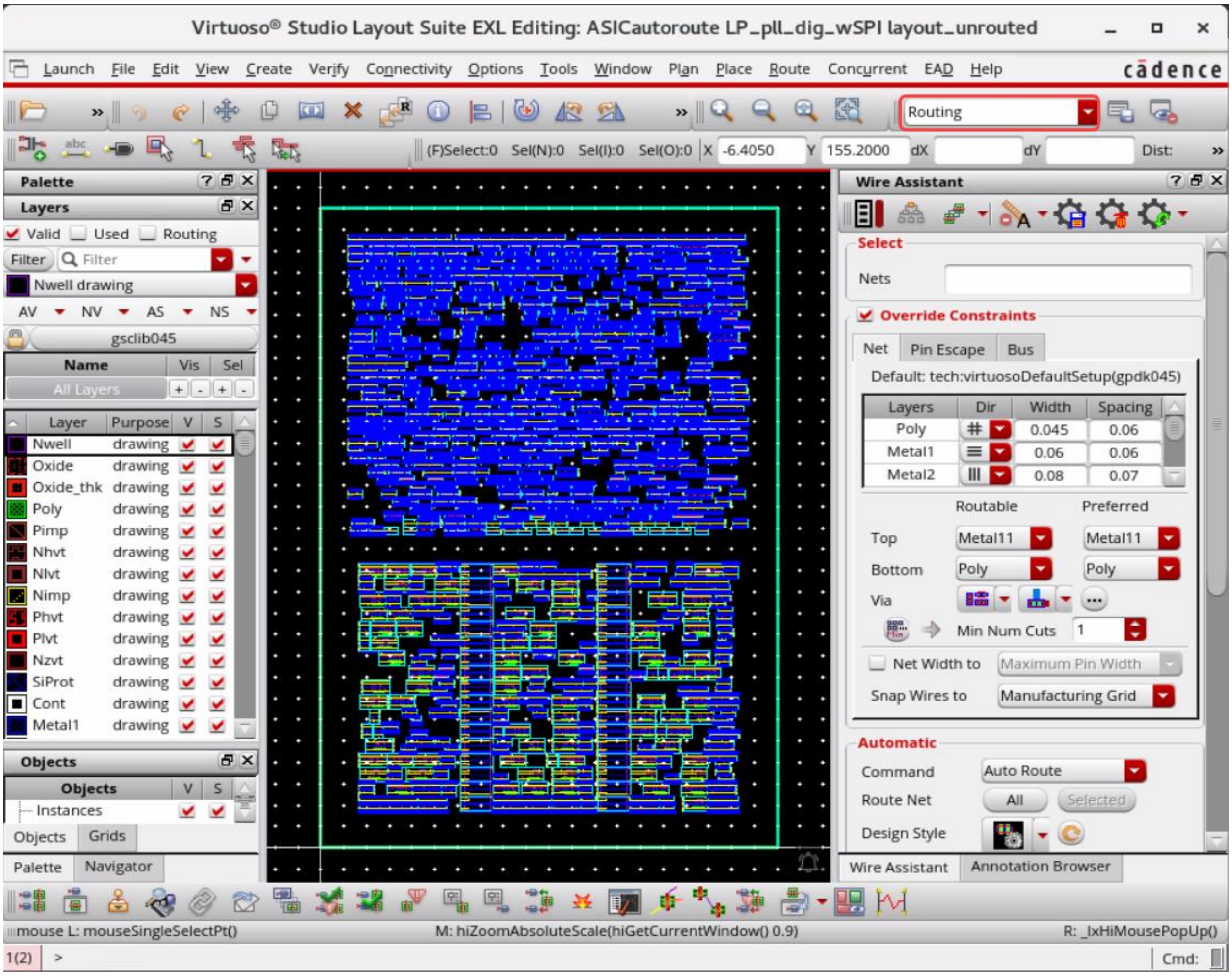
Note: In the **Workspace Configuration**, you can use the predefined workspace *Routing*.

This brings up the Wire Assistant, Annotation Browser, Navigator, and Palette assistants.

5. Press the **Shift+F** bindkey to view all the levels of the hierarchy in the layout canvas if it is not already enabled.



Note: Now you want to set the Design Style and the connectivity extractor setup. This is important if the hierarchy of your standard cell power rails may make it impossible for the cell power rails to route properly because you have shapes below that have no net name that are physically connected to pins with names.

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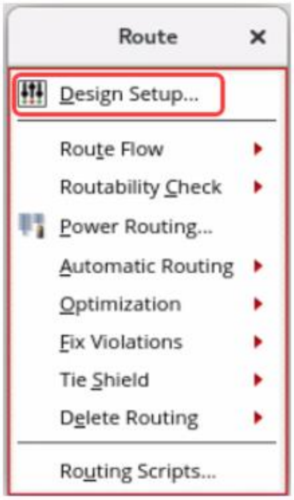
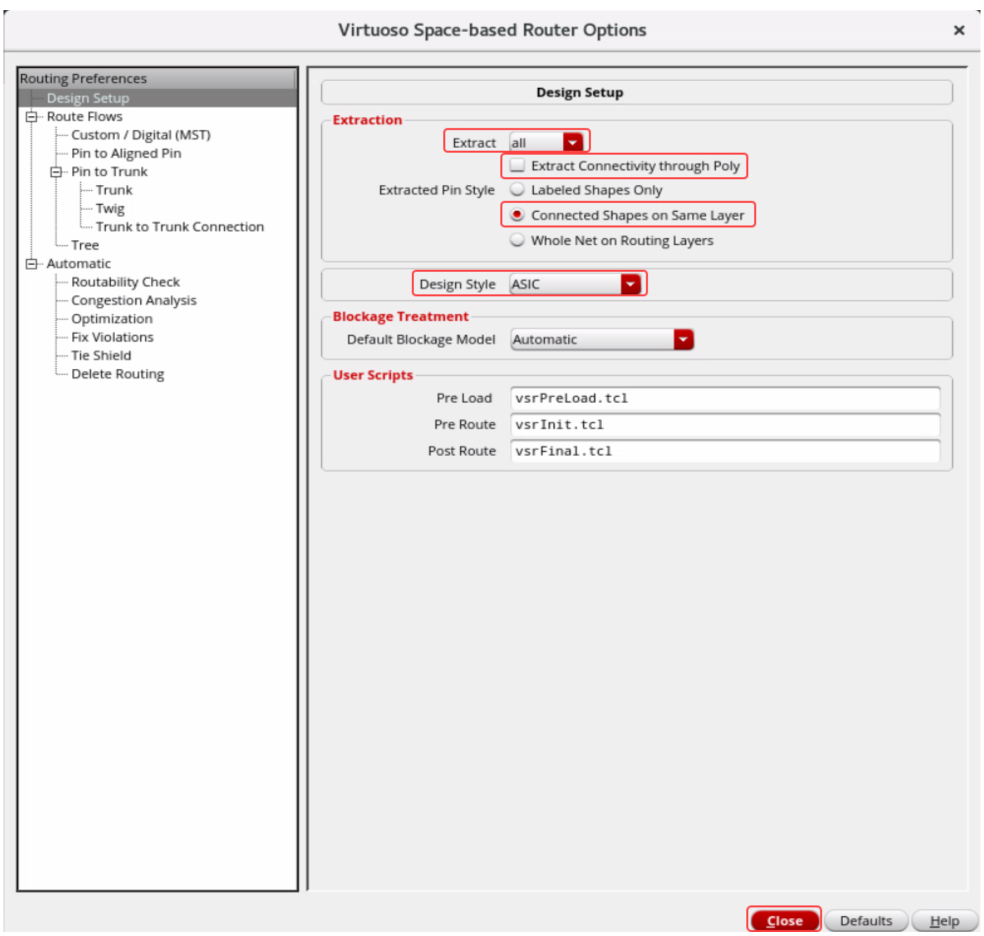
6. In the layout canvas, choose the **Route – Design Setup** menu command or use the **Design Setup** icon in the Virtuoso Space-based Router toolbar to open the Virtuoso Space-based Router Options form.



Note: In the design canvas, choose the **Window – Toolbars – Virtuoso Space-based Router** menu entry to display the *Virtuoso Space-based Router* toolbar if it is not enabled already.

The Virtuoso Space-based Router Options form opens.



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7. In the Virtuoso Space-based Router Options form, select the options as shown here:

   a. **Extract**: *all*

   b. **Extract Connectivity through Poly**: *off*

   c. **Extracted Pin Style**: *Connected Shapes on Same Layer*

   d. **Design Style**: *ASIC* (Sets layer densities in global route so that it will try to push routing up from the lower layers.)

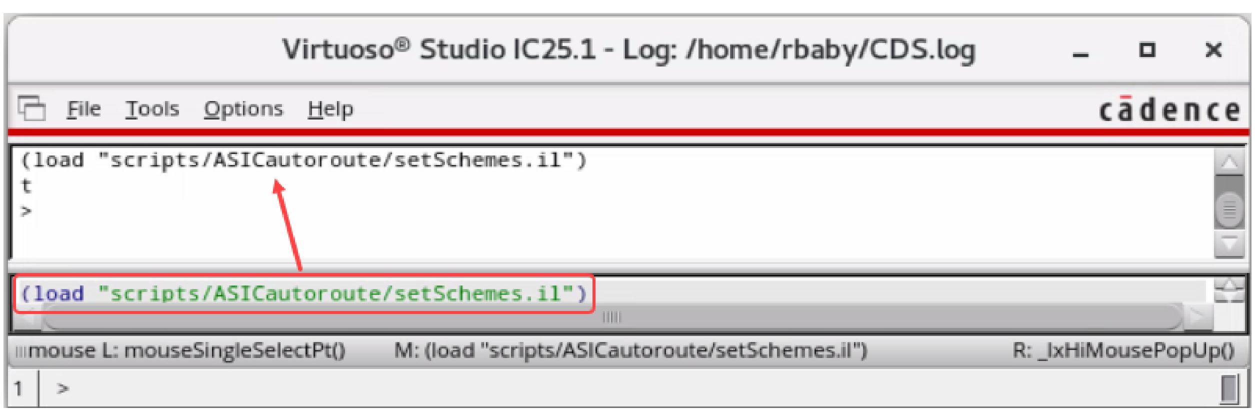
   e. Close the form.

Power Distribution

For the power distribution, you want several structures as shown here:

- Feedthrus for VSS, VDD, and VDDsw
- VDD and VSS core rings around design
- VDDsw ring around lower block
- Horizontal VDD and VSS stripes between the blocks
- Horizontal VDD stripes across lower “block” region for ExtVDD connections
- Two sets of vertical VDD, VDDsw, and VSS stripes
- Cell row power rail stripes (follow-pin)

1. Execute the following command in the CIW window:




```
(load “scripts/ASICautoroute/setSchemes.il”)
```

It will create all the schemes and source the procedures for pin connections for you to use.

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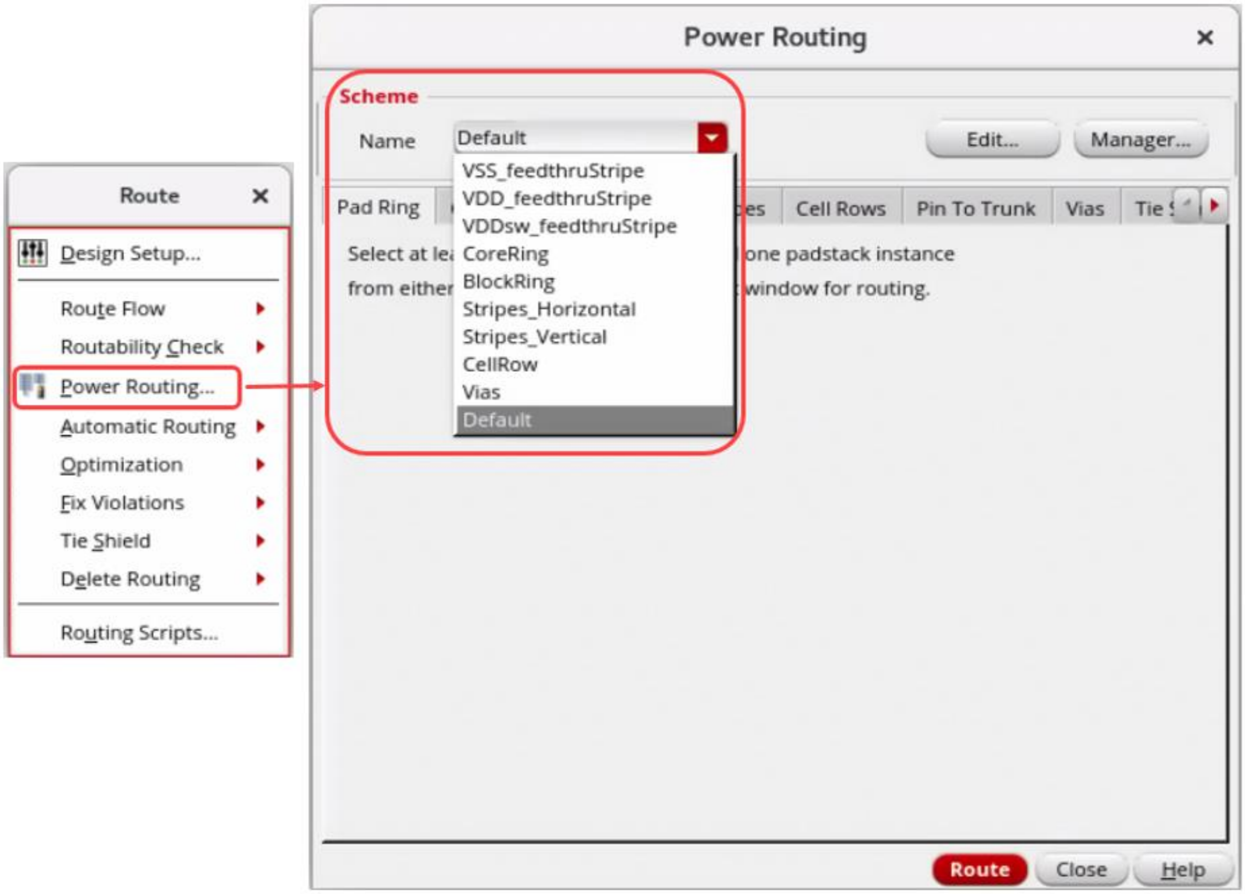
Invoking the Power Router

1. In the layout canvas, choose the **Route – Power Routing** menu command.

   The Power Routing form opens, as shown here.



Vertical Feedthru Stripes for VSS, VDD, and VDDsw Pins

Measure the feedthru pin widths and locations (left edge) in the layout canvas. You will make vertical stripes for them. Since the widths and space are not regular, you will do it with 3 separate *create stripe* commands.

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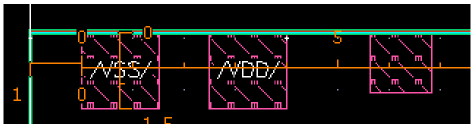
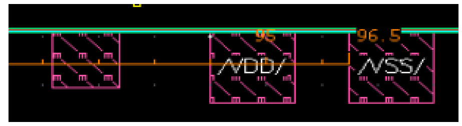
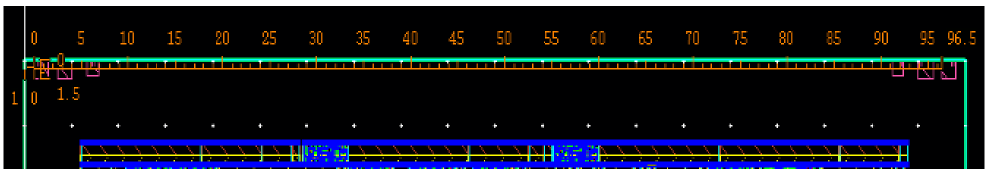
Vertical Feedthru Stripe Creation for the VSS Pin

1. First, you will measure the **VSS** pin. Observe the layer the pin is on and what its width and its left coordinate are. Also, measure the space between the left side of the left **VSS** pin to the left side of the right **VSS** pin.





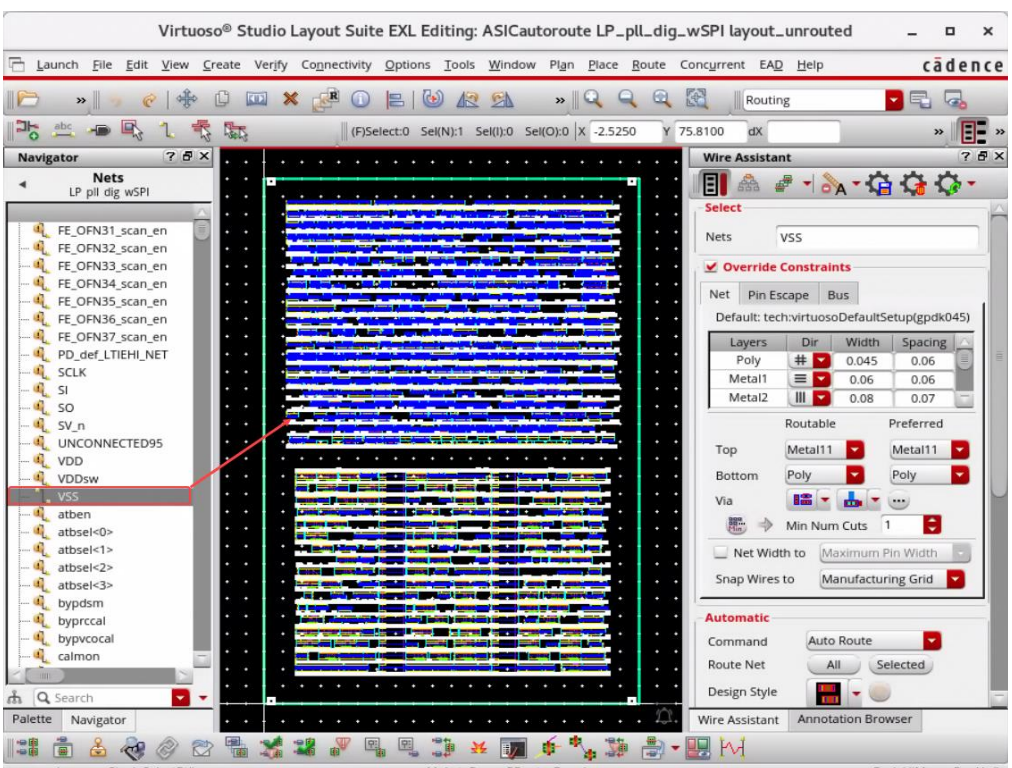


   a. You will observe that the **VSS** pin is on **Metal8**, it has a width of **1.5um**, the left side of the leftmost **VSS** pin starts at **1.0um**, and there is **96.5um** between the left edges of both pins as shown here.

      With this information, you should be able to check/fill the form values in the Power Routing form.

2. Click the **Stripes** tab in the Power Routing form.

3. Select the net **VSS** in the Navigator assistant.



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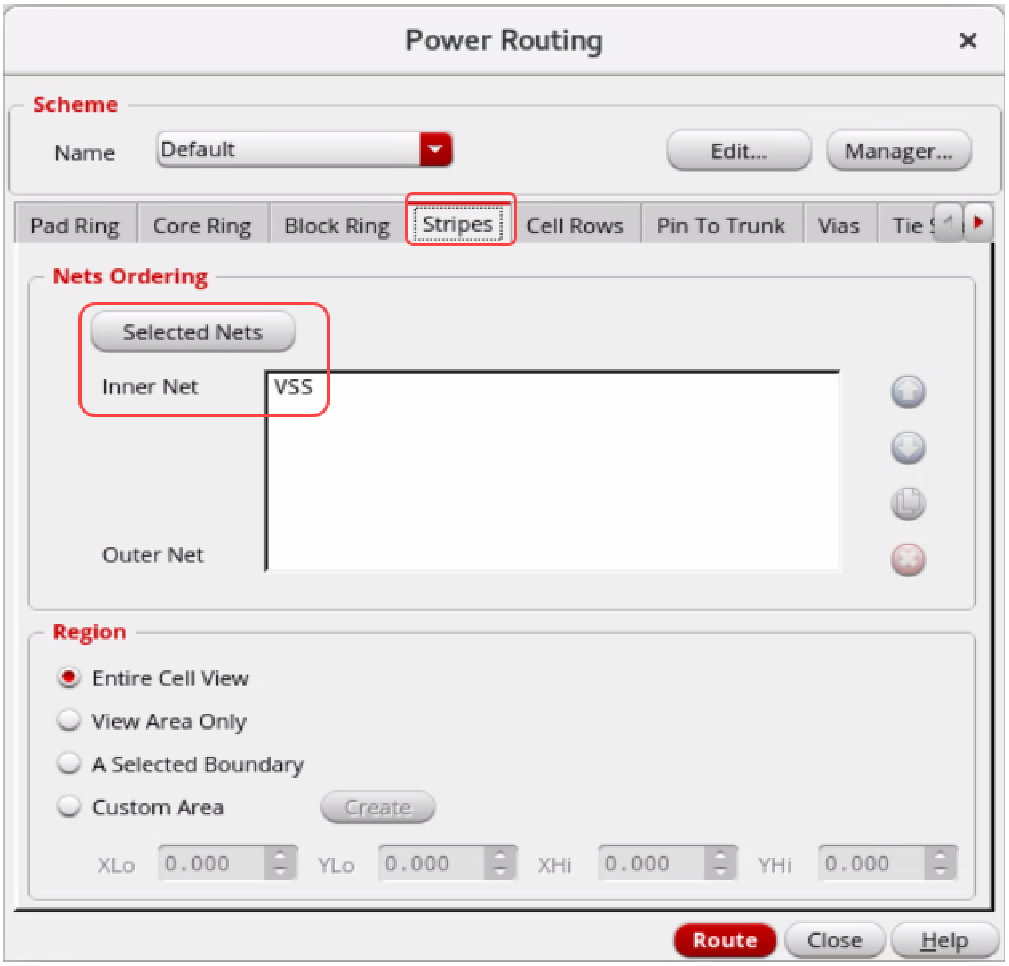
a. Click the **Selected Nets** button in the Power Routing form.

This action adds the net **VSS** to the *Nets Ordering* list in the Power Routing form.



b. Select **VSS_feedthruStripe** in the *Scheme* section in the Power Routing form.

c. Then click the **Edit…** button in the Power Routing form.

The Power Routing Options form opens as shown here.

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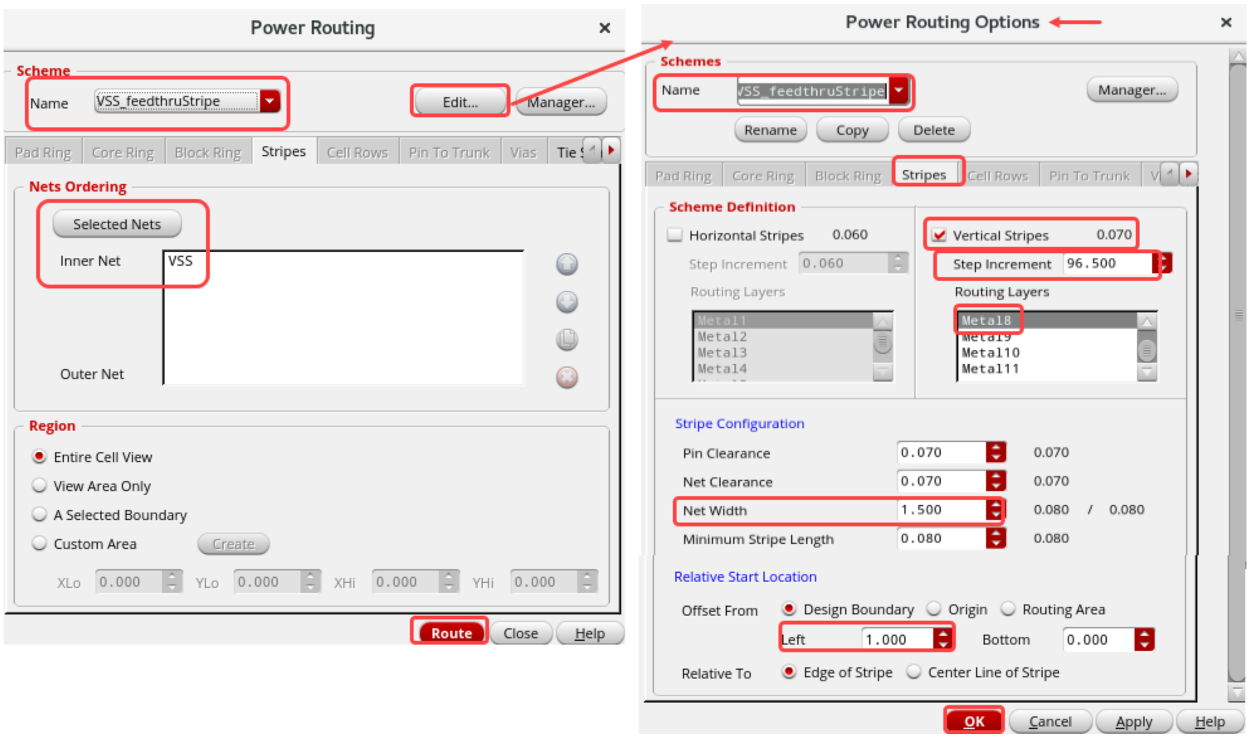
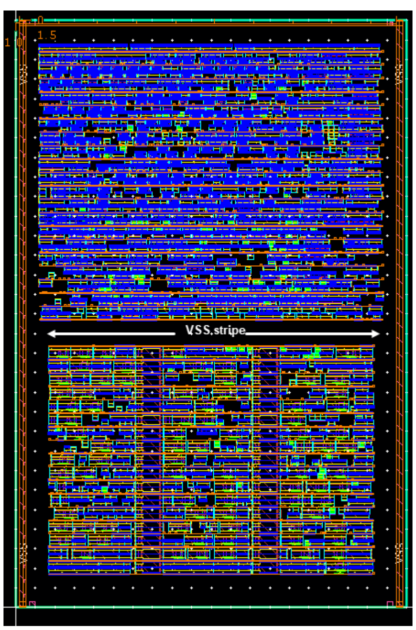
d. Verify whether the **Power Routing Options** form shows the values which you have measured in the previous steps, as shown here.



e. Click **OK** in the Power Routing Options form.

f. Click the **Route** button in the Power Routing form.

The Vertical Feedthru Stripe connection for the **VSS** pin is created.



g. Observe the **CIW/CDS.log** for the Power Routing log information.

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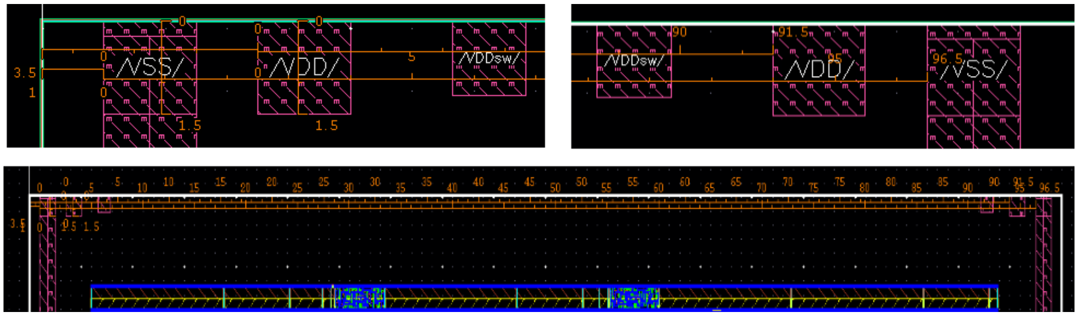
Vertical Feedthru Stripe Creation for the VDD Pin

1. Next, measure the **VDD** pin. Observe the layer the pin is on and what its width and its left coordinate are. Also, measure the space between the left side of the left **VDD** pin to the left side of the right **VDD** pin.



   a. You will observe that the **VDD** pin is on **Metal8**, it has a width of **1.5um**, the left side of the leftmost **VDD** pin starts at **3.5um**, and there is **91.5um** between the left edges of both pins as shown here.

      With this information, you should be able to check/fill the form values in the Power Routing form.

2. Select the net **VSS** in the *Nets Ordering* list in the Power Routing form and click the **X** button to remove it.

   a. Select the net **VDD** in the Navigator assistant.

   b. Click the **Selected Nets** button in the Power Routing form.

      The net **VDD** is added to the *Nets Ordering* list in the Power Routing form.

   c. Select **VDD_feedthruStripe** in the *Scheme* section in the Power Routing form.

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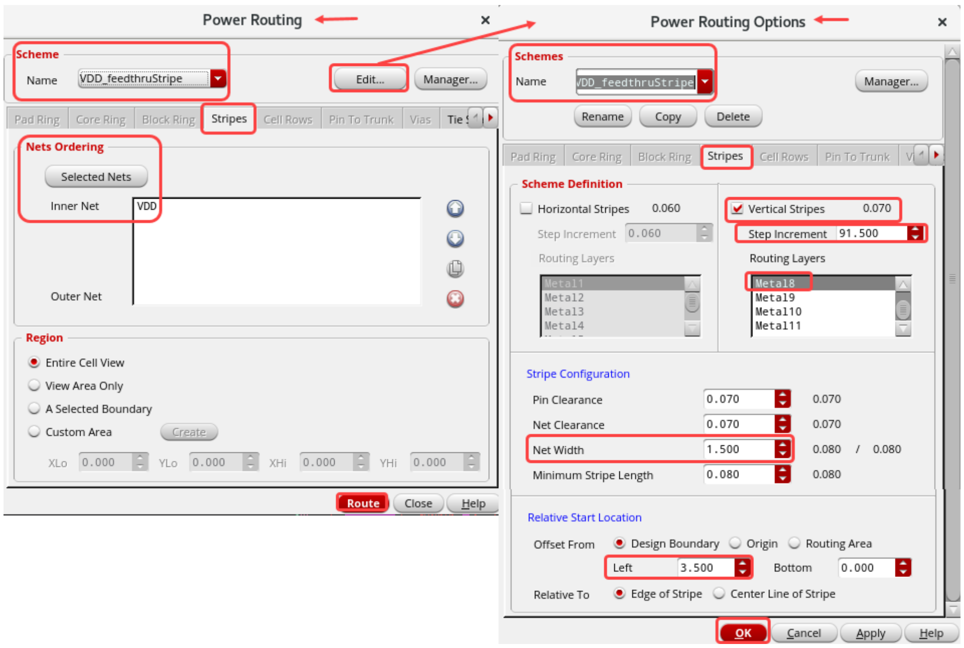
d. Now, click the **Edit…** button in the Power Routing form.

The Power Routing Options form opens, as shown here.

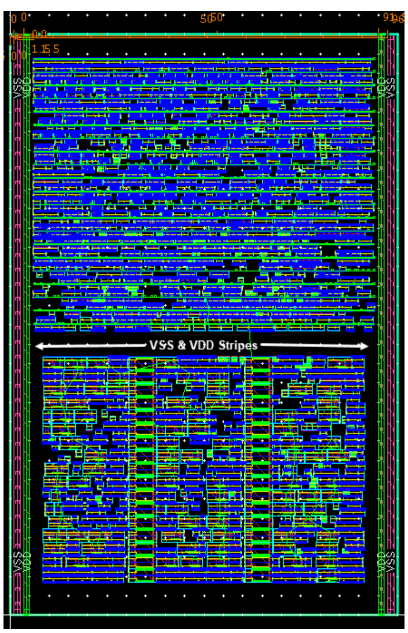
e. Verify whether the **Power Routing Options** form shows the values that you have measured in the previous steps, as shown here.



f. Click **OK** in the Power Routing Options form.

g. Click the **Route** button in the Power Routing form.

The Vertical Feedthru Stripe connection for the **VDD** pin is created.



h. Observe the **CIW/CDS.log** for the Power Routing log information.

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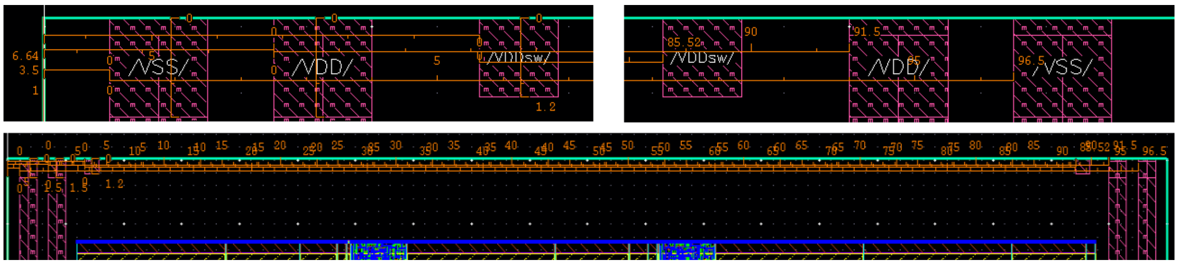
Vertical Feedthru Stripe Creation for the VDDsw Pin

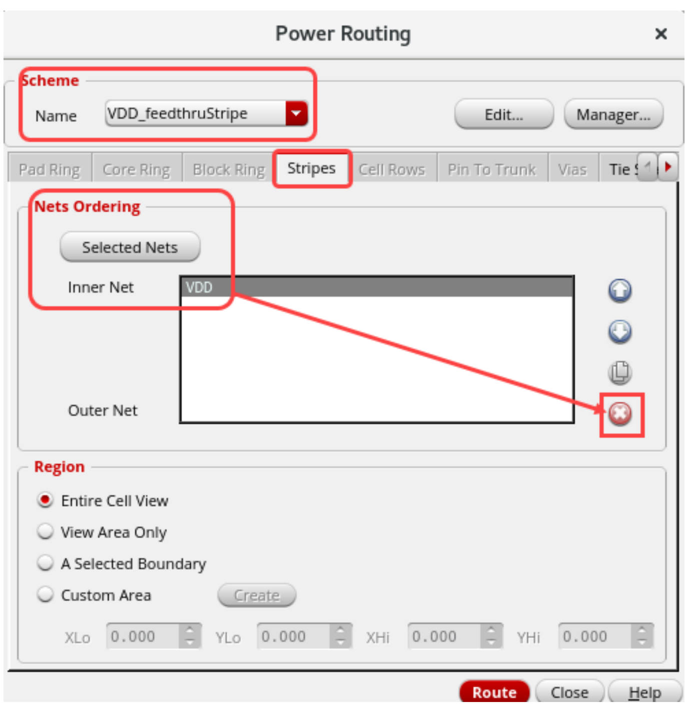
1. Next, measure the **VDDsw** pin. Observe the layer the pin is on and what its width and its left coordinate are. Also, measure the space between the left side of the left **VDDsw** pin to the left side of the right **VDDsw** pin.



   a. You will observe that the **VDDsw** pin is on **Metal8**, it has a width of **1.2um**, the left side of the leftmost **VDDsw** pin starts at **6.64um**, and there is **85.52um** between the left edges of both pins as shown here.

   With this information, you should be able to check/fill the form values in the Power Routing form.

2. Select the net **VDD** in the *Nets Ordering* list in the Power Routing form and click the **X** button to remove it.



   a. Select the net **VDDsw** in the Navigator assistant.

   b. Click the **Selected Nets** button in the Power Routing form.

   The net **VDDsw** is added to the *Nets Ordering* list in the Power Routing form.

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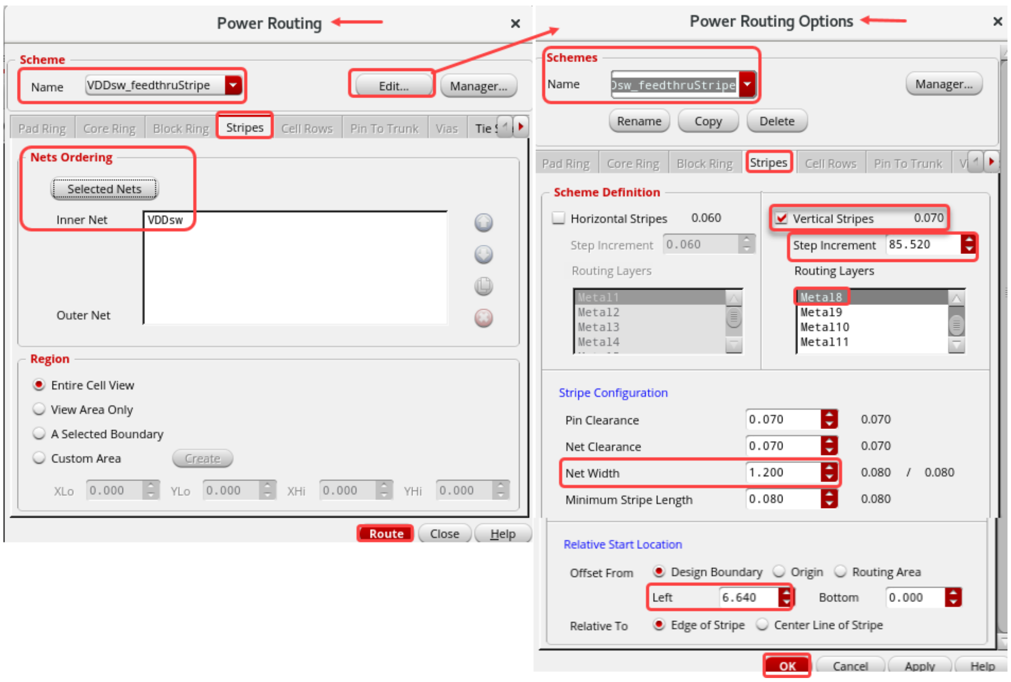
c. Select **VDDsw_feedthruStripe** in the *Scheme* section in the Power Routing form.

d. Now, click the **Edit…** button in the Power Routing form.

The Power Routing Options form opens, as shown here.

e. Verify whether the **Power Routing Options** form shows the values that you have measured in the previous steps, as shown here.



f. Click **OK** in the Power Routing Options form.

g. Click the **Route** button in the Power Routing form.

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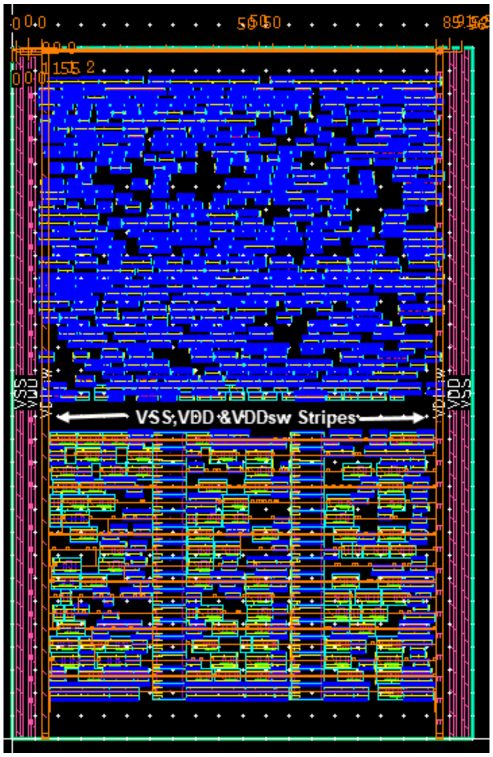
The Vertical Feedthru Stripe connection for the **VDDsw** pin is created. You have finished with the feedthru connections. Your layout should look something like, as shown here.



h. Observe the **CIW/CDS.log** for the Power Routing log information.

Core Rings Creation for VDD and VSS Pins

Now you are ready to create power rings. You want **VDD** and **VSS** to ring the core with **Metal6** vertically and **Metal7** horizontally. You want the core ring to line up with the **VDD** and **VSS** feedthrus so that the vertical segments (**Metal8** and **Metal6**) are coincident (**Metal8** and **Metal6** width and spacing to match).

1. Measure the spacing from the core to the right edge of the **VDD** feedthru to get the core spacing.

   a. After completing the measurement, you will observe that you should have the core ring width to be **1.5um** (to match the feedthru width) and be spaced **1.0um** outside the core region to line up with the feedthrus, so it will be necessary to route **VSS** as the outer ring and **VDD** as the inner ring. You also want **1.0um** spacing between the rings to line up with the feedthrus.

2. Click the **Core Ring** tab in the Power Routing form.

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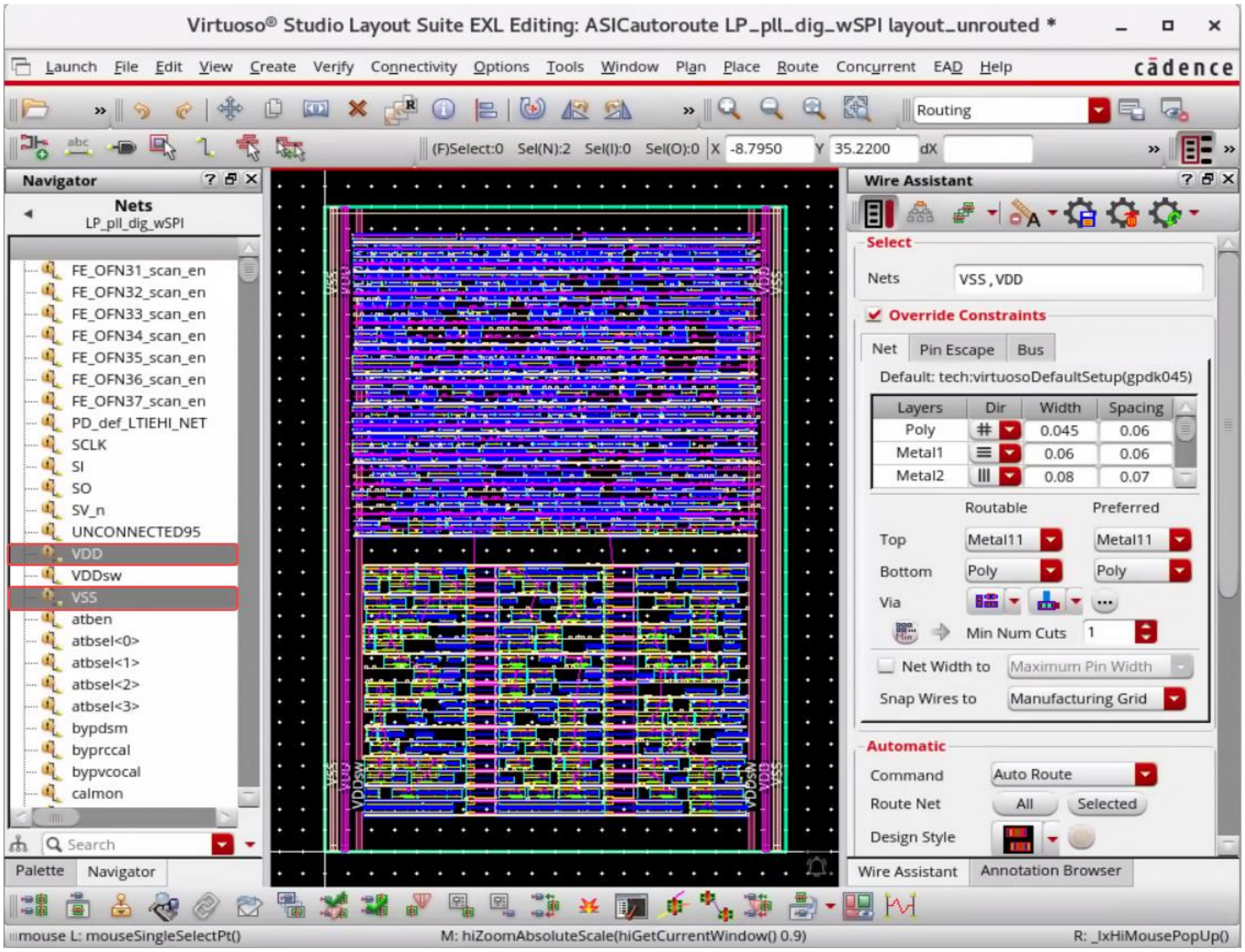
Note: You may have to go to the **Default** *Scheme* to activate the **Power Routing** tabs.

a. Select the nets **VDD** and **VSS** in the Navigator assistant.



b. Click the **Selected Nets** button in the Power Routing form.

The nets **VDD** and **VSS** are added to the *Nets Ordering* list in the Power Routing form.

c. Select **CoreRing** in the *Scheme* section in the Power Routing form.

d. Then click the **Edit…** button in the Power Routing form.

The Power Routing Options form opens as shown here.

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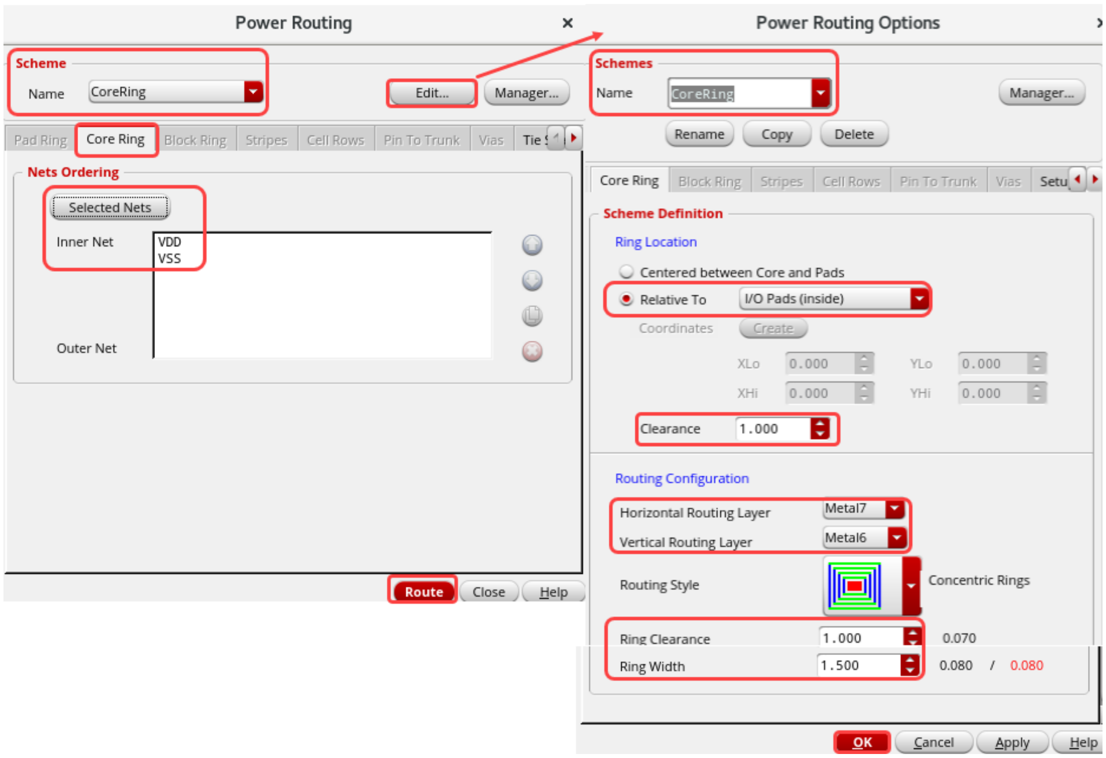
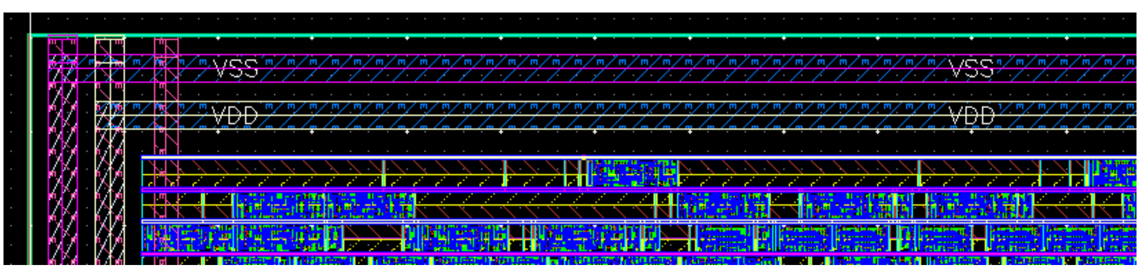
e. Verify whether the **Power Routing Options** form shows the values that you have measured in the previous steps, as shown here.



f. Click **OK** in the Power Routing Options form.

g. Click the **Route** button in the Power Routing form.

The Core Rings for **VDD** and **VSS** are created. You have finished with the **Core Ring** connections. Your layout should look like this. The zoomed-in view is shown here.



h. Observe the **CIW/CDS.log** for the Power Ring Creation log information.

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Block Ring Creation for the VDDsw Pin

You will create the **VDDsw** block ring around the lower block with **Metal6** vertically and **Metal7** horizontally.

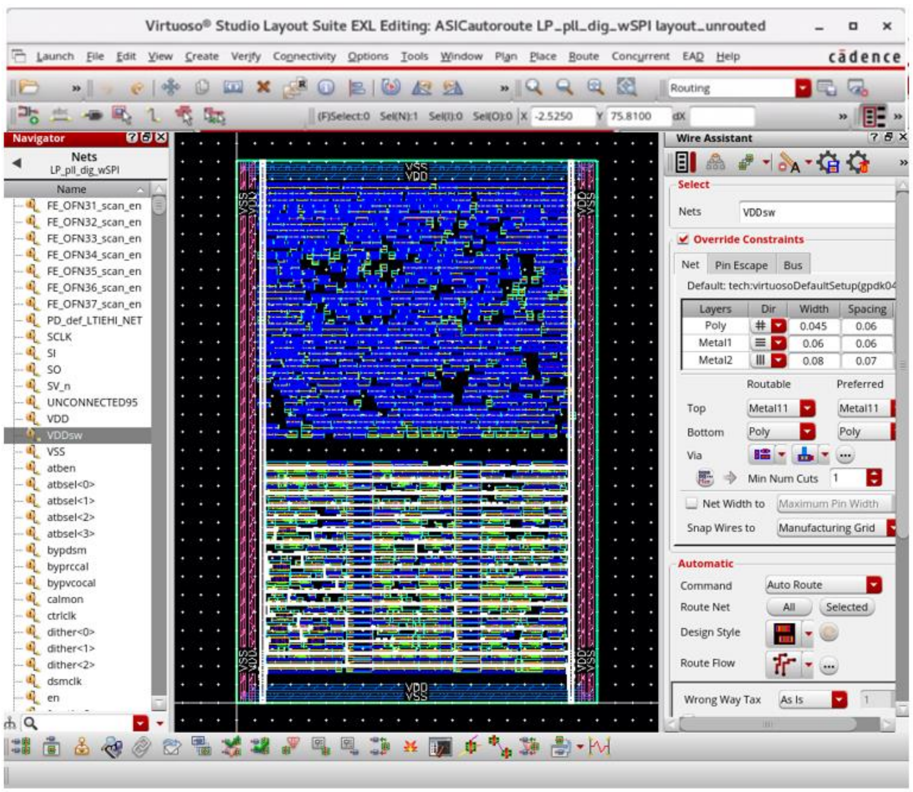
1. For the lower block ring, you want it to also match its feedthru width. Measure the distance from the core to the right edge of the **VDDsw** feedthru.

   For the lower block ring, you want it to be **1.2um** to match its feedthru width and space it **0.56um** outside the core to line up with its feedthru.

2. Click the **Block Ring** tab in the Power Routing form.

   Note: You may have to go to the **Default** *Scheme* to activate the **Power Routing** tabs.

   a. Select the net **VDDsw** in the Navigator assistant.

   b. Click the **Selected Nets** button in the Power Routing form.

      The net **VDDsw** is added to the *Nets Ordering* list in the Power Routing form.

   c. Select **BlockRing** in the *Scheme* section in the Power Routing form.

   d. Then click the **Edit…** button in the Power Routing form.

      The Power Routing Options form opens, as shown here.

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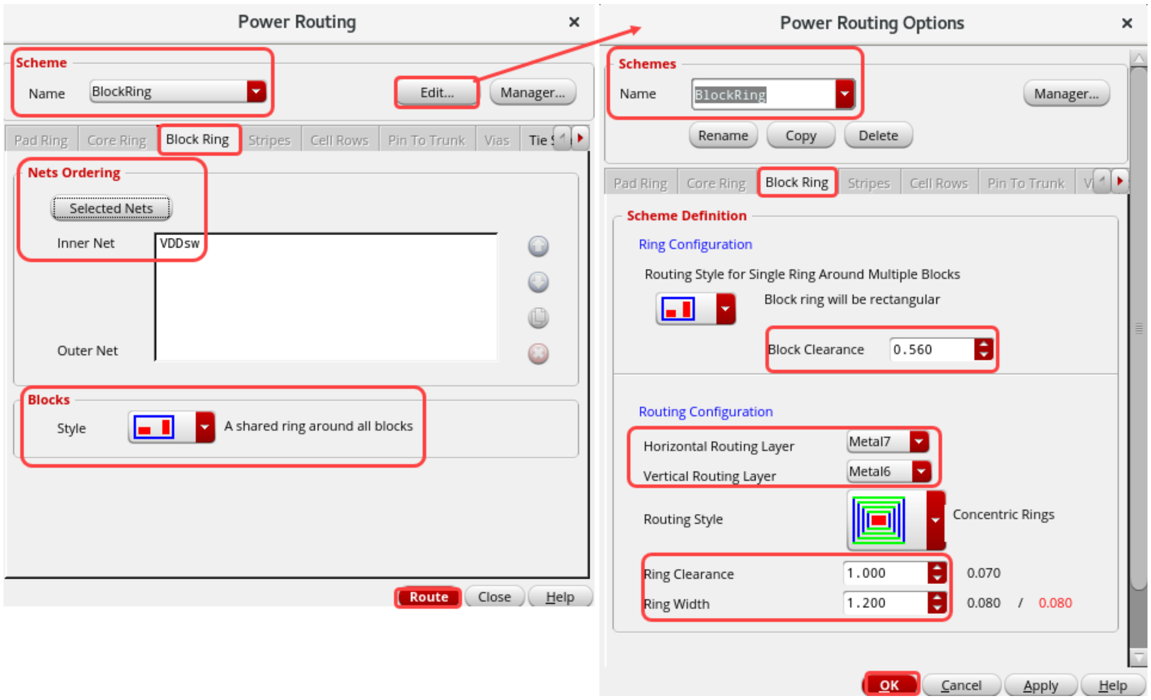
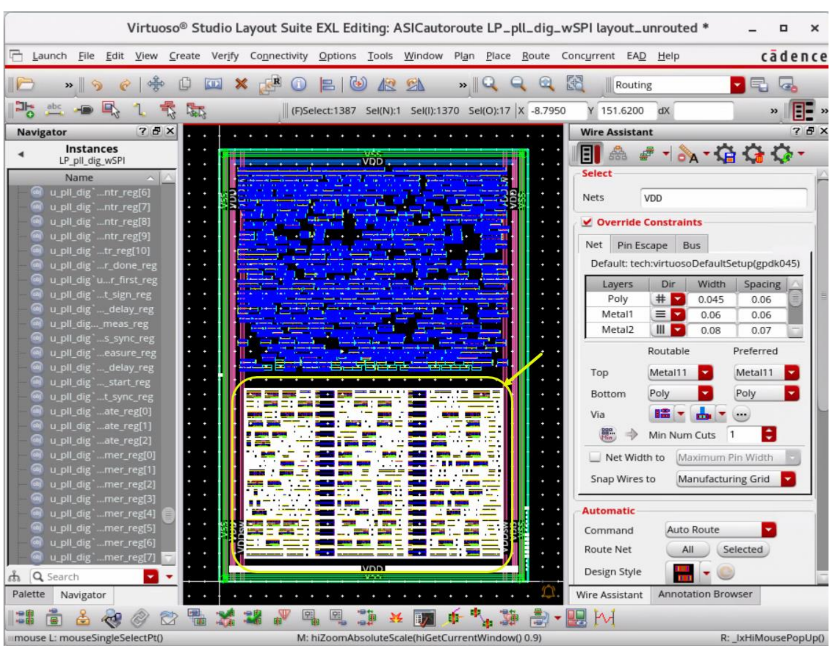
e. Verify whether the **Power Routing Options** form shows the values that you have measured in the previous steps.



Additionally, in the layout canvas, you will need to select the instances that you want to create the block ring around.

f. After setting up the Power Routing Options form, **drag-area-select** the instances in the bottom group for the **VDDsw** block ring, as shown here.



g. Click **OK** in the Power Routing Options form.

h. Click the **Route** button in the Power Routing form.

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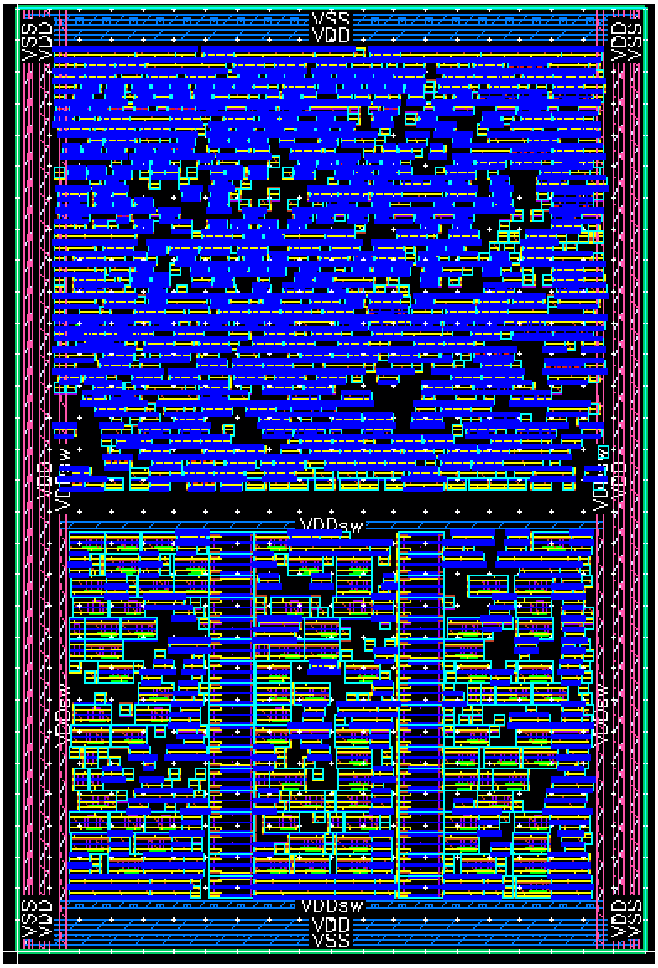
The Block Ring for **VDDsw** is created. You have finished with the **Block Ring** connections. Your layout should look like this.



i. Observe the **CIW/CDS.log** for the Block Ring Creation log information.

Horizontal Stripes Creation for VDD and VSS Pins

Now you will add stripes to help with the current distribution. You want to add 2 horizontal **Metal7** stripes (**VDD** and **VSS**) between the 2 stdcell regions above the **VDDsw** block ring. You want the stripes to be **1.2um** to match the V**DDsw Metal7** width, and have **0.6um** clearance from the **VDDsw** ring and between the **VDD** and **VSS** rings.

1. Measure up **0.06um** from the top of the **VDDsw** ring and note the y-coordinate. You will only want 1 set of stripes, so also note the approximate y-dimension of the prBoundary for the **Step Increment** to make sure that the second stripe will not be drawn within the design region.

   With this information, you should be able to check/fill the form values in the Power Routing form.

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2. Click the **Stripes** tab in the Power Routing form.

   Note: You may have to go to the **Default** *Scheme* to activate the **Power Routing** tabs.

3. Select the net **VDDsw** in the *Nets Ordering* list in the Power Routing form and click the **X** button to remove it.

   a. Select the nets **VDD** and **VSS** in the Navigator assistant.

   b. Click the **Selected Nets** button in the Power Routing form.

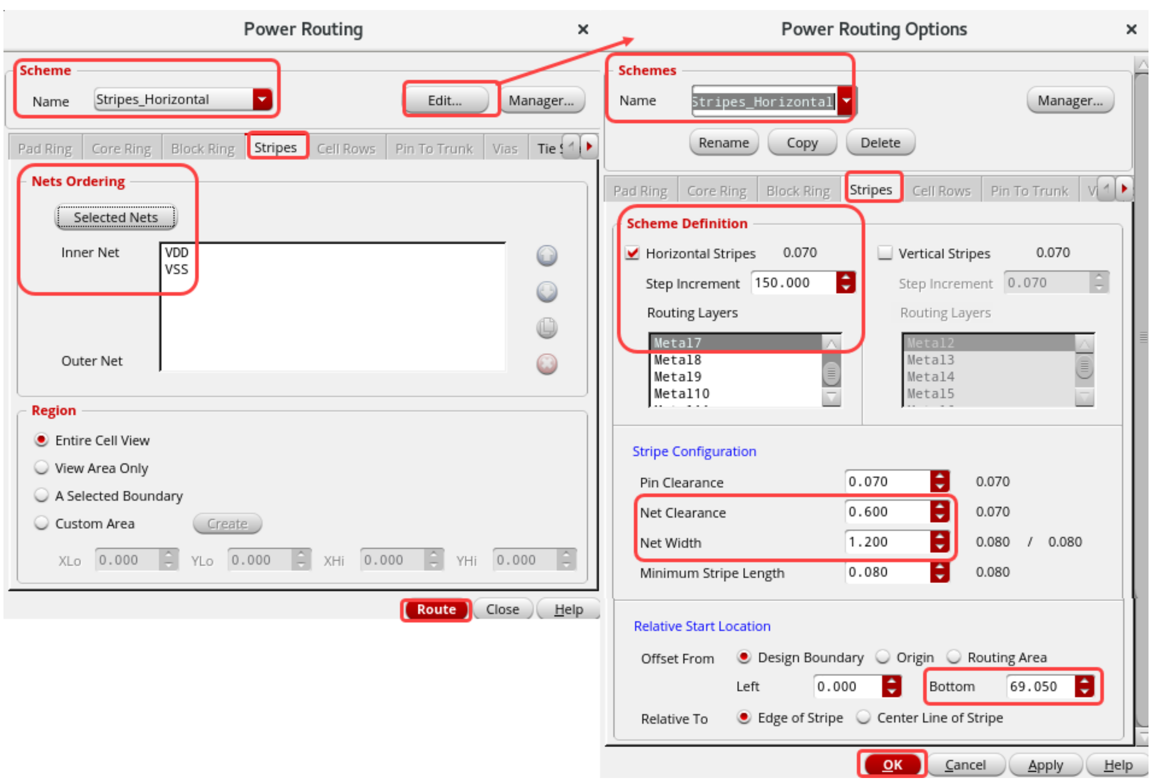
      The nets **VDD** and **VSS** are added to the *Nets Ordering* list in the Power Routing form.

   c. Select **Stripes_Horizontal** in the *Scheme* section in the Power Routing form.

   d. Then click the **Edit…** button in the Power Routing form.

      The Power Routing Options form opens, as shown here.

   e. Verify whether the **Power Routing Options** form shows the values that you have measured in the previous steps, as shown here.



   f. Click **OK** in the Power Routing Options form.

   g. Click the **Route** button in the Power Routing form.

      The Horizontal Stripes are created for the **VDD** and **VSS** Pins.

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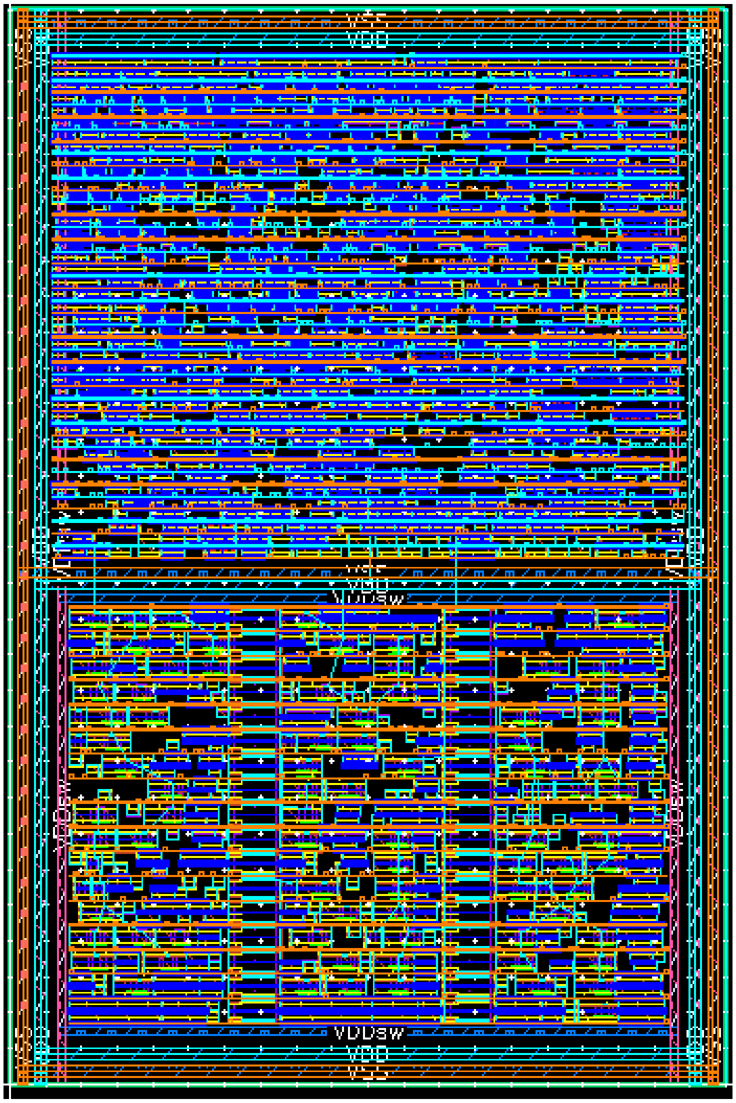
You have finished with the **Horizontal Stripes** connections. Your layout should look like this.



h. Observe the **CIW/CDS.log** for the Horizontal Stripes Creation log information.

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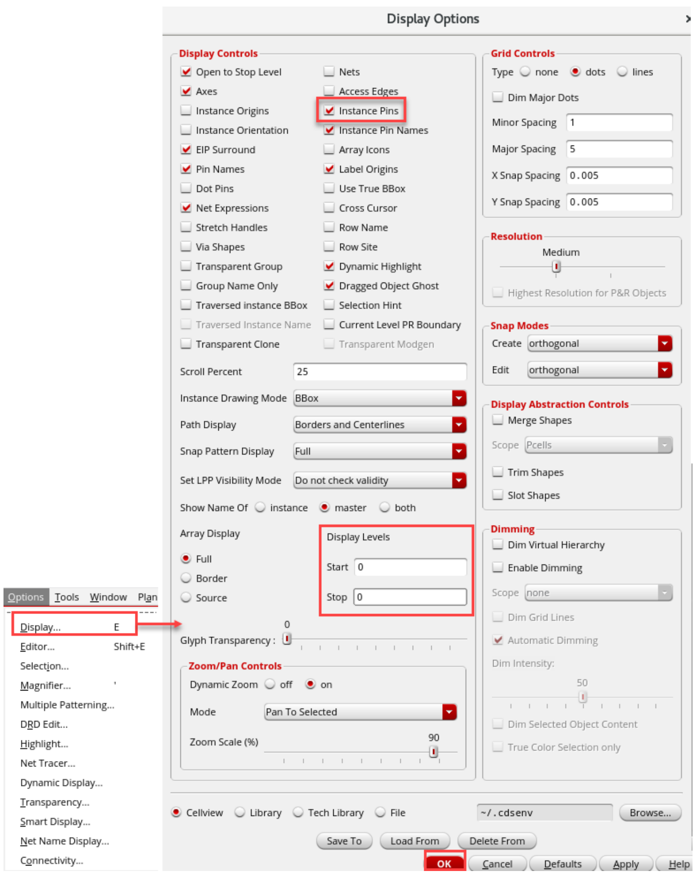
Horizontal Stripes Creation to Route the VDD Power to ExtVDD Pins

The next section is to route the **VDD** power to the **ExtVDD** pins on the standard cells in the region with the V**DDsw** ring around it. You need to create horizontal stripes on **Metal7** and drop via stacks to the pins for connectivity to **VDD**.

> **Note:** The stripes must be over the **ExtVDD** pins for the vias to be inserted.

Measure the pin sizes to determine the width of the strap. It might be helpful to set the view depth to **0** and turn on **Instance Pins** in the Display Options form.

1. In the layout canvas, choose the **Options – Display** menu entry or press the bindkey **E** to open the Display Options form.



   a. Set the view depth to **0** if not set already.

   b. Turn on the **Instance Pins** radio button.

   c. Click **OK** in the Display Options form.

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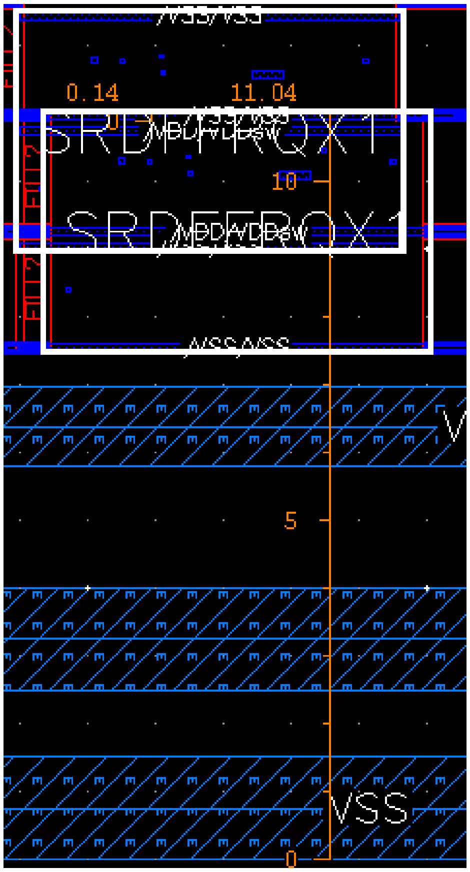
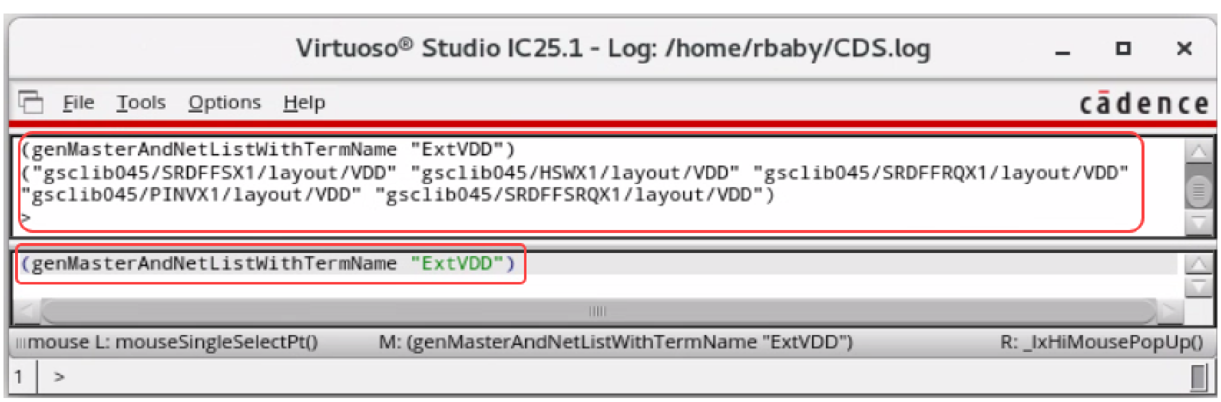
The **PINVX1** cell has a little wider pin, so you will use its width. The measurement of **SRDFFRQX1** on the bottom can be used to find the offset. The measurement of **SRDFFRQX1** above it can be used to find the repeat distance.



All the pins that need to be connected to **VDD** have terminal names **ExtVDD**. You can use SKILL® to find all the instance masters with pins named **ExtVDD** to make sure that your power stripes go through them. You have loaded the **scripts/ASICautoroute/setSchemes.il** file in the beginning of your lab setup. This file has the SKILL code with the procedure **genMasterAndNetListWithTermName**.

2. Enter the following command in the CIW window:

```
(genMasterAndNetListWithTermName "ExtVDD")
```



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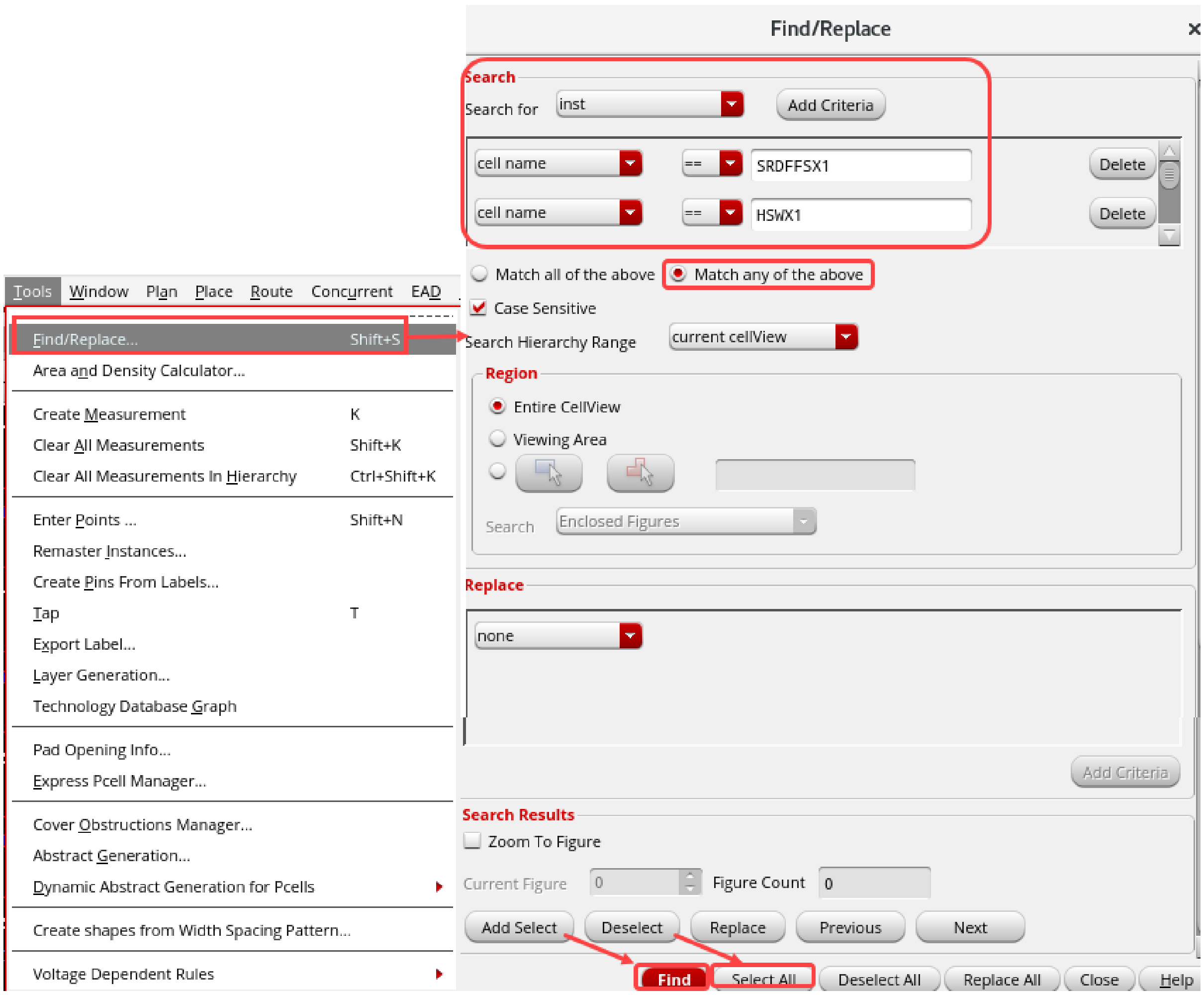
You get the following result in the CIW output field.

("gsclib045/SRDFFSX1/layout/VDD" "gsclib045/HSWX1/layout/VDD" "gsclib045/SRDFFRQX1/layout/VDD" "gsclib045/PINVX1/layout/VDD" "gsclib045/SRDFFSRQX1/layout/VDD")

3. In the layout canvas, choose the **Tools – Find/Replace** menu command or press the bind key **Shift+S** to open the Find/Replace form.

   Use the **Find/Replace** utility to look at all the instances to see that the pins line up.



   a. Enter the **5** cell master names from your execution of **(genMasterAndNetListWithTermName “ExtVDD”)** in the Find/Replace form.

   b. Now click the **Find** radio button.

   c. Then click on the **Select All** button.

   d. Close the **Find/Replace** form.

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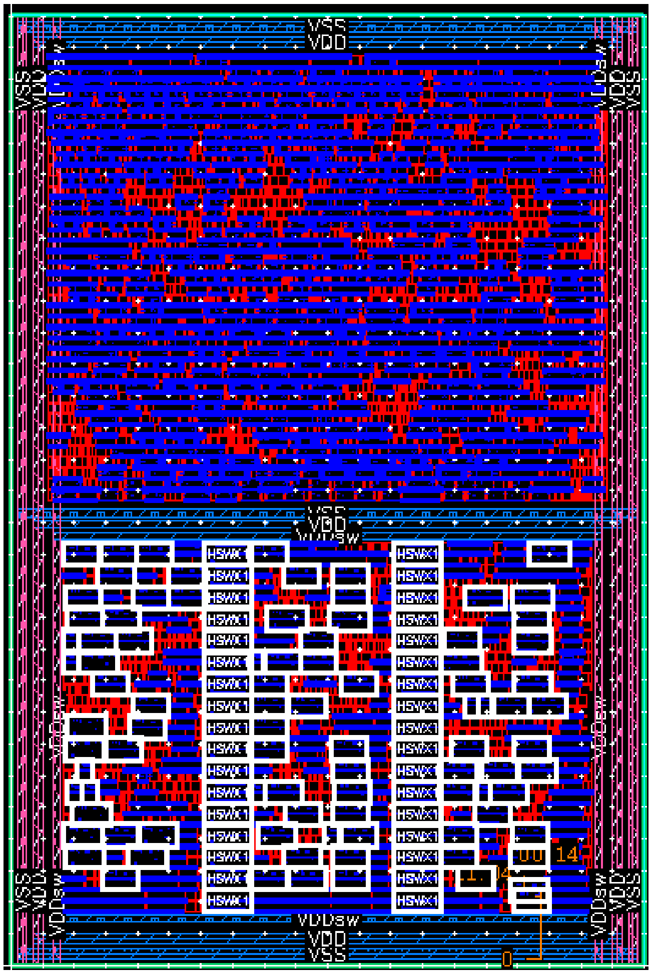
All the cells with the **5** cell master instances are selected, as shown here.



The GUI has no way to restrict the creation of the stripes to just the region over the lower group of standard cells. Measure a bBox around that area, including the **VDD** power ring.

4. Execute the following command in the CIW window:

```
(rdeEval "proute_stripes -layers Metal7 -nets VDD -net_width 0.14 -
   y_step 3.42 -direction horizontal -bottom_offset 11.04 -routing_area
   [list 3.5 0.0 96.5 66.75]")
```

Horizontal **VDD** power stripes to **ExtVDD** pins are created.

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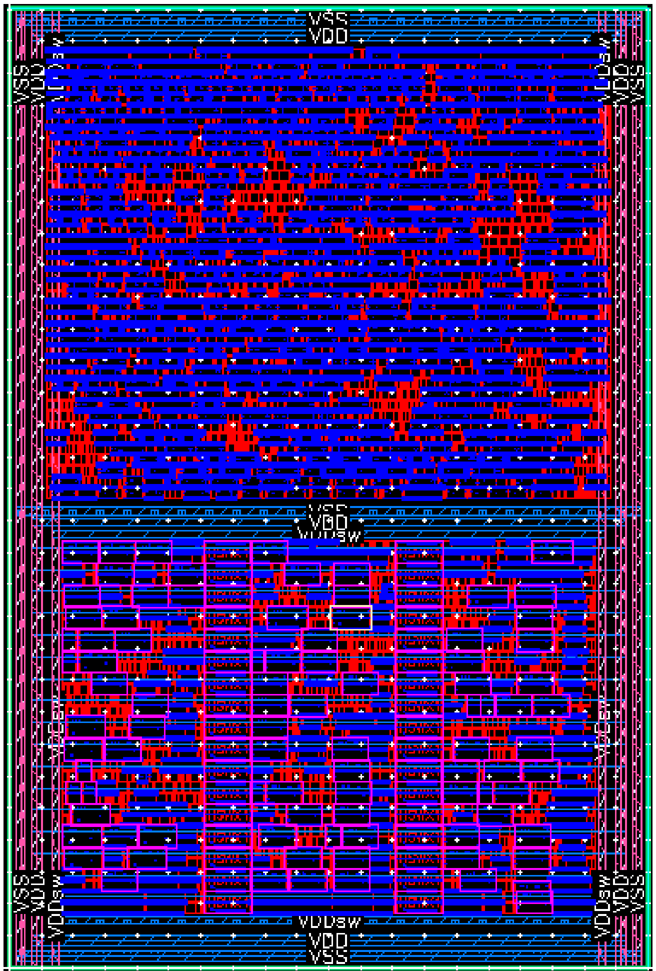
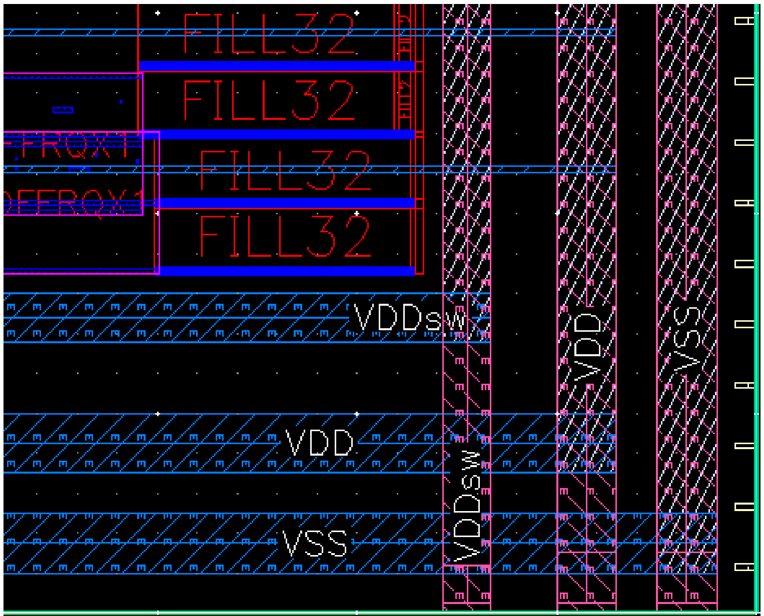
Your layout should look like the graphic shown here.



5. Zoom in and view the **VDD** power stripes, as shown here.



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Vertical Stripes Creation for VSS, VDD and VDDsw Pins

Now you will create the vertical stripes. You will create **2** sets of stripes for **VDD**, **VSS**, and **VDDsw** and want them on **Metal6** with **30um** apart. You want the stripes **1um** wide with a spacing of **1um**. Find the x-coordinate about one-third of the way into the block from the left.

Note: The **VDD** stripe must be over the instances of master **gsclib045/HSWX1/layout** to connect the stdcell pin to the stripe.

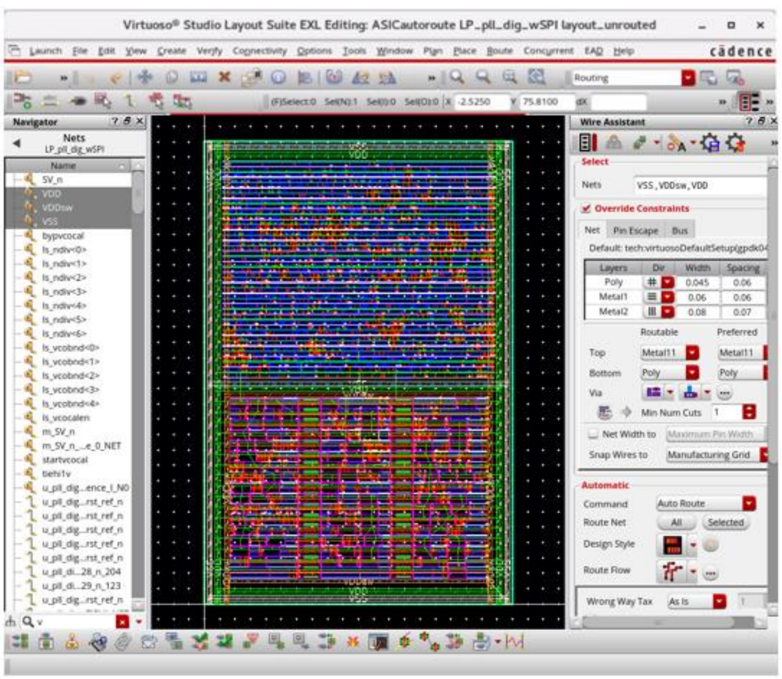
With this information, you should be able to check/fill the form values in the Power Routing form.

1. Make sure that you are on the **Stripes** tab in the Power Routing form.

   Note: You may have to go to the **Default** *Scheme* to activate the **Power Routing** tabs.

2. Select the nets **VDD** and **VSS** in the *Nets Ordering* list in the Power Routing form and click the **X** button to remove it.

   a. Select the nets **VSS**, **VDD**, and **VDDsw** in the Navigator assistant.

   b. Click the **Selected Nets** button in the Power Routing form.

   The nets **VSS**, **VDD**, and **VDDsw** are added to the *Nets Ordering* list in the Power Routing form.

   c. Select **Stripes_Vertical** in the *Scheme* section in the Power Routing form.

   d. Then click the **Edit…** button in the Power Routing form.

   The Power Routing Options form opens as shown here.

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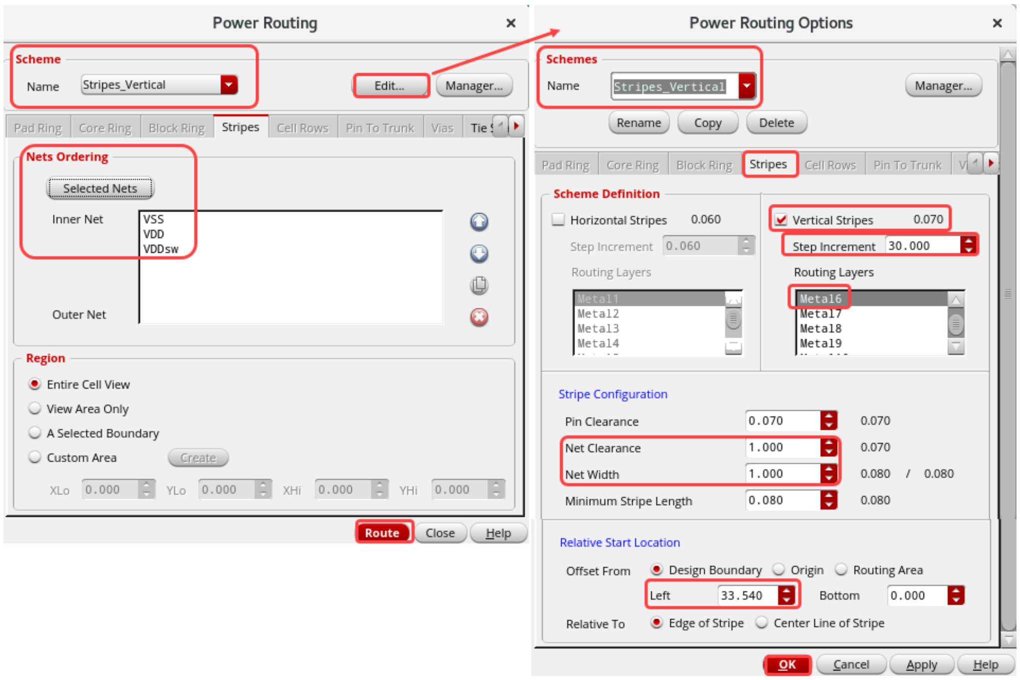
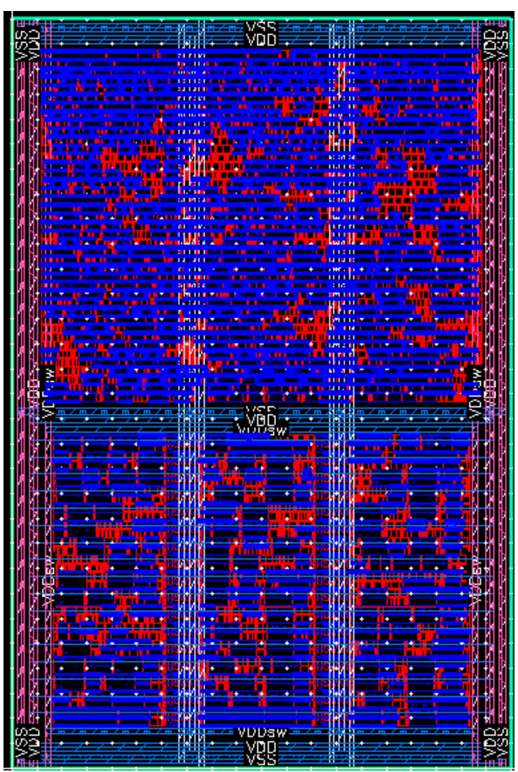
e. Verify whether the **Power Routing Options** form shows the values that you have measured in the previous steps, as shown here.



f. Click **OK** in the Power Routing Options form.

g. Click the **Route** button in the Power Routing form.

The Vertical Stripes are created for the **VSS**, **VDD**, and **VDDsw** pins. You have finished with the **Vertical Stripes** connections. Your layout should look like this.



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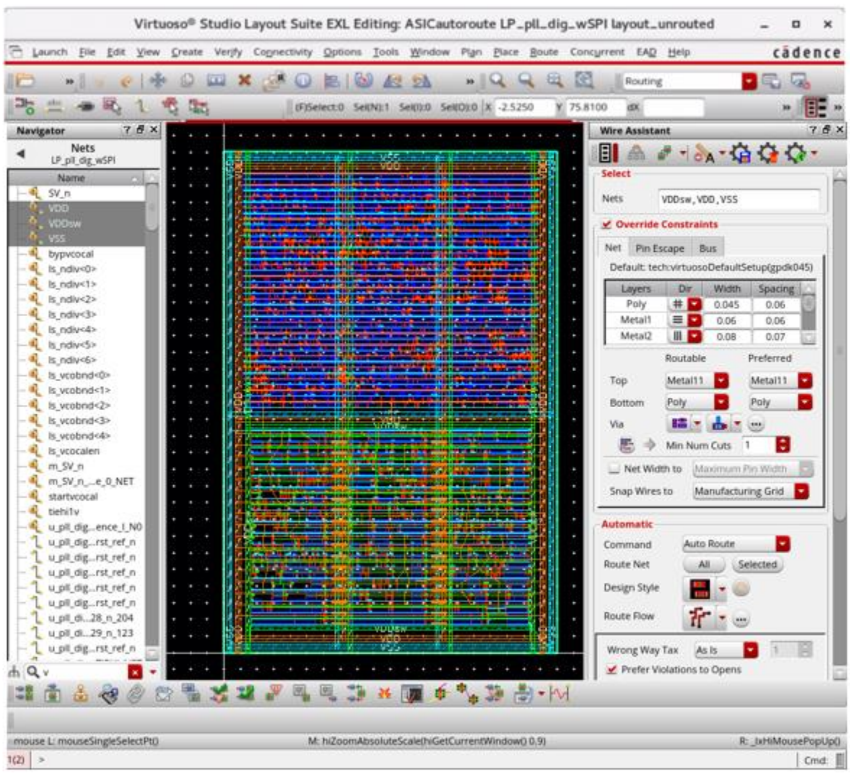
Cell Rows Core Power Rail Routing

The next power structure that you want to add is the core power rail routing. This step is different; here, the nets need to be selected in the Navigator assistant.

Select the nets **VDD**, **VDDsw,** and **VSS** in the Navigator assistant and specify **Metal1** as the routing layer, extending the power rail routing to the **End of Row**. Also, select the **Extend Straps to Rails** option to make the connection from the stdcell rows to the rings.

1. Select the nets **VDD**, **VDDsw**, and **VSS** in the Navigator assistant.



2. Click the **Cell Rows** tab in the Power Routing form.

Note: You may have to go to the **Default** *Scheme* to activate the **Power Routing** tabs.

a. Select **CellRow** in the *Scheme* section in the Power Routing form.

b. Then click the **Edit…** button in the Power Routing form.

The Power Routing Options form opens as shown here.

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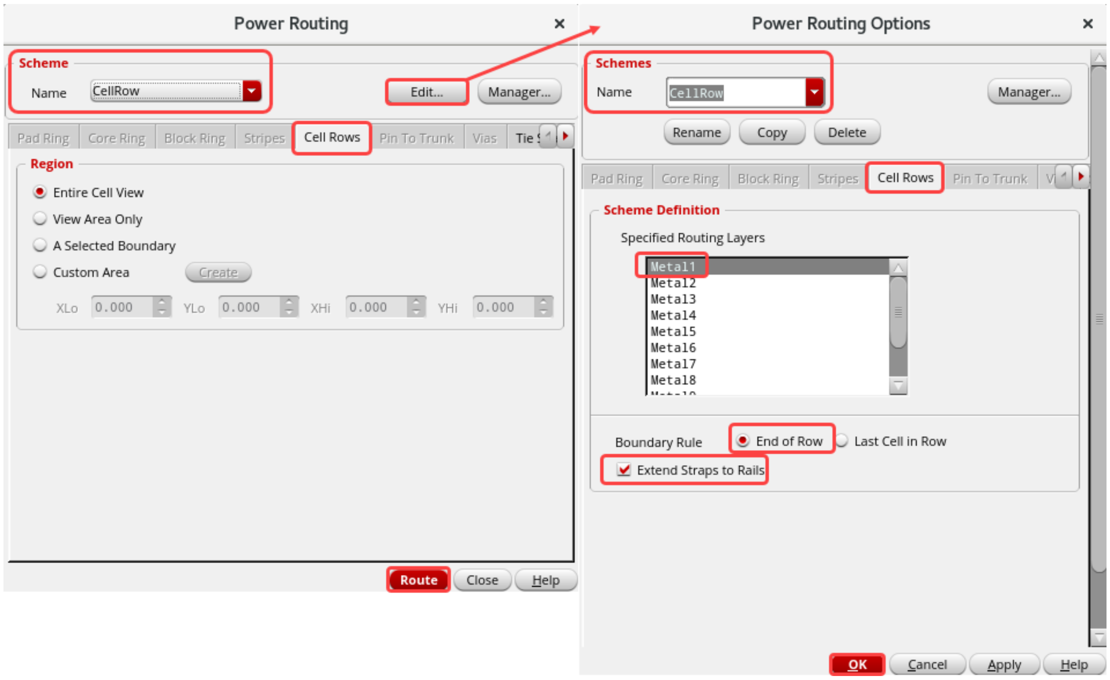
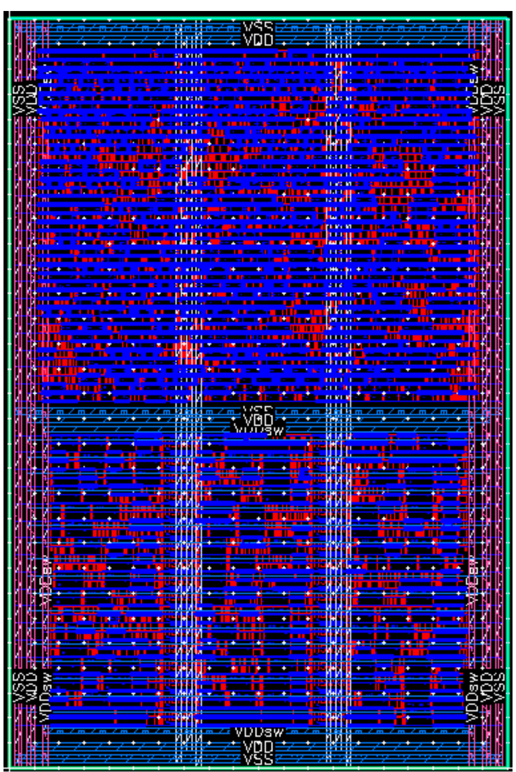
c. Verify whether the **Power Routing Options** form shows the values that you have mentioned in the previous steps.



d. Click **OK** in the Power Routing Options form.

e. Click the **Route** button in the Power Routing form.

The Cell Rows for **VDD**, **VDDsw**, and **VSS** are created. You have finished with the **Cell Rows** connections. Your design should look like this.



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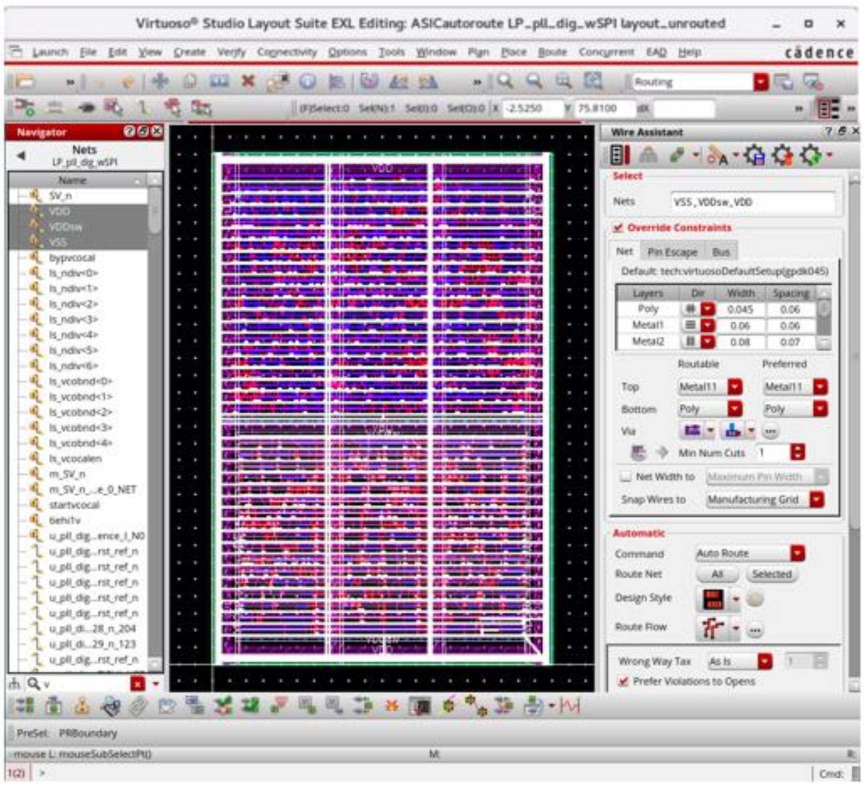
f. Observe the **CIW/CDS.log** for the Cell Rows Creation log information.

Inserting Vias

Now you need to insert vias to connect up the metal power routing.

1. Select all the **3** power nets **VDD**, **VDDsw**, and **VSS,** in the Navigator assistant.



2. Click the **Vias** tab in the Power Routing form.

   Note: You may have to go to the **Default** *Scheme* to activate the **Power Routing** tabs.

   a. Select **Vias** in the *Scheme* section in the Power Routing form.

   b. Then click the **Edit…** button in the Power Routing form.

      The Power Routing Options form opens, as shown in the next step.

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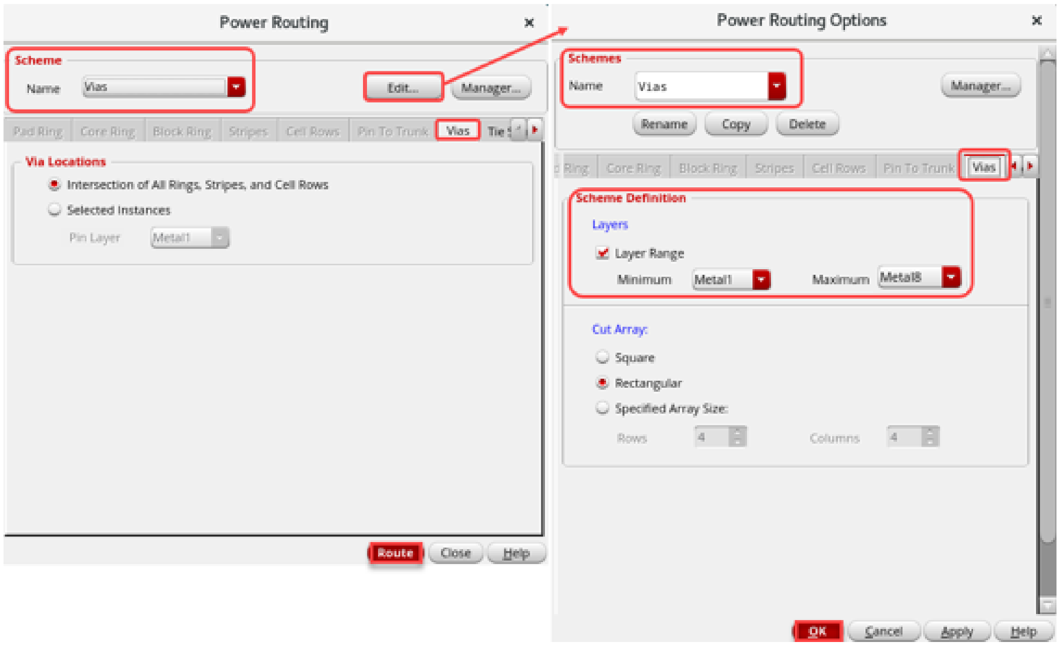
c. Set the **Layer Range** from **Metal1** through **Metal8** in the Power Routing Options form.



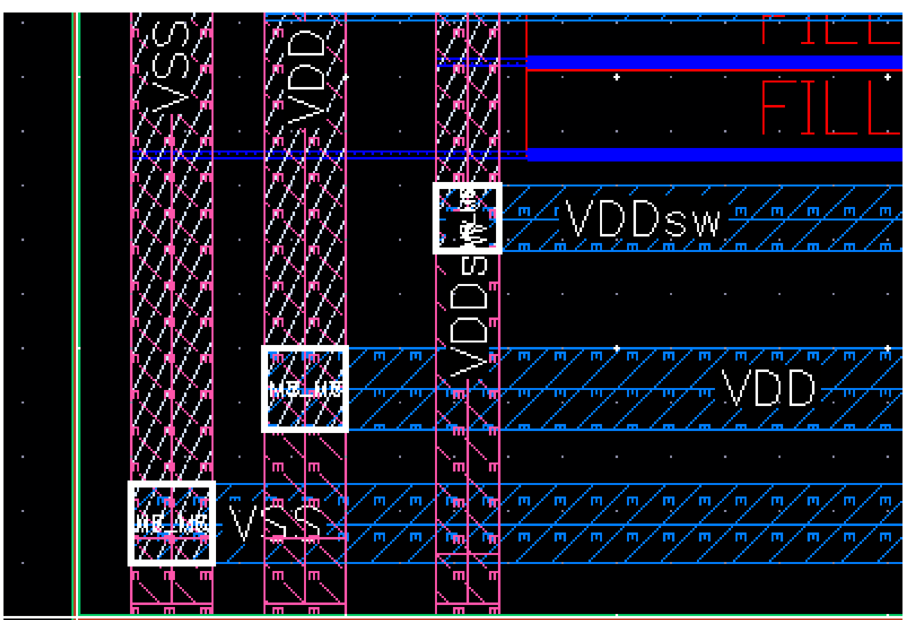
d. Click **OK** in the Power Routing Options form.

e. Click the **Route** button in the Power Routing form.

The Vias are inserted to connect up the metal power routing.

f. Observe the **CIW/CDS.log** for the via insertion log information.

g. Zoom in and notice that there are vias in all the intersections of metal horizontal and vertical power routing structures.



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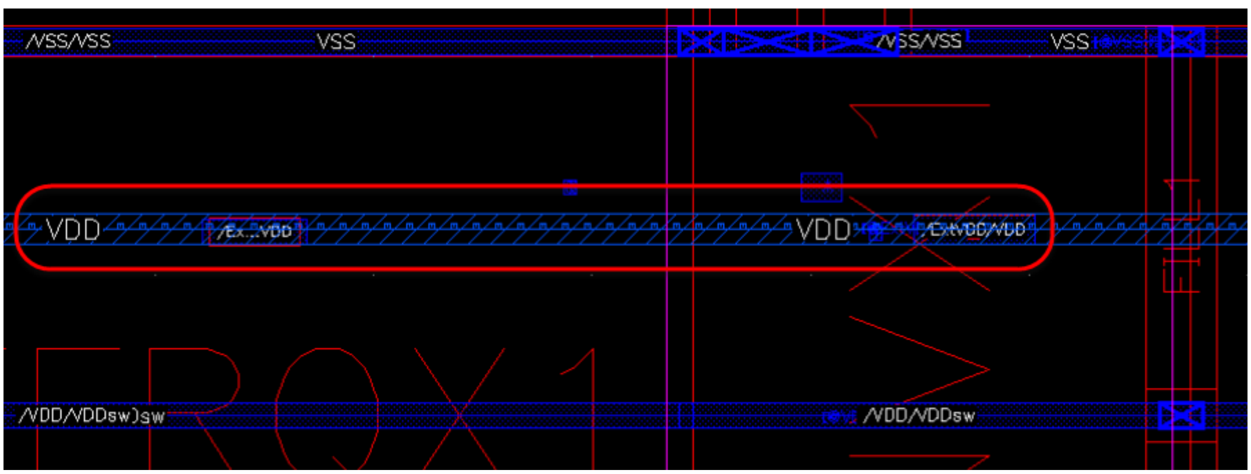
Now the last via connection you need are from the **VDD** pins on the instances of masters with terminals named **ExtVDD**. The SKILL was provided for you previously with the procedure **genMasterAndNetListWithTermName** to get the list of masters with terminals. You can use that SKILL with this to create via connections from **Metal7** horizontal straps, and **Metal6** vertical straps to the instance pins that need to be connected to **VDD**. You loaded the **scripts/ASICautoroute/setSchemes.il** file in the beginning of your lab setup. This file has the SKILL code with the procedure **viaInsertionToTerm** that uses the SKILL procedure **genMasterAndNetListWithTermName** to execute all the *proute_via_insertion* commands to connect up all the proper instances.

3. Enter the following command in the CIW window:

```
(viaInsertionToTerm "ExtVDD")
```

The procedure selects all the instances of the master with terminals named **ExtVDD**, and gathers their names in a list (instNamesList). It passes that list of instances into another *proute_via_insertion* command with net connected to the instTerm **ExtVDD** (**VDD** in all of these cases) and pin layer **Metal1**. You should now see that the pin is connected to the stripe. You will see spurious parsing errors in the CIW window. You can ignore them. Check to see that all the instance pins are connected.

The zoomed in graphic view is shown here.



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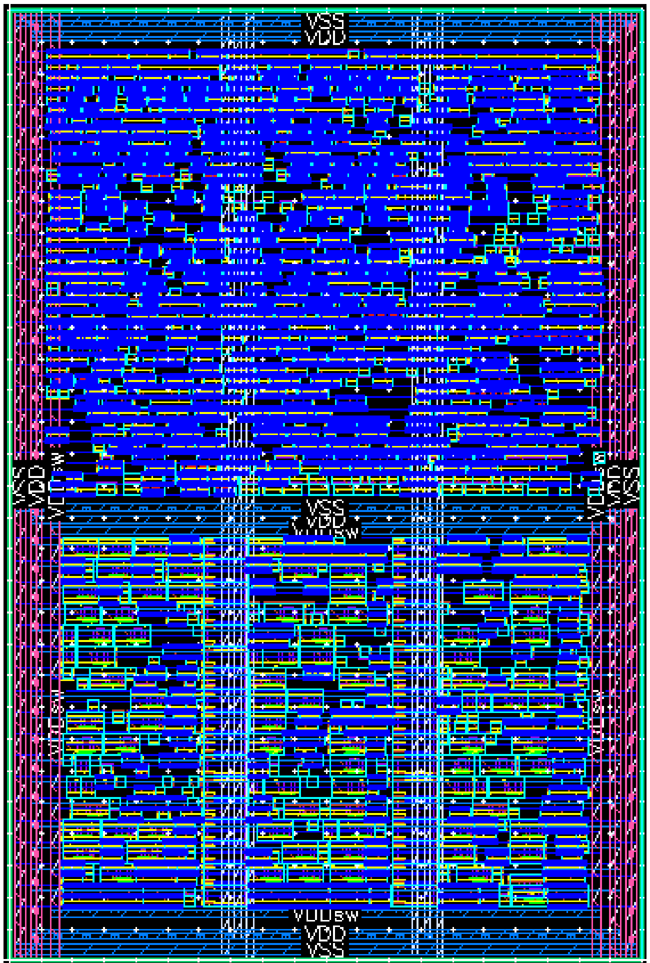
Your layout should look like the graphic shown here.



Trimming the Dangling Stripes

The next step is to trim the dangling stripes.

1. Execute the following command in the CIW window:

   ```
   (rdeEval "proute_trim_stripes -nets {VDD VSS VDDsw}")
   ```

   The nets **VDD**, **VSS**, and **VDDsw** are trimmed.

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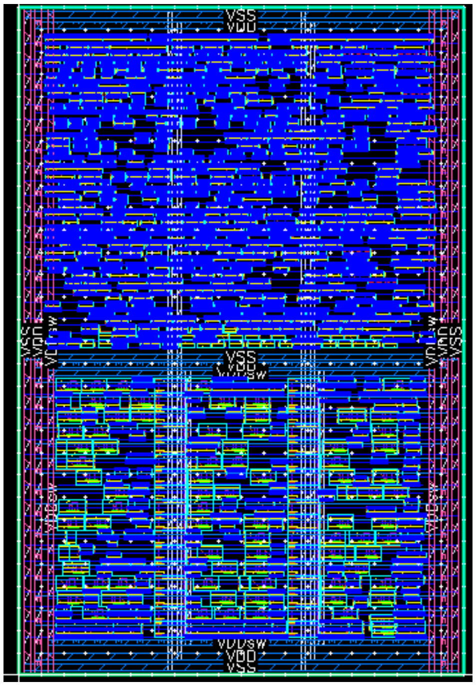
Your layout should look like the graphic, as shown here.



You have now successfully created all the power routing structures for your design.

2. Save the design to *ASICautoroute/myLP_pll_dig_wSPI/layout_power_routed*.

3. Close the **Power Routing** form.

4. Close all the cellviews you had opened without saving them.

5. Exit the Virtuoso session.

Summary

In this lab, you learned about the following:

- How to implement a power structure on a given design using VSR.

End of Lab

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Lab 3-2 Signal Routing

Objective: To see how to run automatic routing in the Wire Assistant.

In this lab section, you will:

- See how to run automatic routing in the Wire Assistant.

Typically, you would run your clock routing first in a digital routing tool like EDI/Innovus if your timing is critical. If it is not, and you can use MST routing for your clock tree, you can proceed with this lab with the modifications shown in the Tcl files.

There is a **Common Storage Mechanism** (**CSM**) that sets values for consistency between the routing results for assisted and automatic routing commands. In the terminal window, edit your *.cdsenv* file to have the following:

```
layout enableMaximizeCuts boolean nil
layout useDoubleCutVias bollean nil
```

This will turn off maximizing cuts and putting in double-cut vias during routing. You should do them later in the flow, not during routing.

Opening the Software

1. To start the Cadence software, enter this command in the same terminal window:

   ```
   virtuoso &
   ```

   The Command Interpreter Window (CIW) appears.

2. In the CIW, choose **File – Open** and enter the following values in the form to open the cellview in EXL mode:

Library	ASICautoroute
Cell	LP_pll_dig_wSPI
View	layout_power_routed

   Note: If you have completed the power routing, then you can use **ASICautoroute/myLP_pll_dig_wSPI/layout_power_routed** that you had saved in the previous lab. In case you have not finished, then you can continue with this previously created view **ASICautoroute/LP_pll_dig_wSPI/layout_power_routed**.

   The Virtuoso Layout Suite EXL Palette opens.

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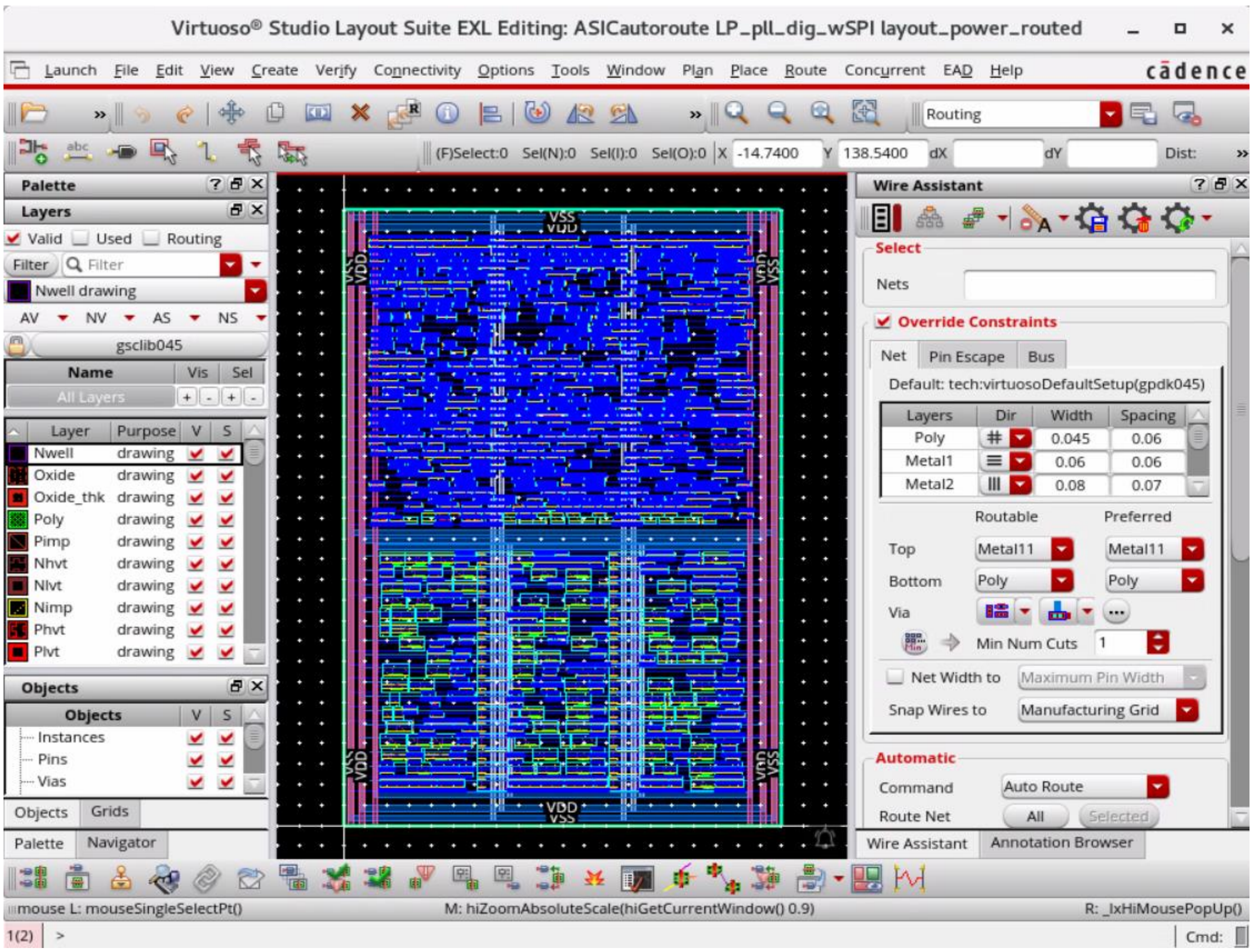
Note: In the **Workspace Configuration**, you can use the predefined workspace *Routing*.

This brings up the Wire Assistant, Annotation Browser, Navigator, and Palette assistants.

3. Press the **Shift+F** bindkey to view all the levels of the hierarchy in the layout canvas if it is not already enabled.

   The layout should look something like the graphic, as shown here.



4. Select all the nets in the Navigator assistant.

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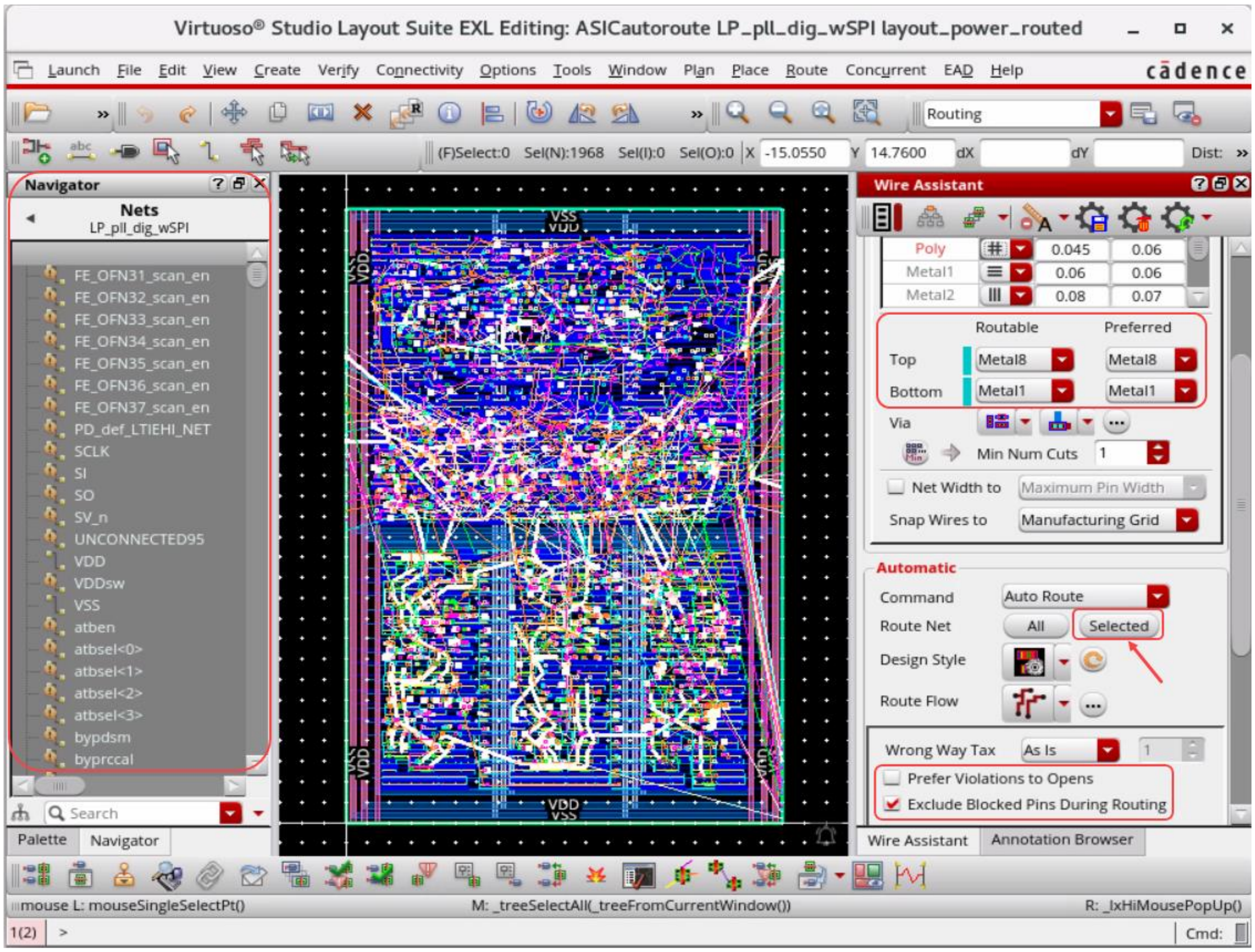
Running Signal Routing

1. In the layout canvas, in the Wire Assistant, set the following:



a. *Routable*: *Top*: **Metal8**; *Bottom*: **Metal1**

b. *Prefer Violations to Opens* – **Off**

c. *Exclude Blocked Pins During Routing* – **On**

d. Click the **Selected** button in the Wire Assistant **Auto Routing** menu.

Note: The reason you are doing this instead of just clicking the **All** button is that the **tiehi** and **tielo** connections will NOT route unless power and ground nets are explicitly selected in the Navigator assistant.

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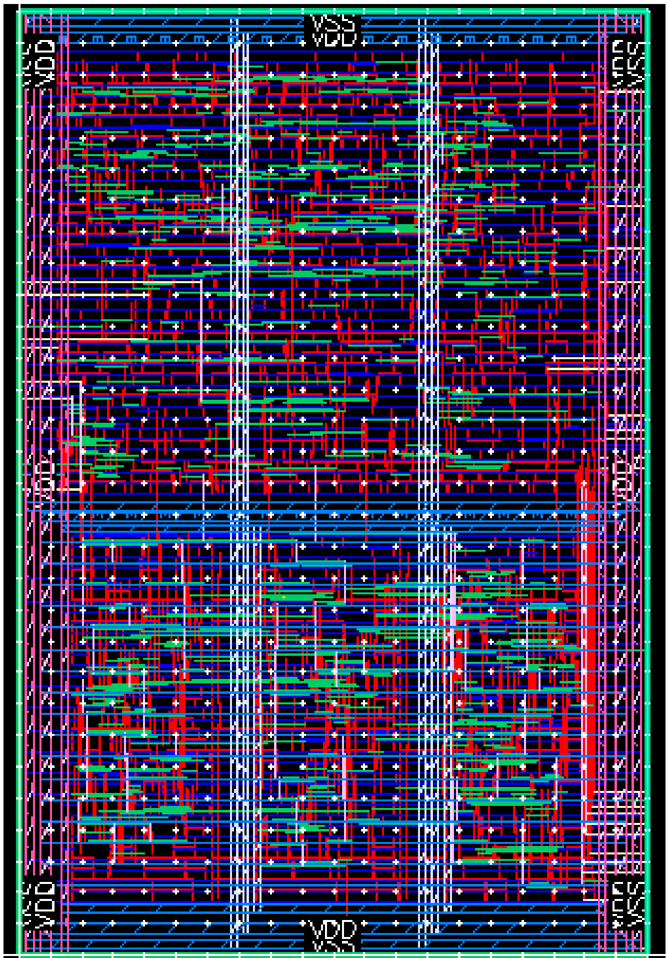
Your design should look something like this when you are finished.



You have now successfully created the signal routing for your design.

2. Save the design to *ASICautoroute/myLP_pll_dig_wSPI/layout_signal_routed*.

3. Observe the **CIW/CDS.log** for the Route Summary log information.

4. Close all the cellviews you had opened without saving them.

5. Exit the Virtuoso session.

Summary

In this lab, you learned about the following:

- How to run signal routing using the Wire Assistant.

End of Lab

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Lab 3-3 Connectivity Verification and DRCs

Objective: To see how to run connectivity checking for opens and shorts and Design Rule Checking on your routing results.

In this lab section, you will:

- See how to run connectivity checking for opens and shorts and Design Rule Checking (DRC) on your routing results.

Opening the Software

1. To start the Cadence software, enter this command in the same terminal window:

   ```
   virtuoso &
   ```

   The Command Interpreter Window (CIW) appears.

2. In the CIW, choose **File – Open** and enter the following values in the form to open the cellview in EXL mode:

Library	ASICautoroute
Cell	LP_pll_dig_wSPI
View	layout_signal_routed

   Note: If you have completed the signal routing, then you can use **ASICautoroute/myLP_pll_dig_wSPI/layout_signal_routed** that you had saved in the previous lab. In case you have not finished, then you can continue with this previously created view **ASICautoroute/LP_pll_dig_wSPI/layout_signal_routed**.

   The Virtuoso Layout Suite EXL palette opens.

   Note: In the **Workspace Configuration**, you can use the predefined workspace *Routing*.

   This brings up the Wire Assistant, Annotation Browser, Navigator, and Palette assistants.

3. Press the **Ctrl+F** bindkey to bring the view depth to **0** in the layout canvas.

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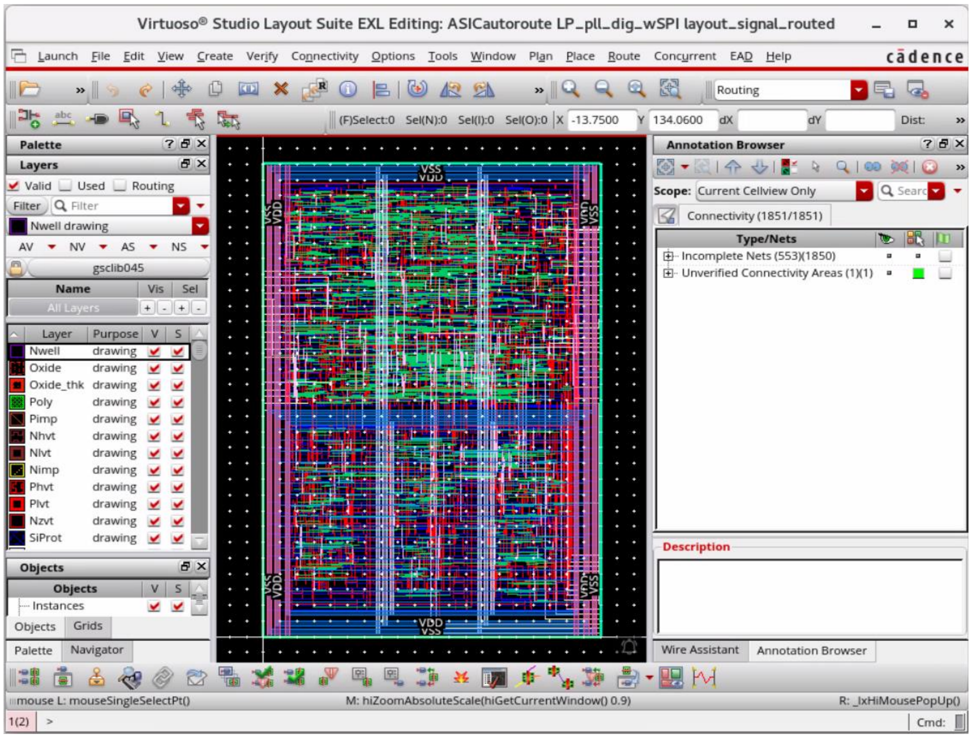
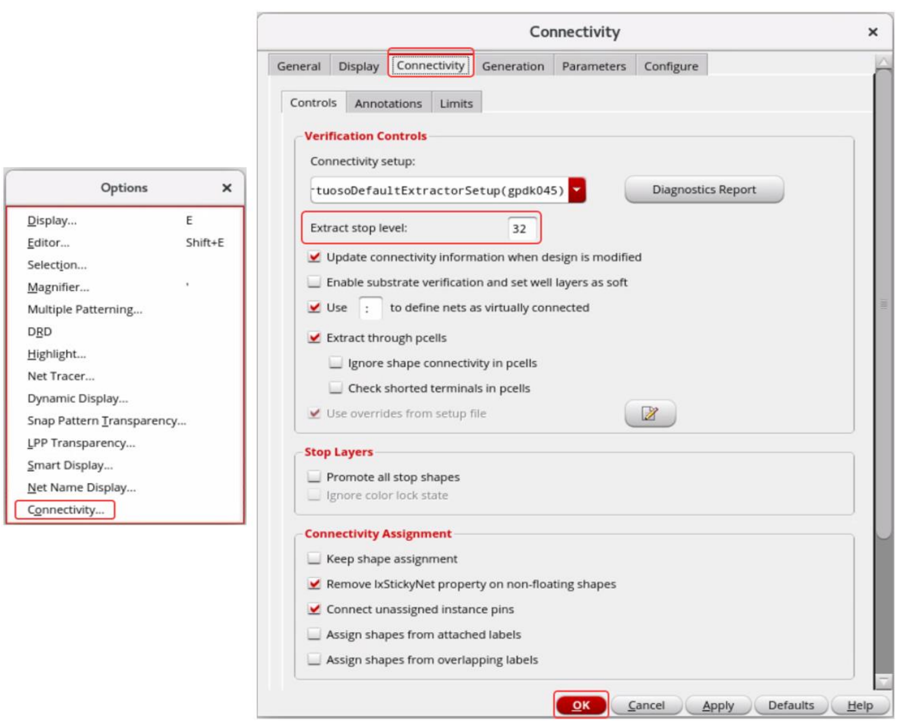
The layout should look something like the graphic, as shown here.



Extracting the Design

1. In the layout canvas, choose the **Options – Connectivity** menu command to open the Connectivity form.



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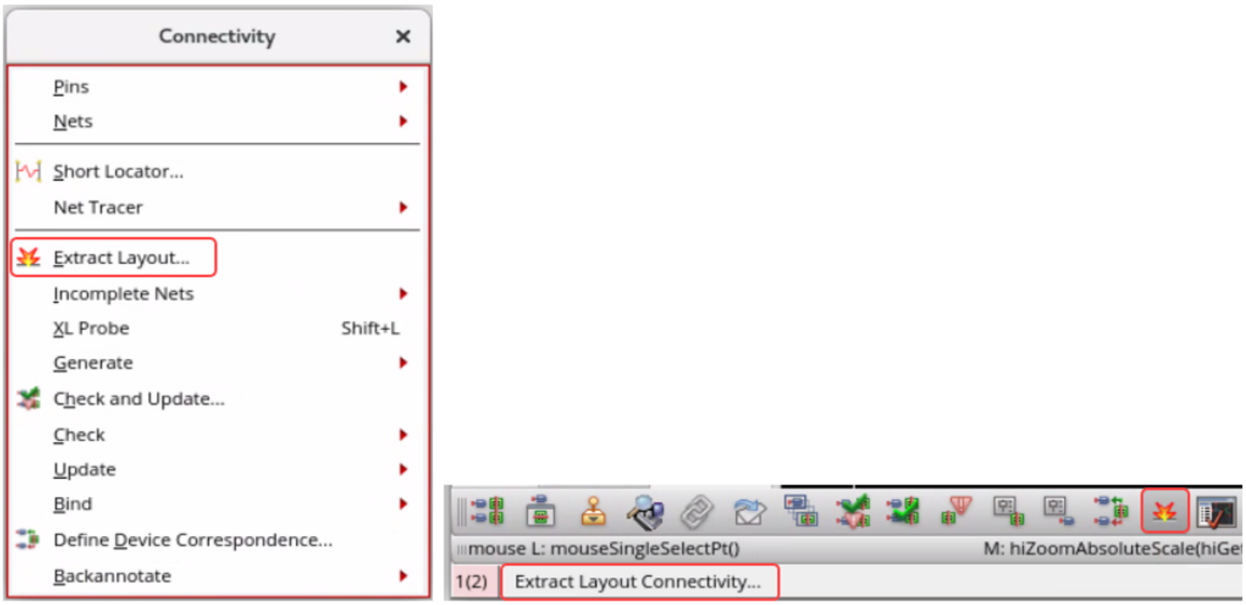
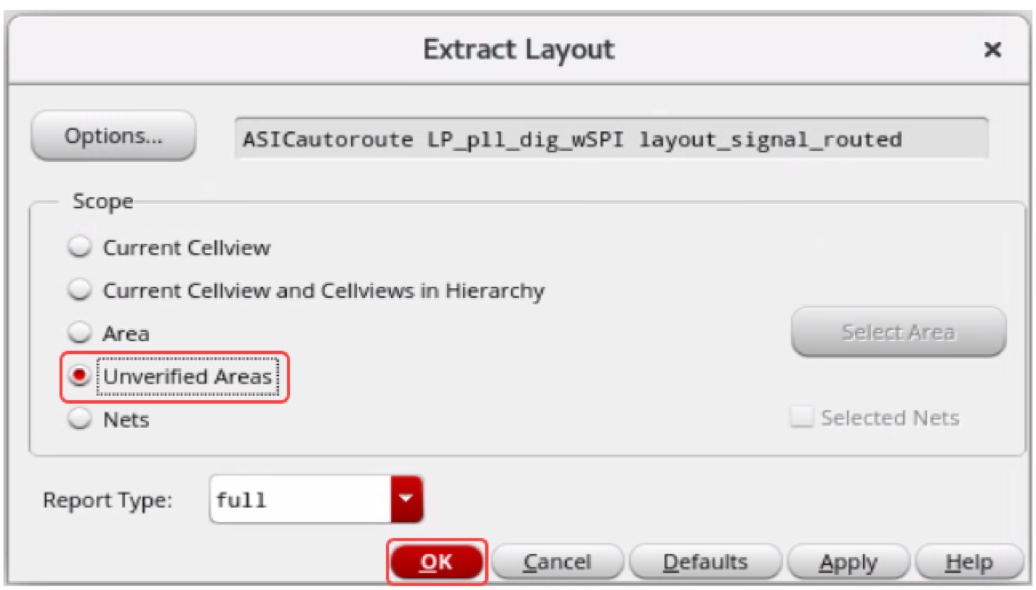
a. Set *Extract stop level:* as **32** in the **Connectivity** tab as shown here.

b. Click **OK** in the Connectivity form.

2. In the layout canvas, choose the **Connectivity – Extract Layout** menu command or use the **Extract Layout Connectivity** icon in the Layout XL Toolbar to open the Extract Layout form.



The Extract Layout form appears, as shown here.



a. Select *Unverified Areas* in the **Scope** section.

b. Click **OK** to run the extraction.

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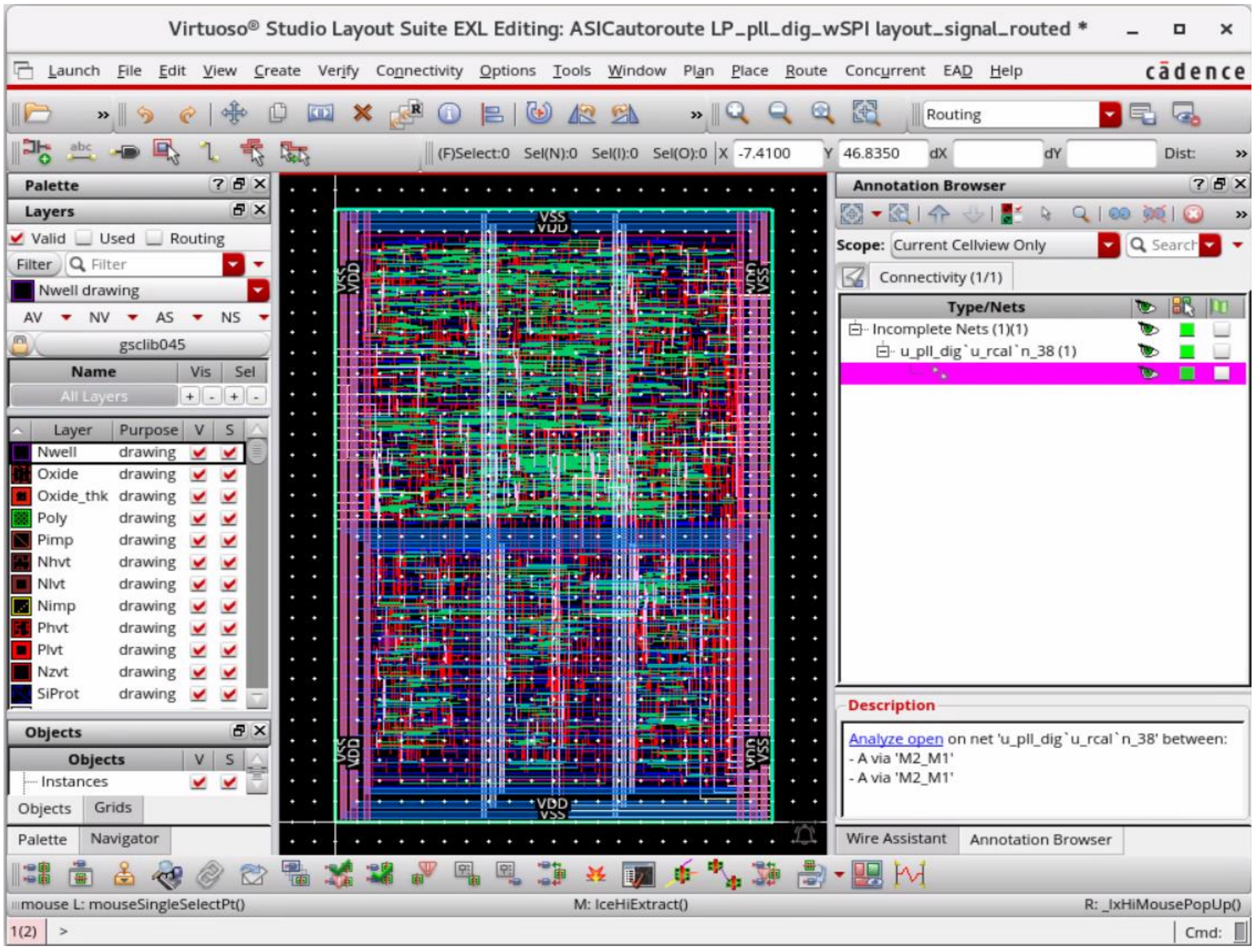
Your design will look like this when you are finished with the extraction.



Fixing the Connectivity Errors

There should be 0 opens and 0 shorts. This build has 1 open and 0 shorts (1 **M2_M1** via shifted to the right). If you have opens, you can select the incomplete nets from the Annotation Browser by selecting the opens and **right**-click to choose **Select Nets**. You may be able to just click the **Auto Route the Selected** button in the Wire Assistant and they may route. In this case, just select the via and move it to the left to complete the routing. Then stretch the **Metal2** jog down to prevent DRCs.

1. To fix the errors, in the layout canvas, do the following:

   a. Select the incomplete net in the Annotation Browser assistant.

   b. Click the **Zoom To Selected** button in the Annotation Browser assistant.

   c. **Right**-click and choose **Select Nets** in the Annotation Browser assistant.

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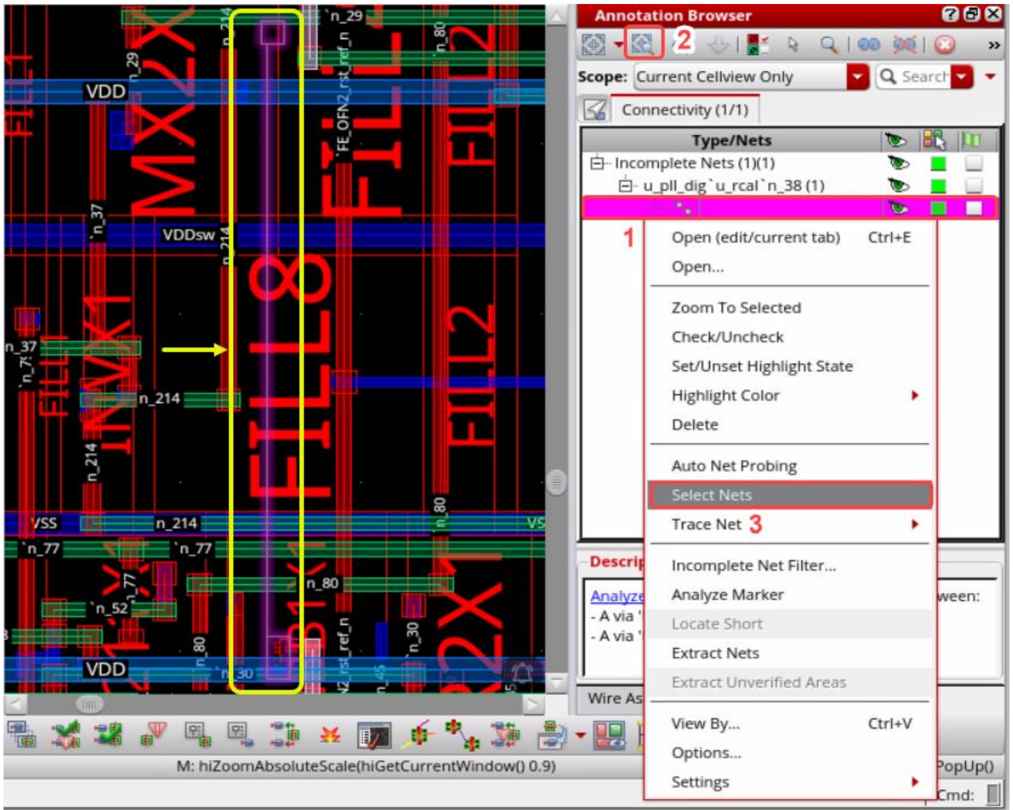
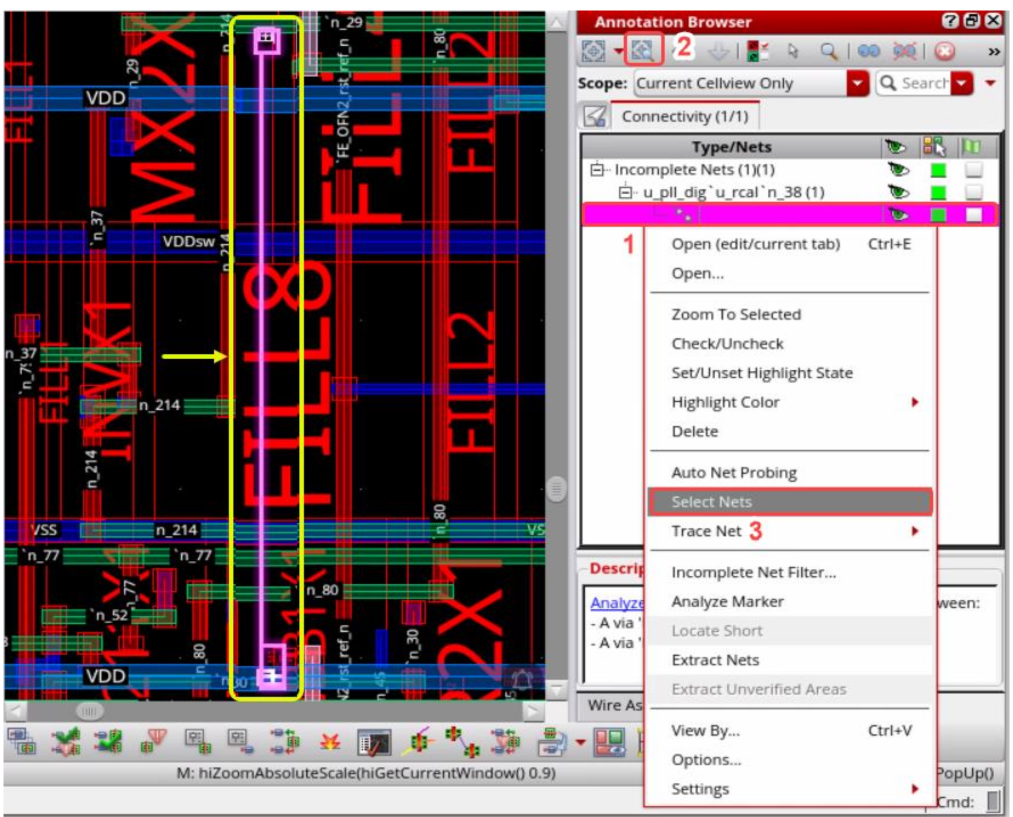
d. Click the **Selected** button in the Wire Assistant **Auto Routing** menu.





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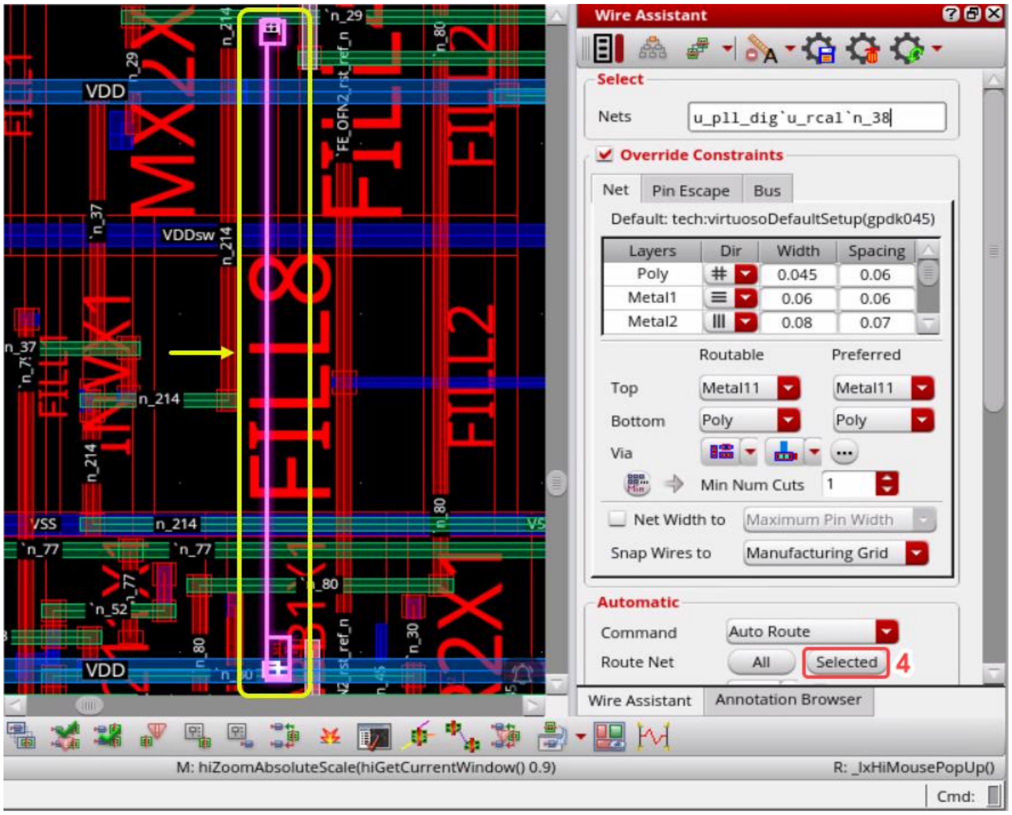
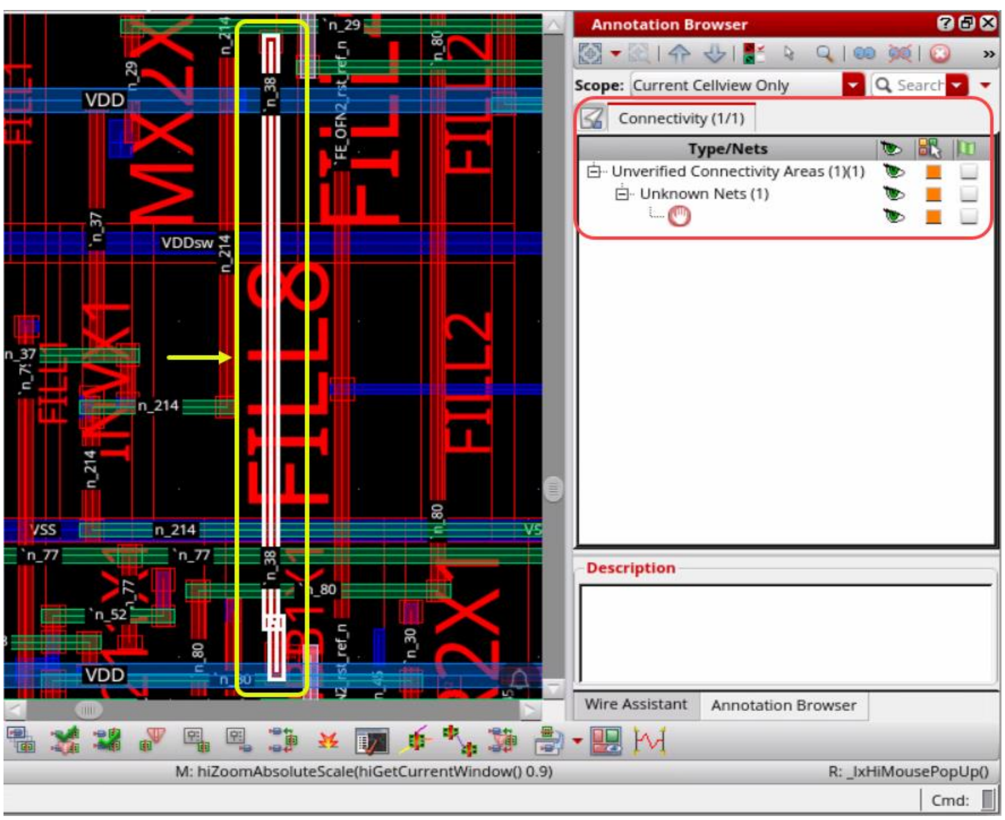


Your design will look something like this when you are finished.



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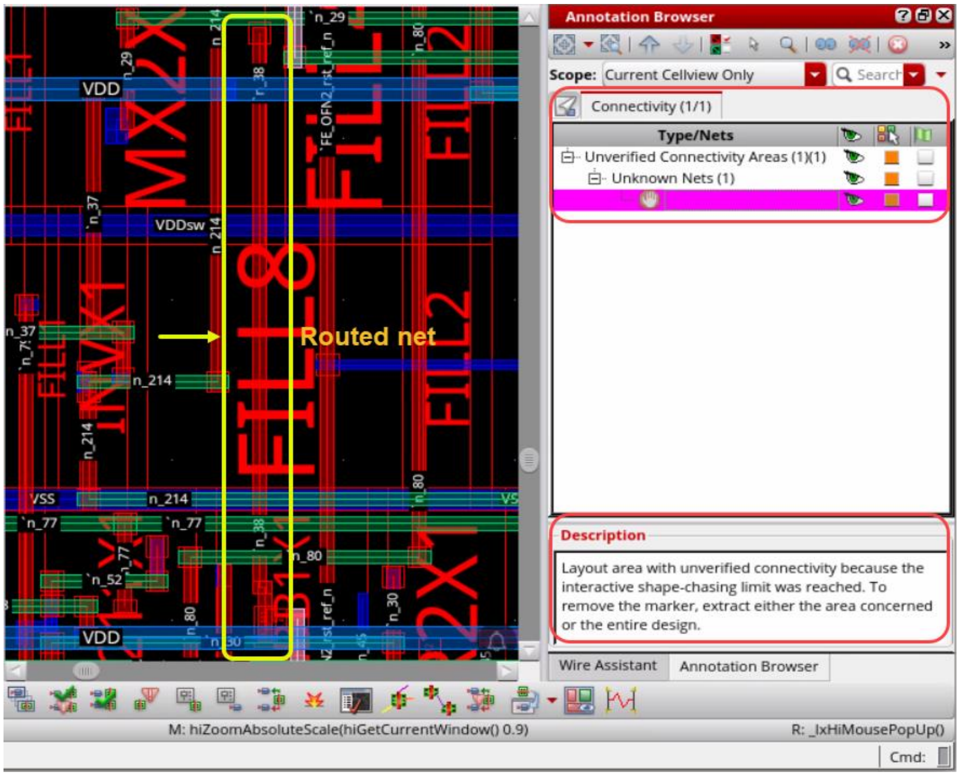
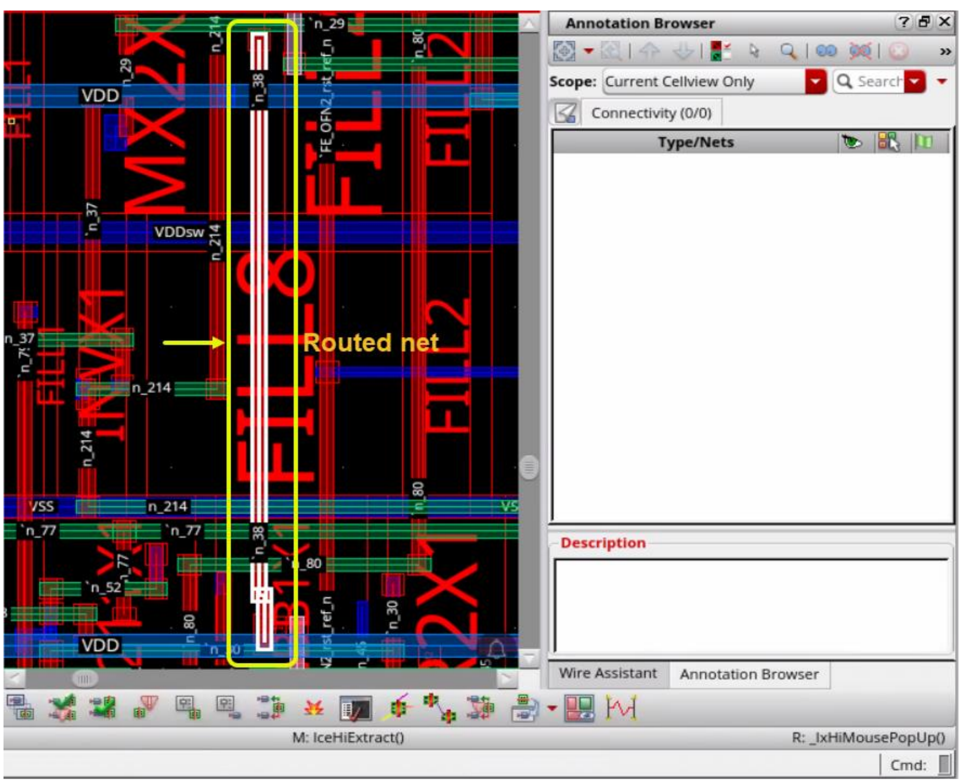
e. Run **Connectivity Extraction** if required as done in the previous step, if you see some unverified connectivity markers in the Annotation Browser assistant.



After running the connectivity extraction, your layout design will look something like the graphic shown here when you are finished.



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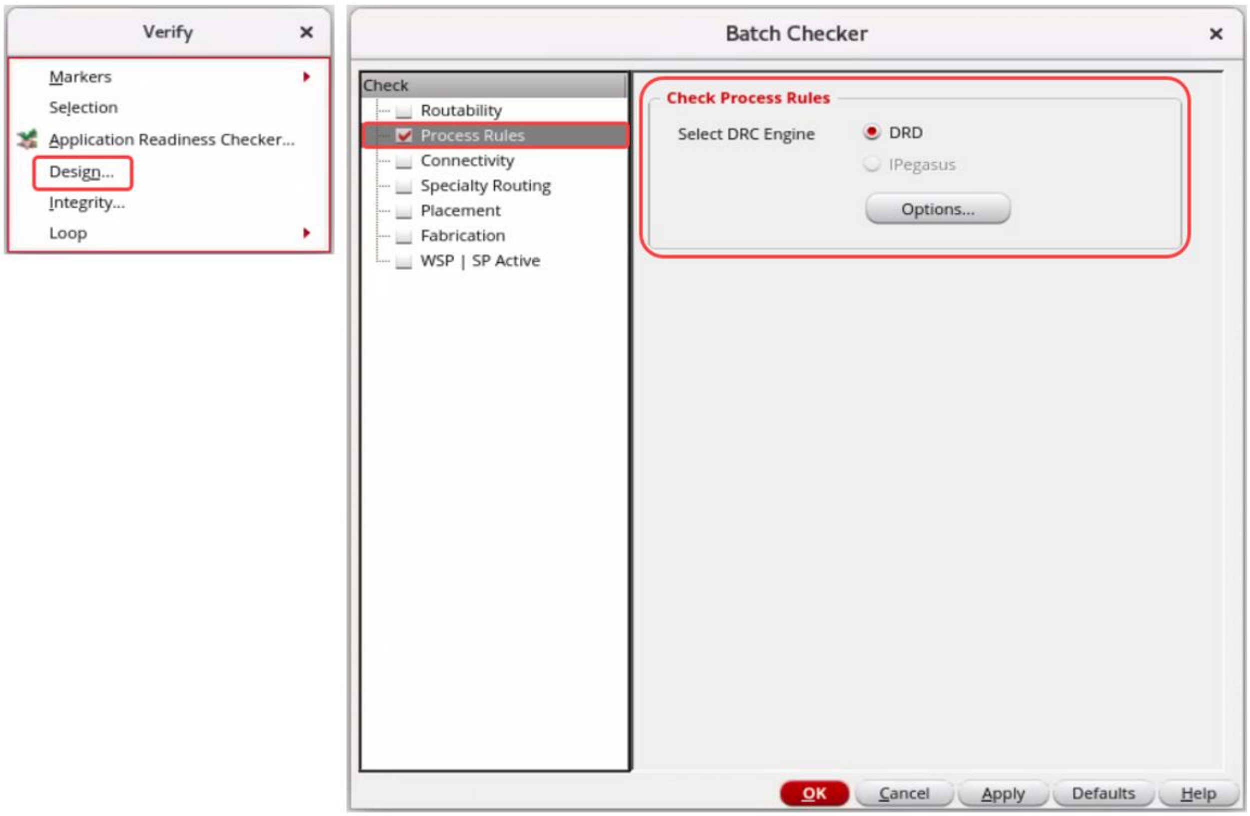
2. Save the design to *ASICautoroute/myLP_pll_dig_wSPI/layout_all_routed* and continue with the next section to verify the design.

Verifying the Design (DRC Check)

Now you will run the DRC check.

1. Choose **Verify – Design** in the layout canvas to open the Batch Checker form.



a. In the **Process Rules** section, under the **Check Process Rules** section, select the *Options* button to open the *DRD Options* form.

b. In the *Batch* tab, change the **Hierarchy Range** to *current to bottom*.

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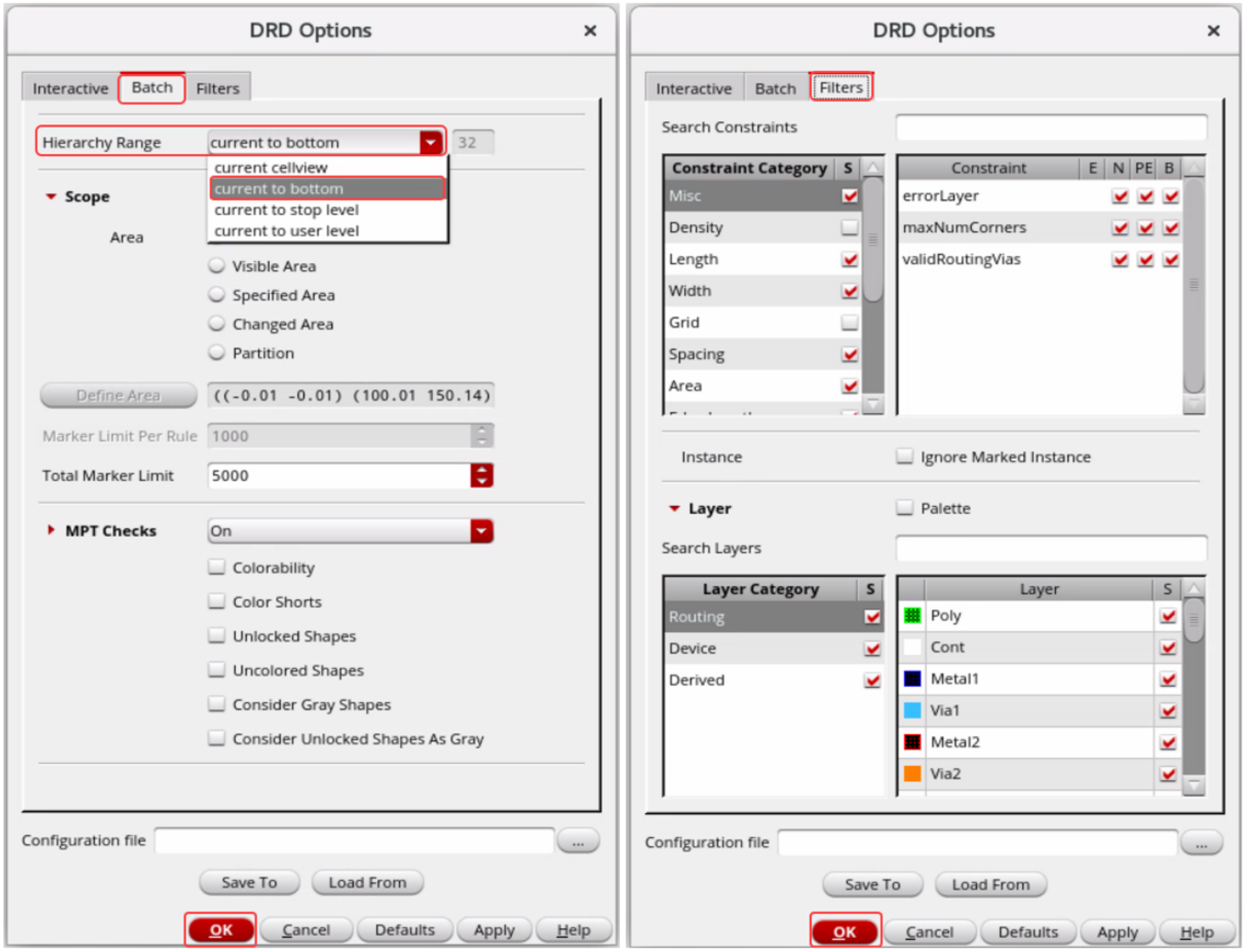
c. In the *Filters* tab, keep the values as default, as shown here.



d. Click **OK** in the DRD Options form.

e. Click **OK** in the Batch Checker form to run DRCs.

There were no errors introduced by the router (only FEOL errors). You can use the Annotation Browser assistant to view and fix errors if you had any.

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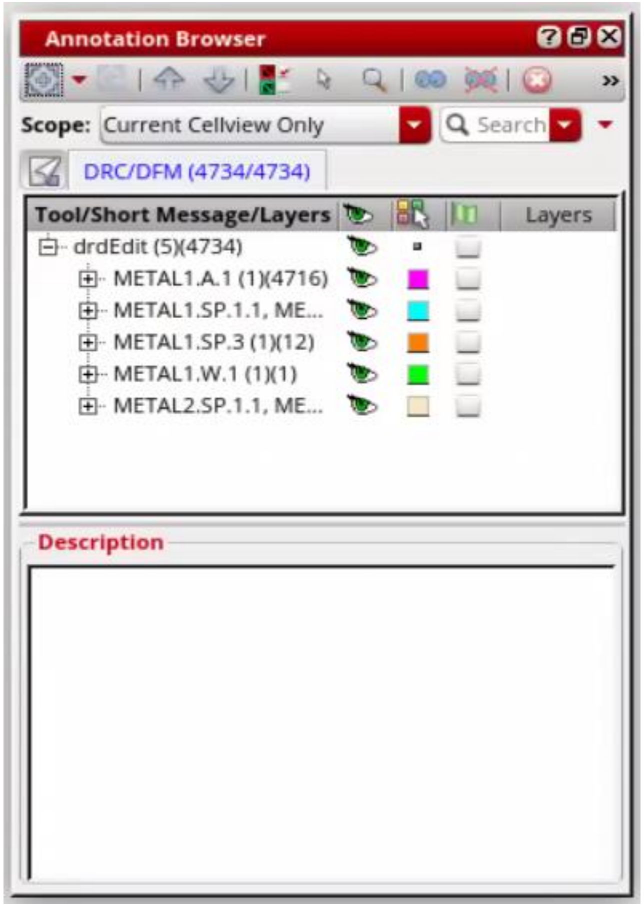
Note: FEOL – Front End of Line (usually below M1, refers to transistor layers)



You have verified your design.

2. Close the design without saving.

3. Exit the Virtuoso session.

Summary

In this lab, you learned about the following:

- How to verify your design.

End of Lab

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